

Metabolite Reorganization Results

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Node/Edge Overlaps

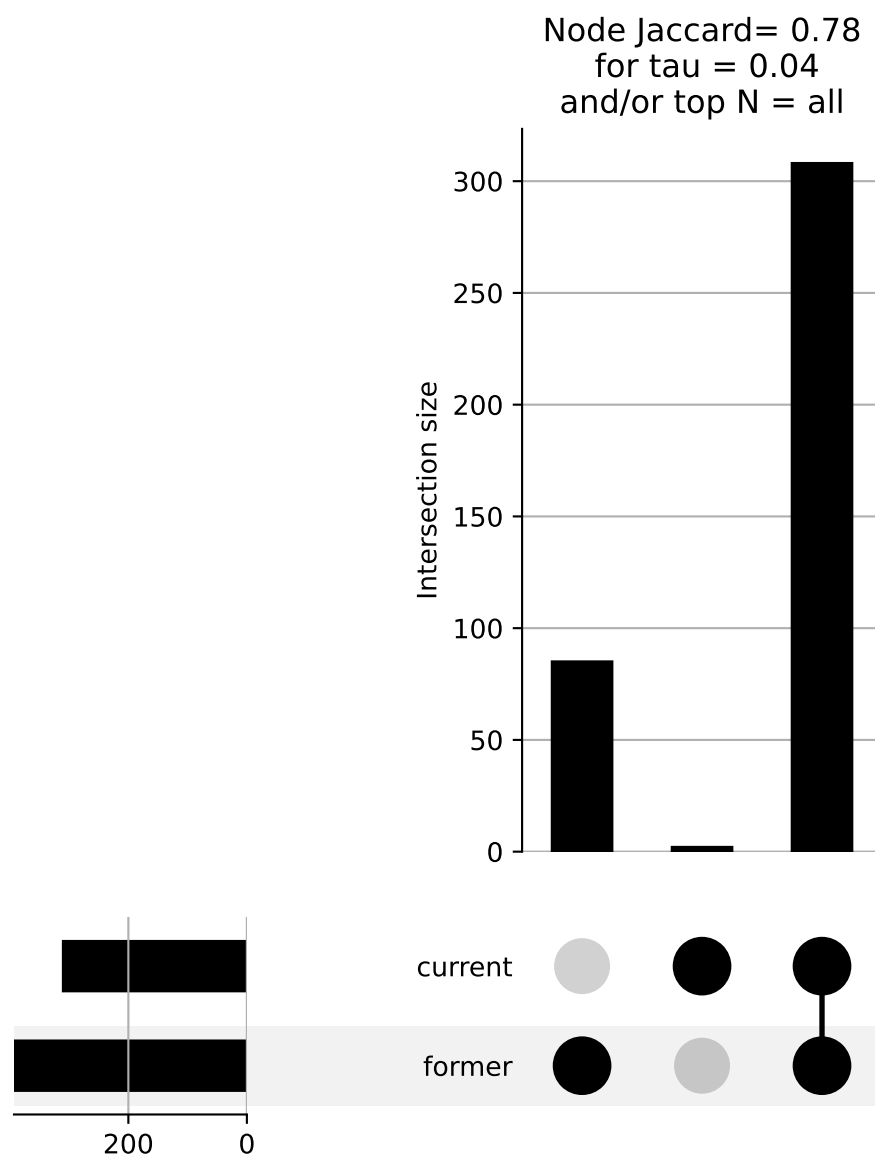


Figure 1: Node intersection and difference

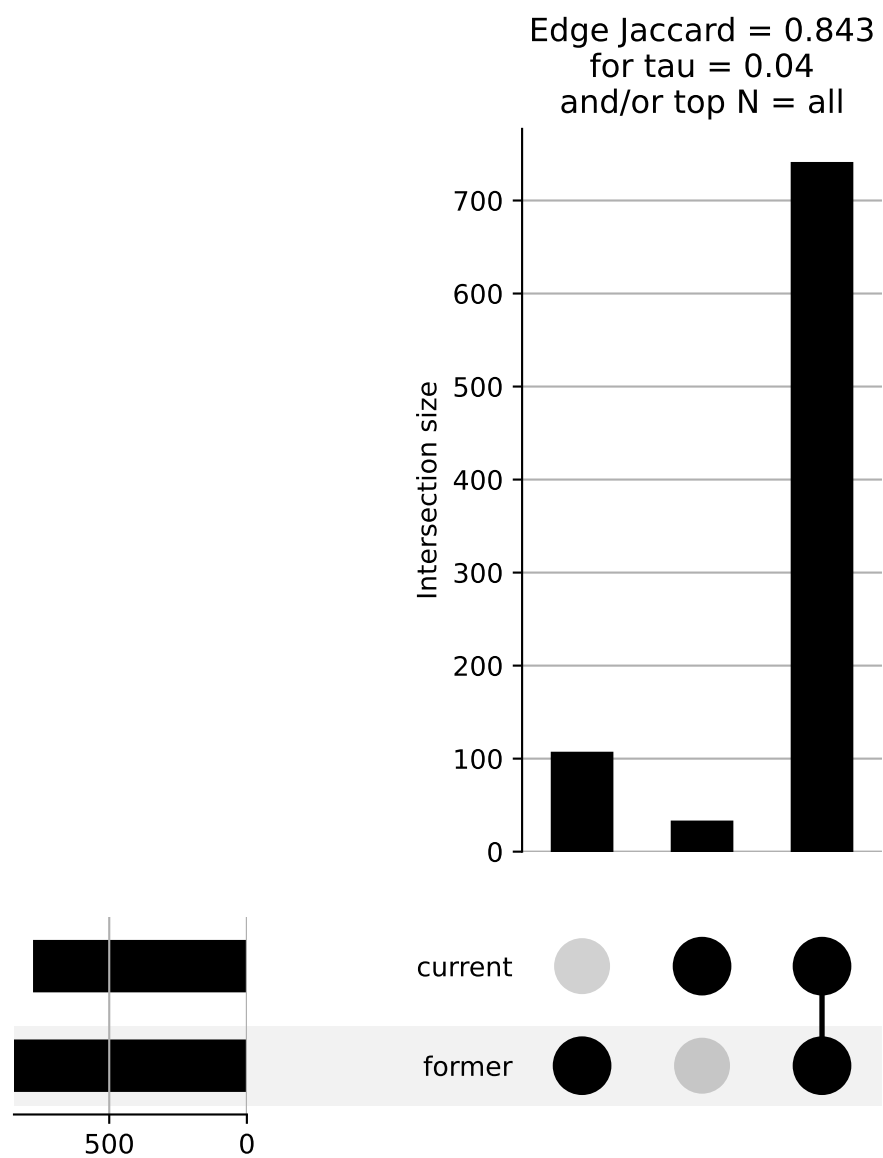


Figure 2: Edge intersection and difference.

Current Smokers

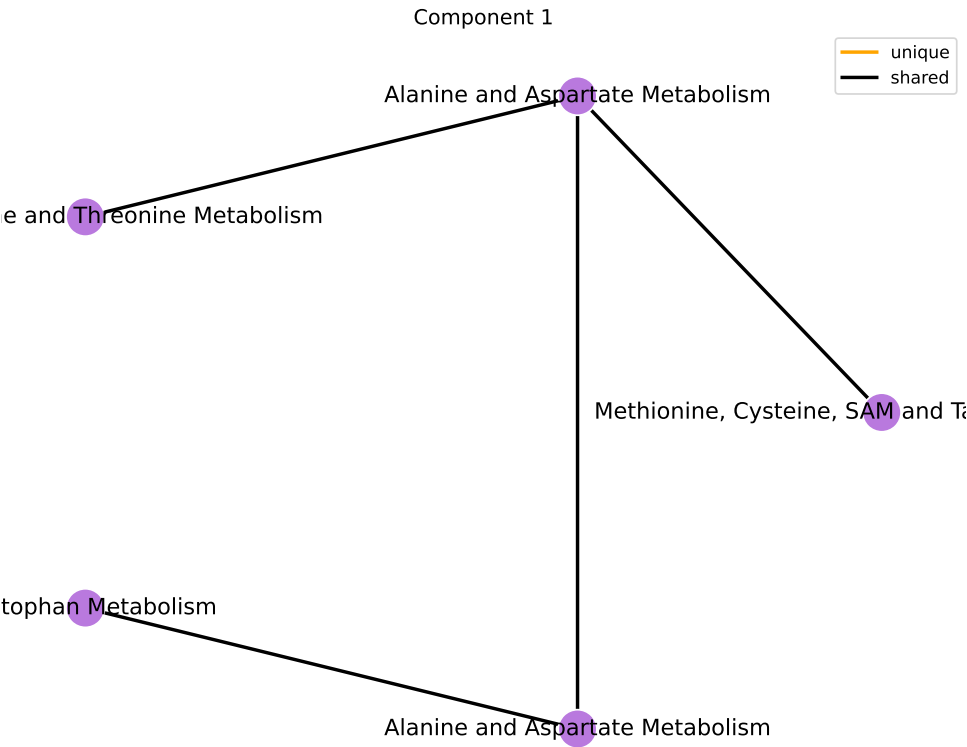
Component 1

Super-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Amino Acid	180	5	8.199	2.104	0.00989	0.089
Nucleotide	32	2	9.536	2.255	0.0318	0.143

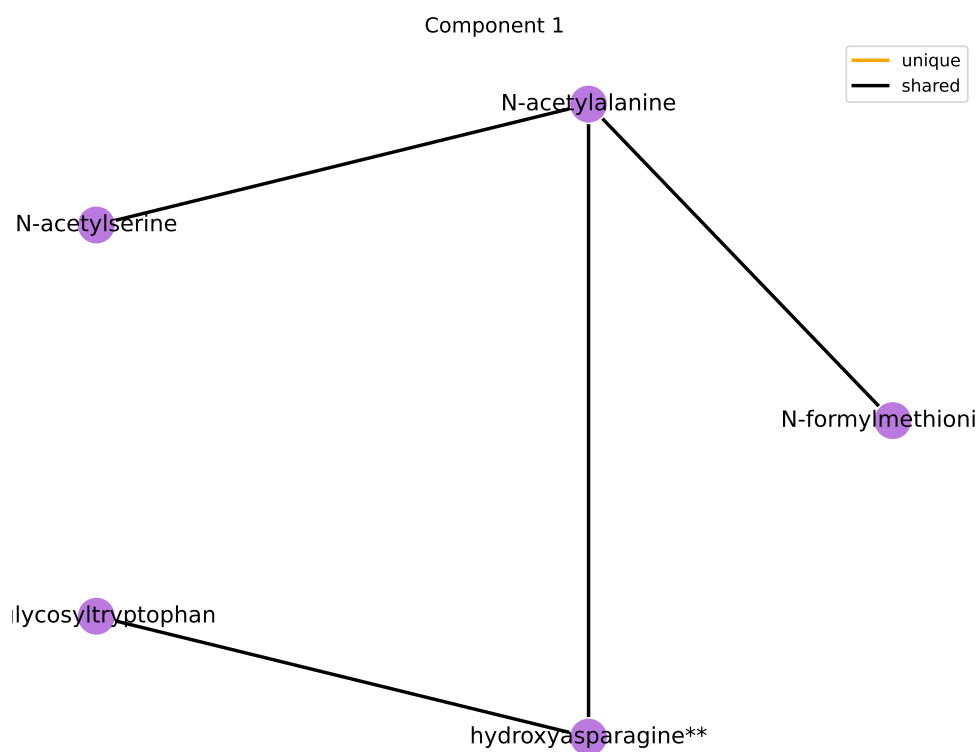
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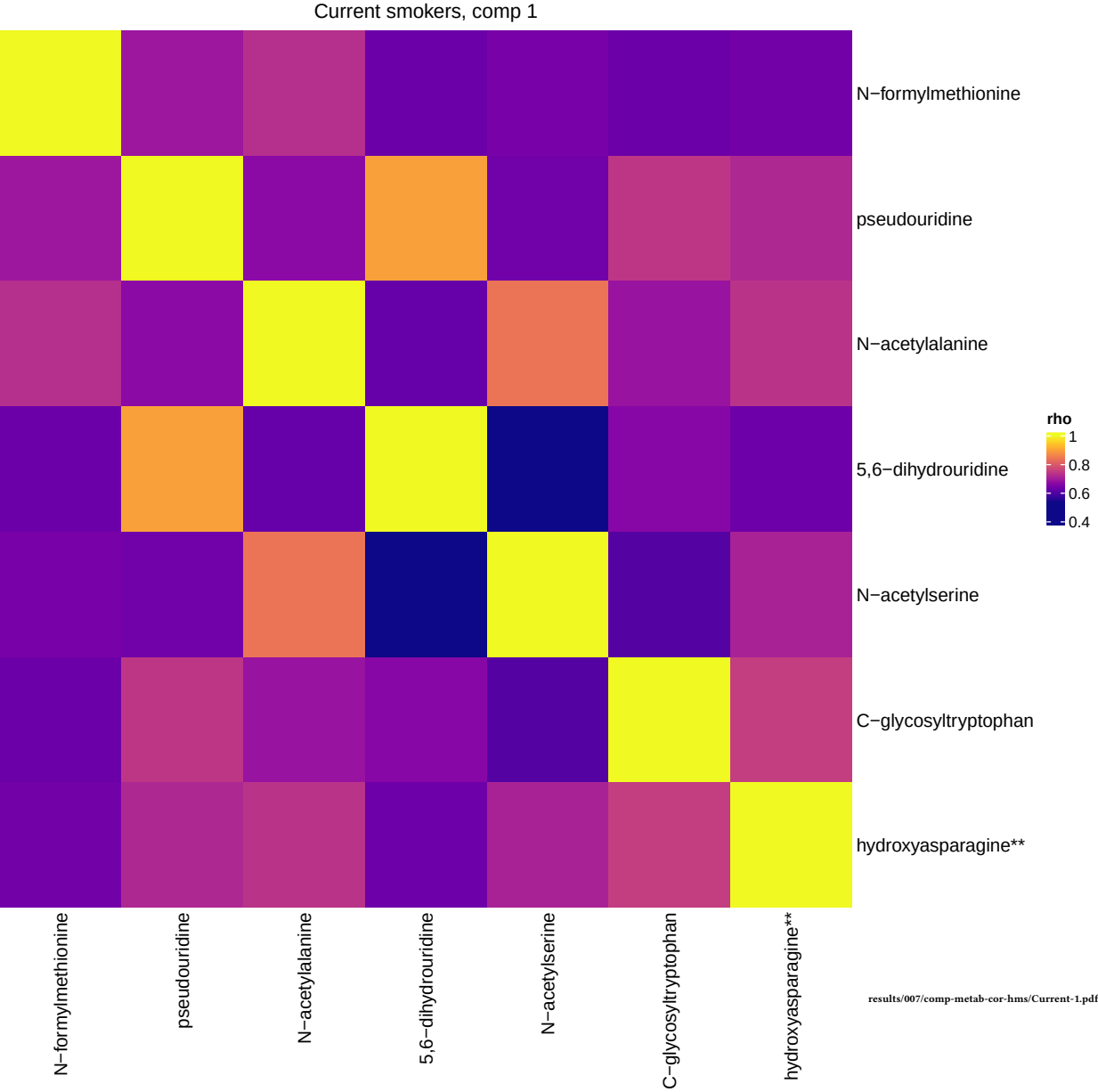
Sub-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Alanine and Aspartate Metabolism	8	2	47.866	3.868	0.002	0.165
Pyrimidine Metabolism, Uracil containing	10	2	36.068	3.585	0.00318	0.165
Glycine, Serine and Threonine Metabolism	10	1	13.562	2.607	0.0891	1
Tryptophan Metabolism	16	1	8.114	2.094	0.139	1
Methionine, Cysteine, SAM and Taurine Metabolism	17	1	7.601	2.028	0.147	1

results/006/enrichment/curr-components/1-subpathway.csv



results/005_1/component-graphs/current/sub_pathway-0.pdf





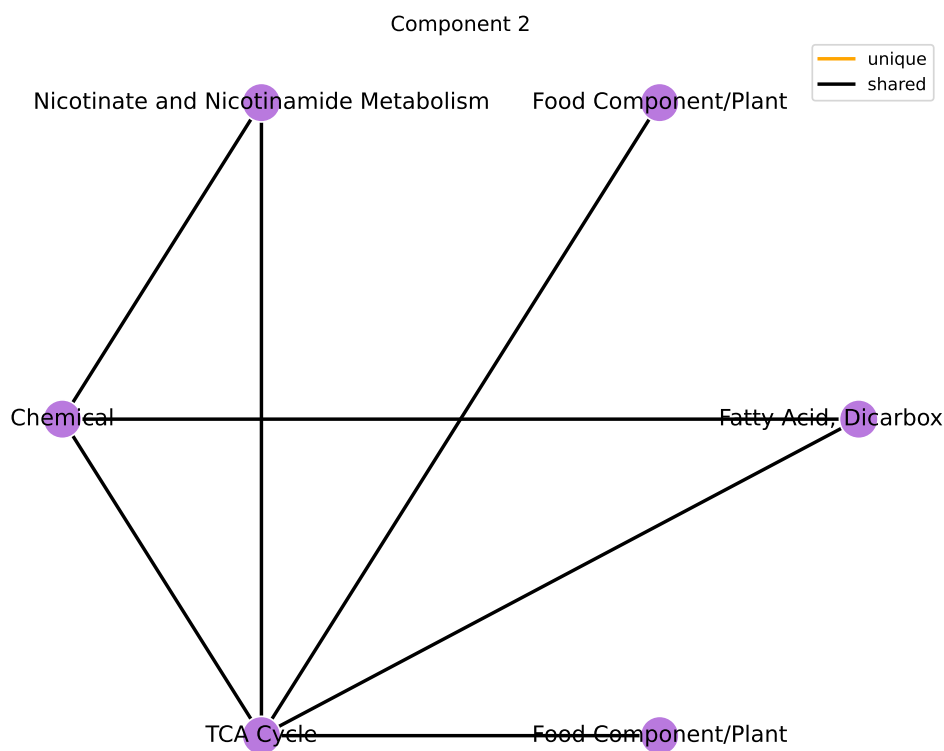
Component 2

Super-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Xenobiotics	96	4	9.501	2.251	0.00627	0.0564
Energy	10	1	13.562	2.607	0.0891	0.401
Cofactors and Vitamins	25	1	5.026	1.615	0.21	0.63

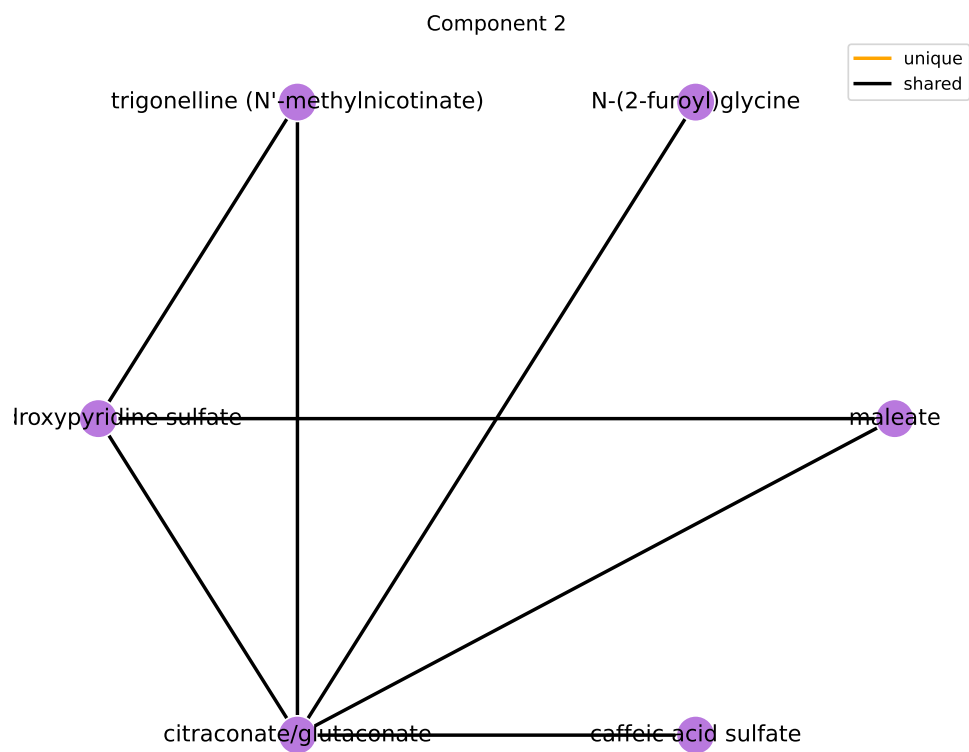
results/006/enrichment/curr-components/2-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Food Component/Plant	38	3	15.174	2.72	0.00355	0.369
Nicotinate and Nicotinamide Metabolism	5	1	30.252	3.41	0.0454	1
TCA Cycle	9	1	15.248	2.724	0.0805	1
Chemical	18	1	7.145	1.966	0.155	1
Fatty Acid, Dicarboxylate	23	1	5.494	1.704	0.195	1

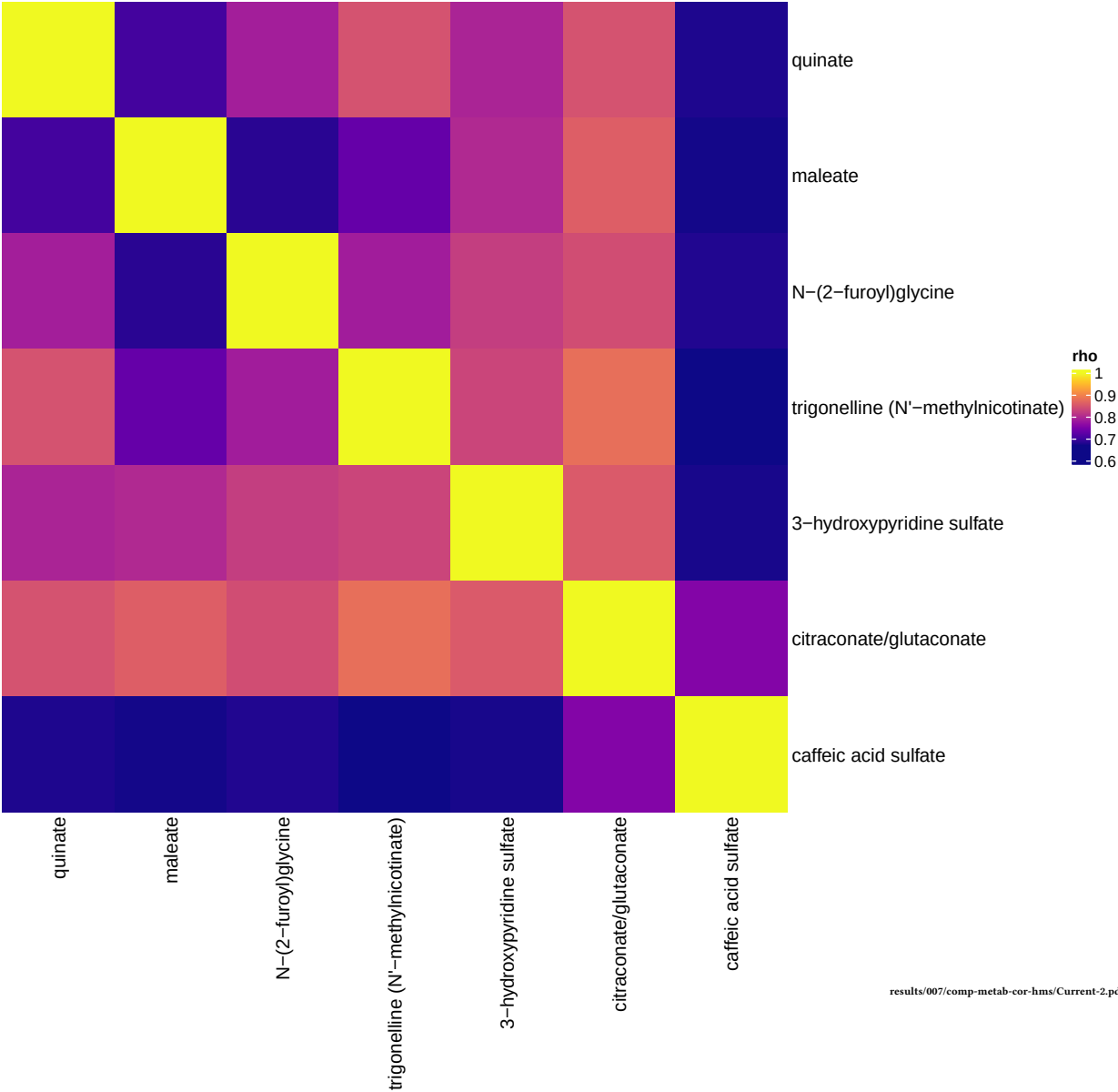
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Current smokers, comp 2



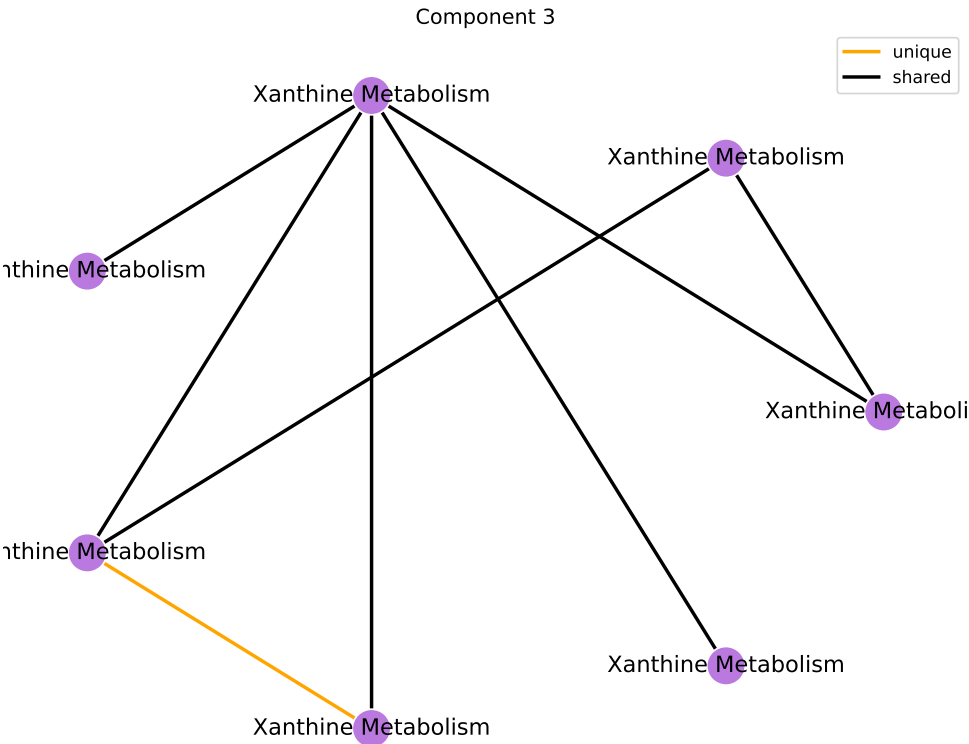
Component 3

Super-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Xenobiotics	96	8	Inf	Inf	5.09e-08	4.58e-07

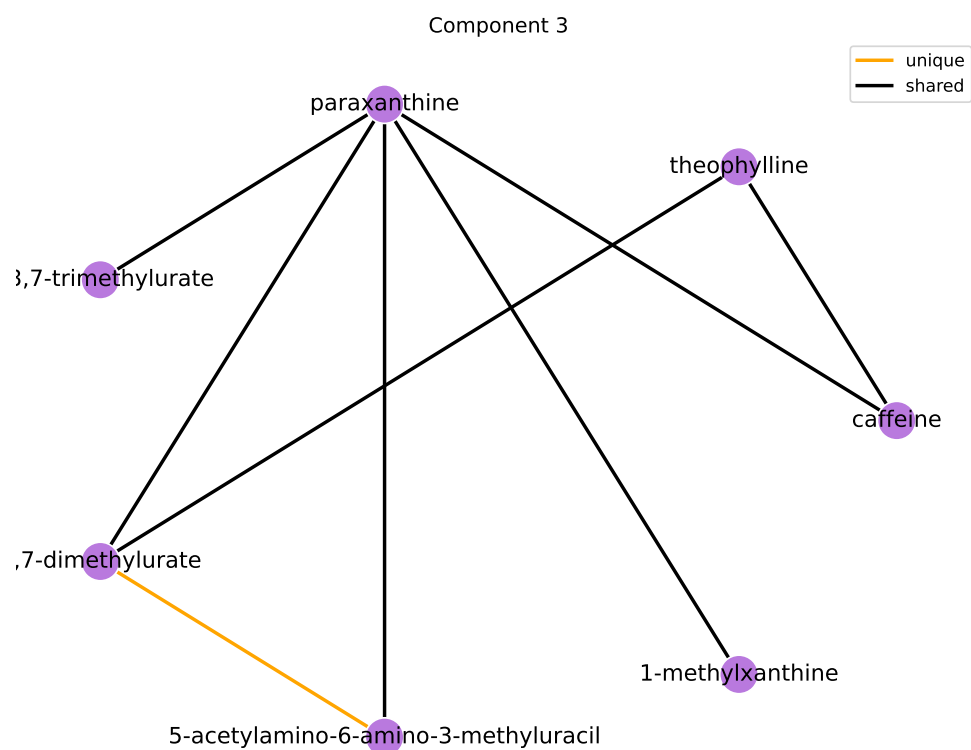
results/006/enrichment/curr-components/3-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Xanthine Metabolism	14	8	Inf	Inf	1.15e-15	1.2e-13

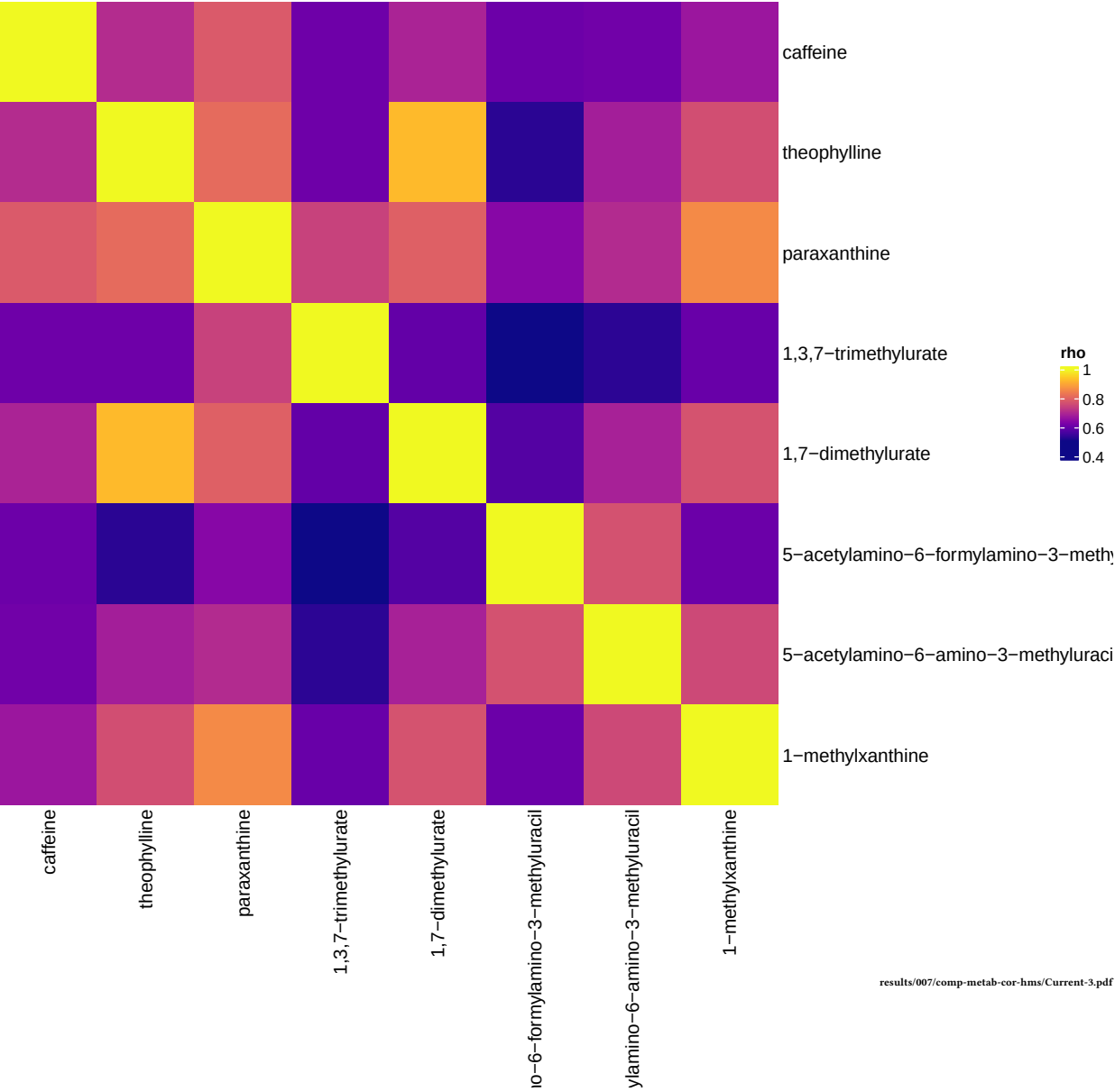
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Current smokers, comp 3



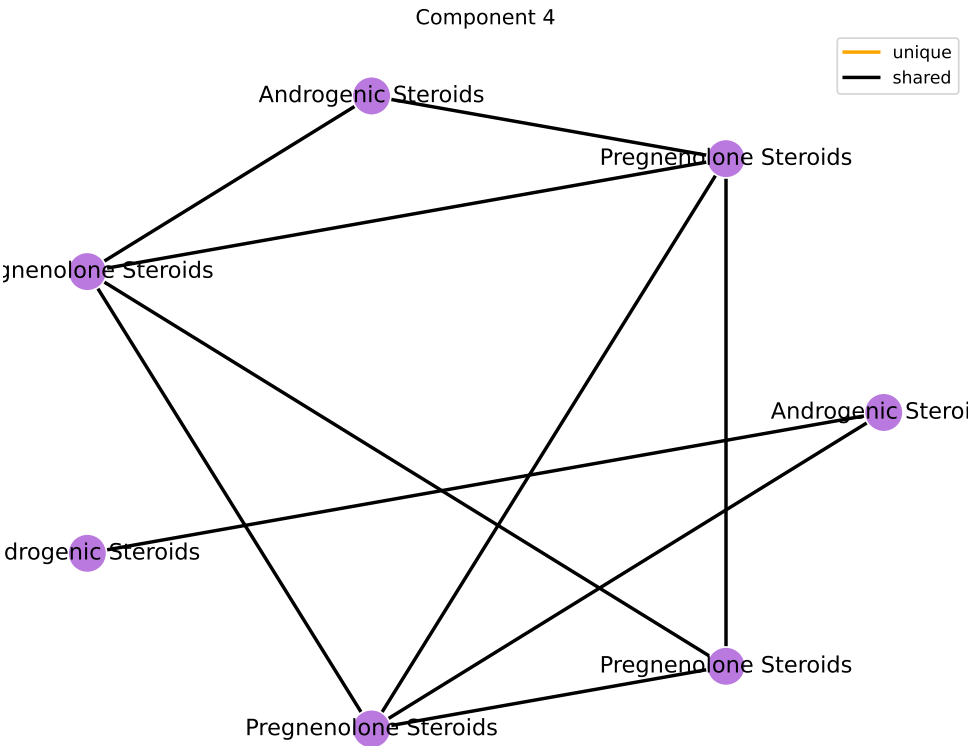
Component 4

Super-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Lipid	363	9	Inf	Inf	0.00126	0.0113

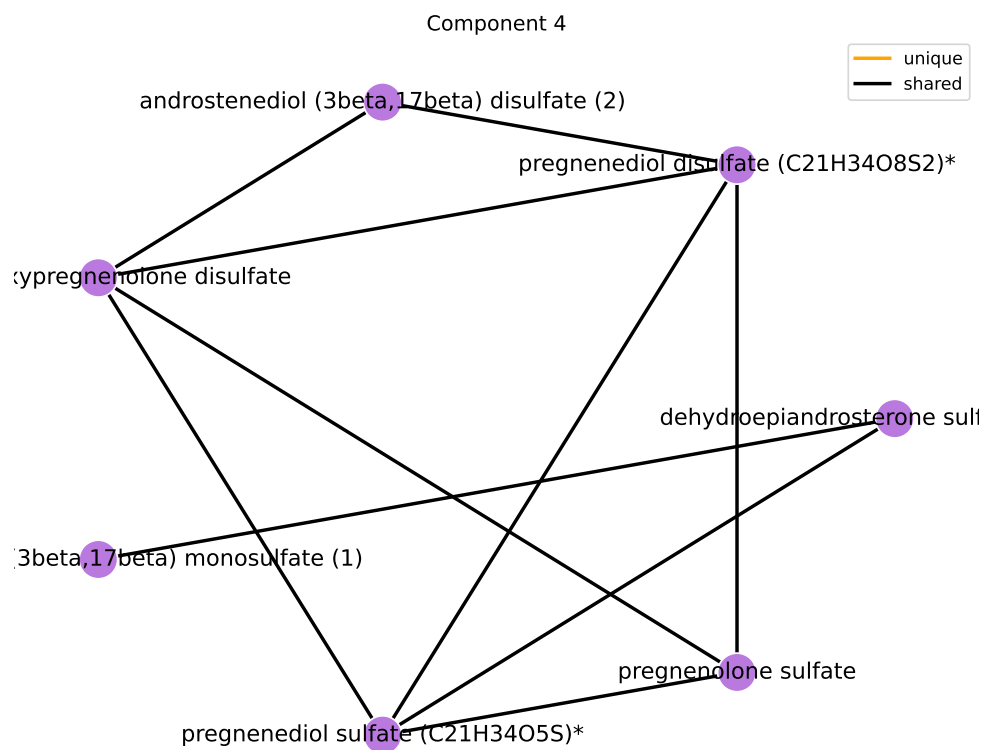
results/006/enrichment/curr-components/4-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Pregnenolone Steroids	4	4	Inf	Inf	9.23e-09	9.6e-07
Androgenic Steroids	14	5	97.826	4.583	1.18e-07	6.14e-06

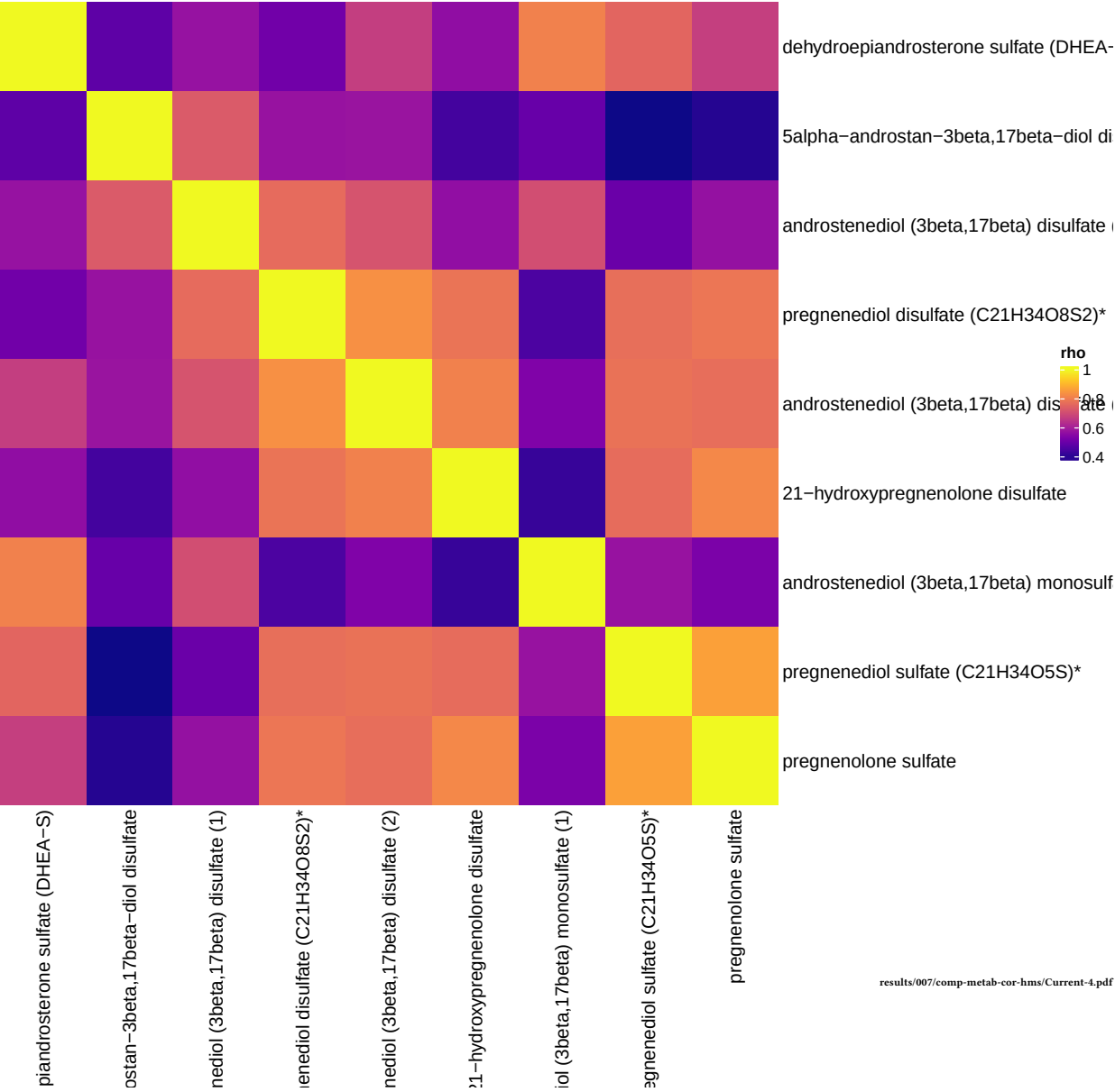
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Current smokers, comp 4



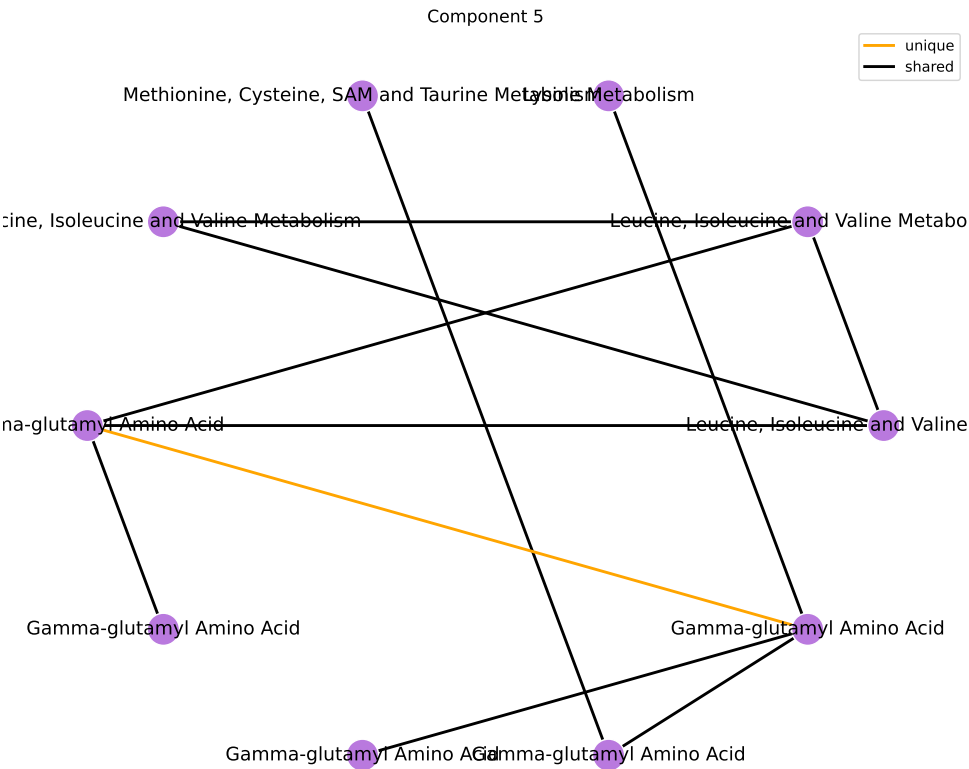
Component 5

Super-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Peptide	25	6	44.83	3.803	2.84e-07	2.56e-06
Amino Acid	180	5	2.719	1	0.094	0.423

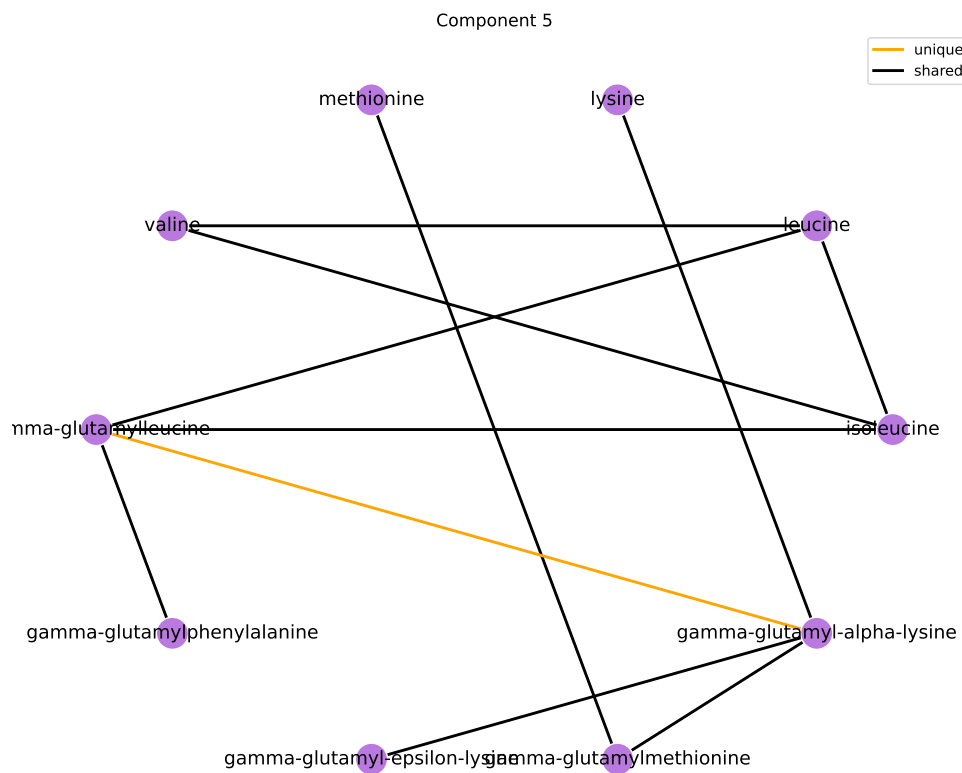
results/006/enrichment/curr-components/5-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Gamma-glutamyl Amino Acid	17	6	77.358	4.348	2.08e-08	2.16e-06
Leucine, Isoleucine and Valine Metabolism	31	3	9.552	2.257	0.00819	0.426
Lysine Metabolism	12	1	6.647	1.894	0.162	1
Methionine, Cysteine, SAM and Taurine Metabolism	17	1	4.549	1.515	0.222	1

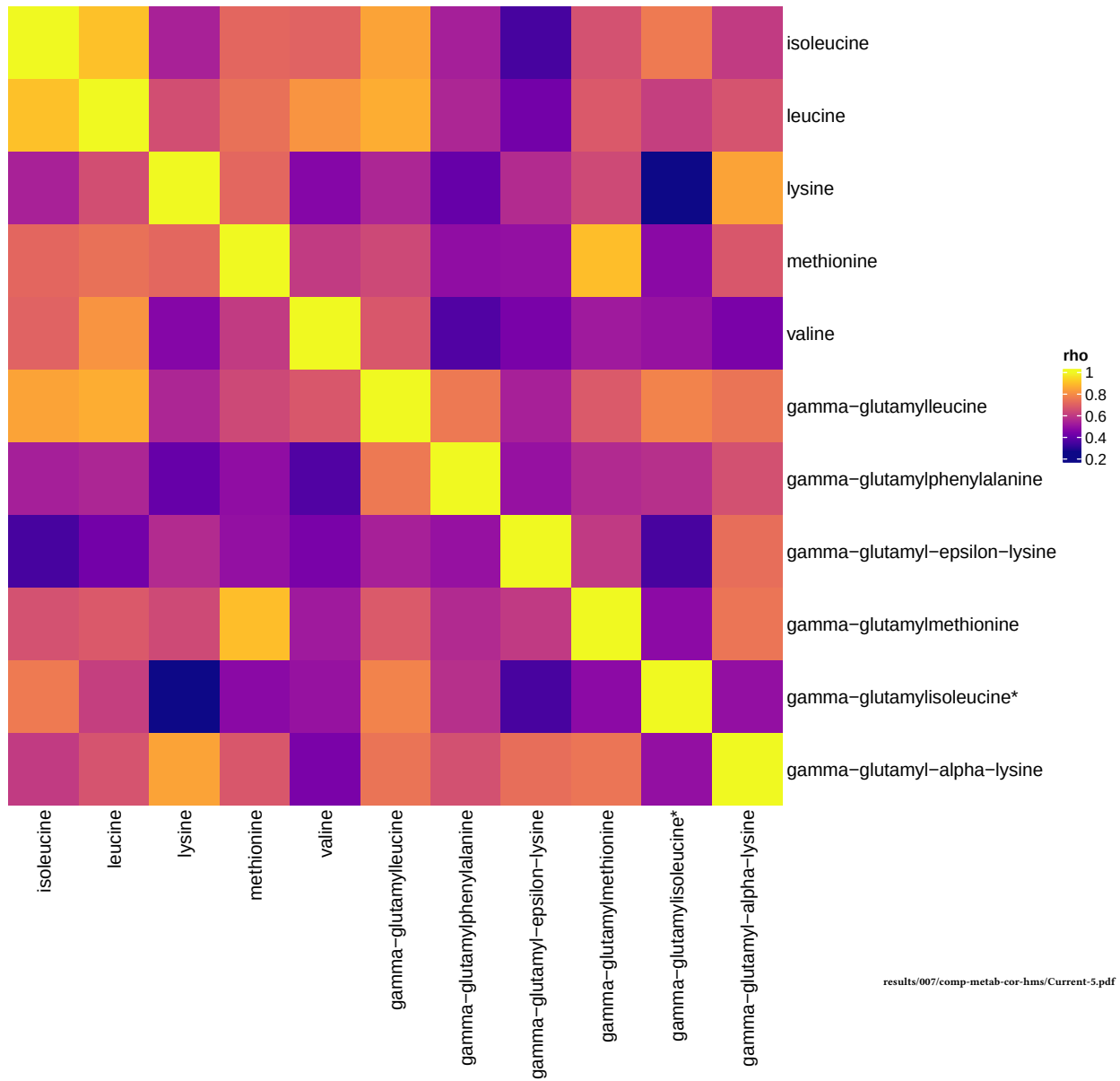
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Current smokers, comp 5



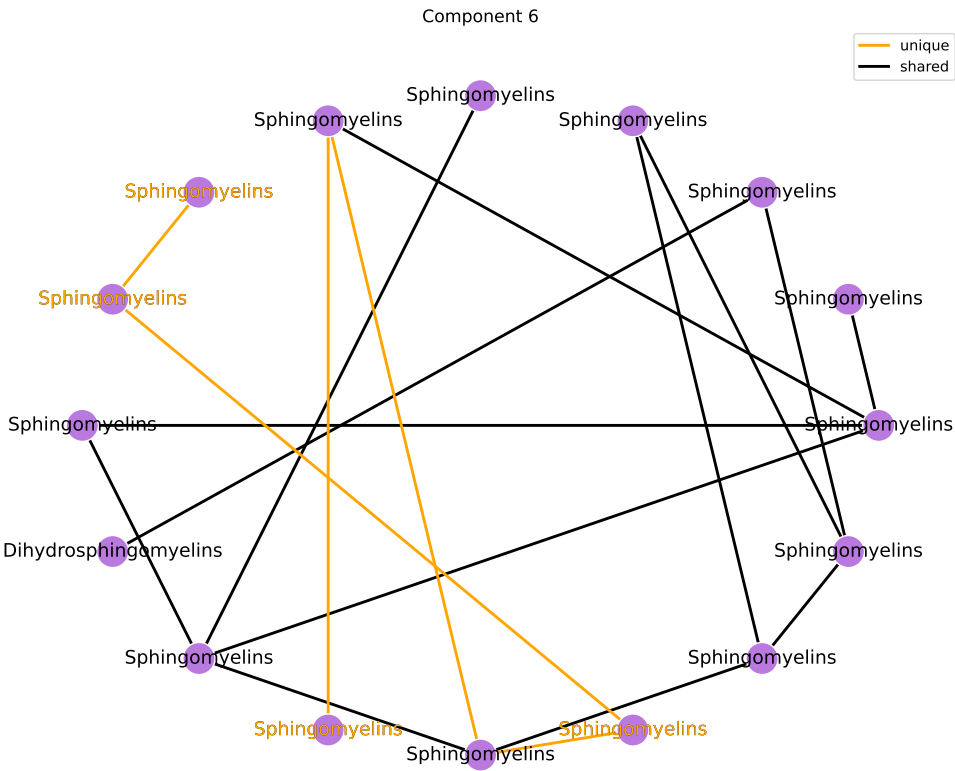
Component 6

Super-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Lipid	363	17	Inf	Inf	3e-06	2.7e-05

results/006/enrichment/curr-components/6-superpathway.csv

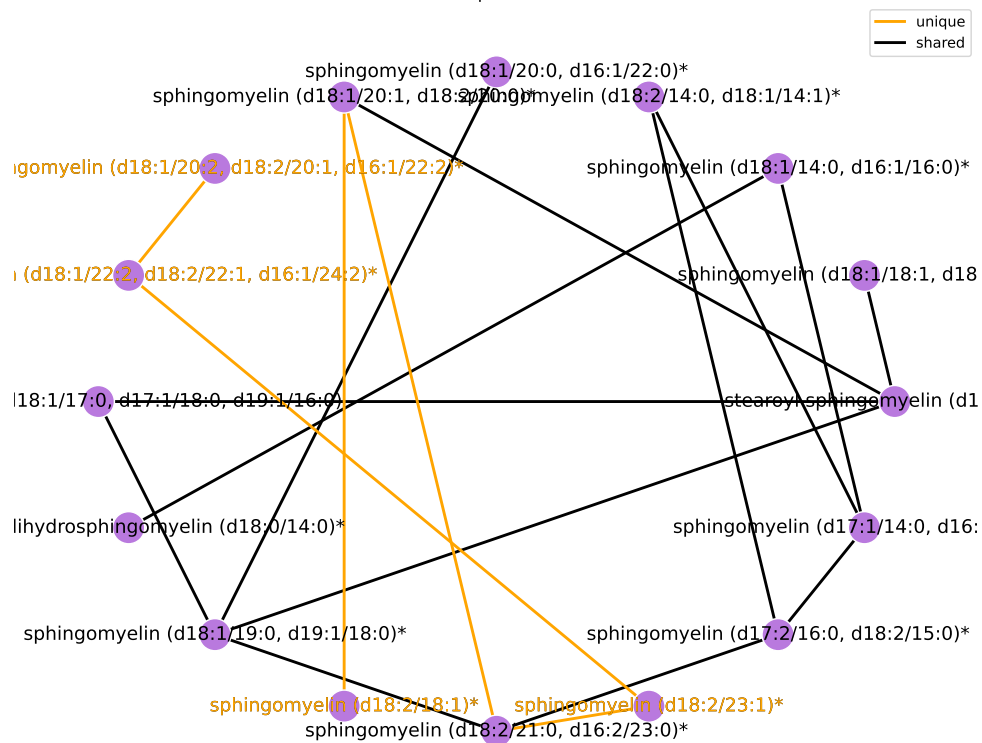
Sub-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Sphingomyelins	28	16	913.538	6.817	1.05e-24	1.09e-22
Dihydrosphingomyelins	5	1	11.394	2.433	0.107	1

results/006/enrichment/curr-components/6-subpathway.csv

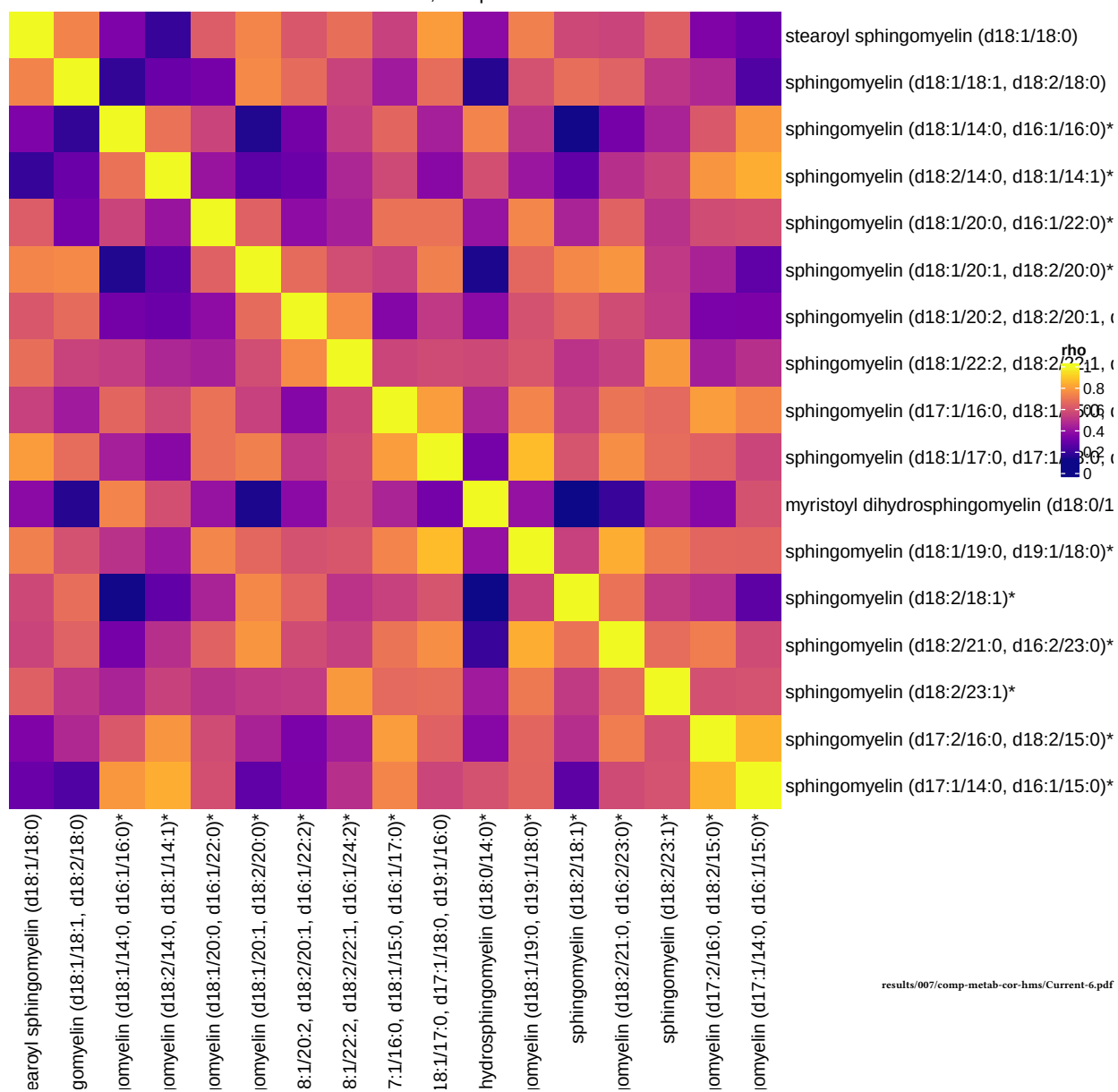


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Component 6



Current smokers, comp 6



Component 7

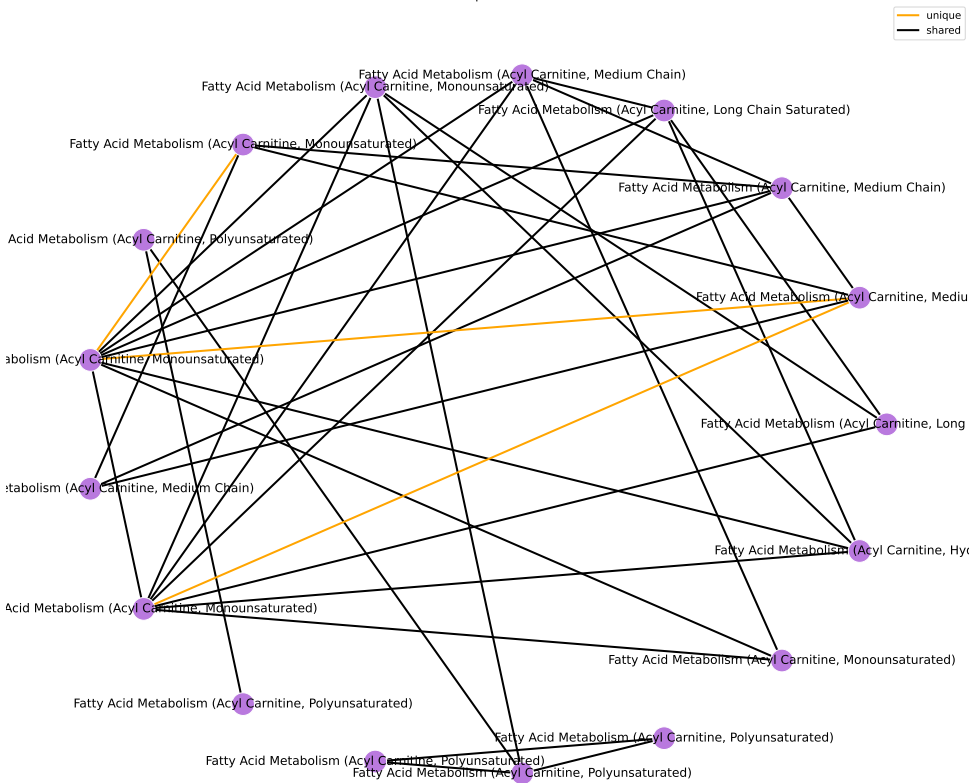
Super-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Lipid	363	18	Inf	Inf	1.4e-06	1.26e-05

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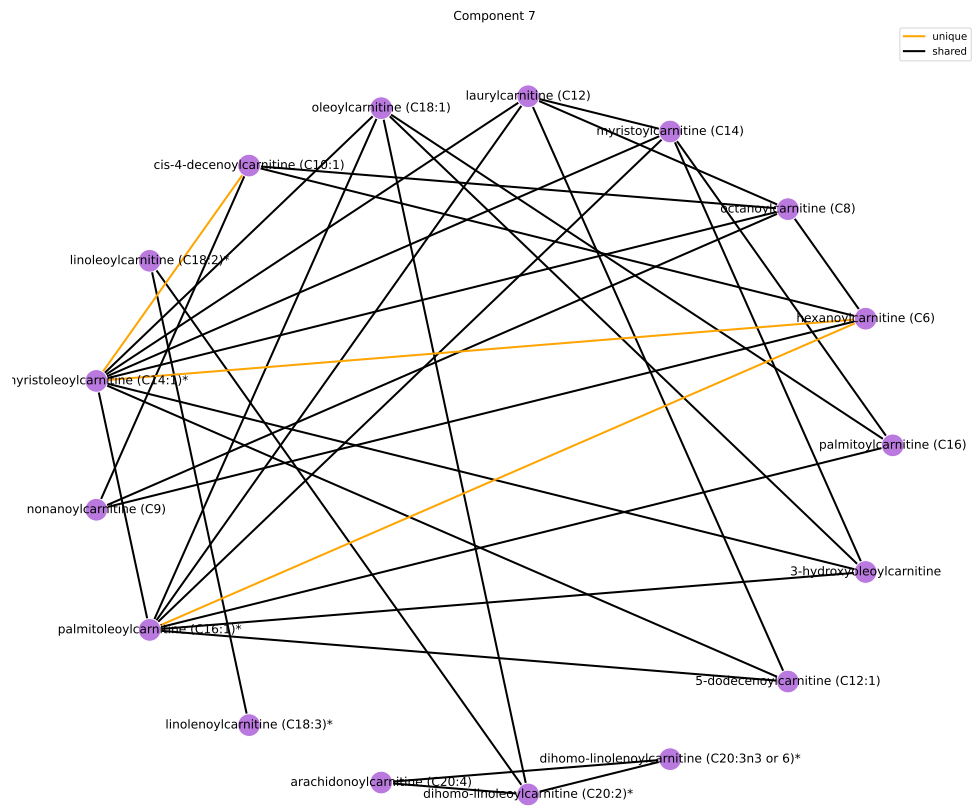
Sub-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Fatty Acid Metabolism (Acyl Carnitine, Medium Chain)	5	5	Inf	Inf	4.16e-09	2.16e-07
Fatty Acid Metabolism (Acyl Carnitine, Polyunsaturated)	5	5	Inf	Inf	4.16e-09	2.16e-07
Fatty Acid Metabolism (Acyl Carnitine, Monounsaturated)	8	5	90.526	4.506	2.23e-07	7.73e-06
Fatty Acid Metabolism (Acyl Carnitine, Long Chain Saturated)	8	2	15.084	2.714	0.0137	0.356
Fatty Acid Metabolism (Acyl Carnitine, Hydroxy)	3	1	21.318	3.06	0.0697	1

results/006/enrichment/curr-components/7-subpathway.csv

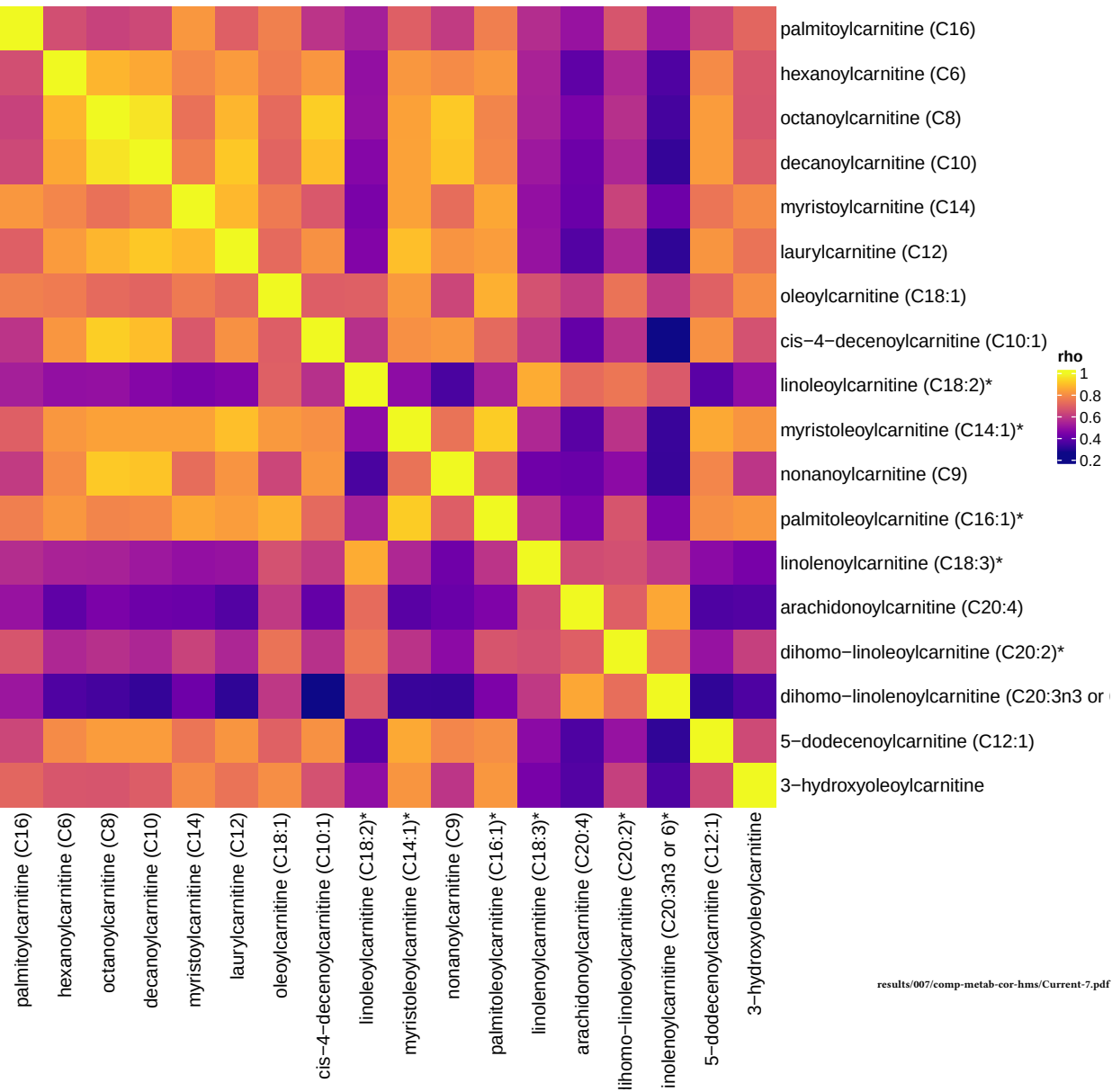
Component 7



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Current smokers, comp 7



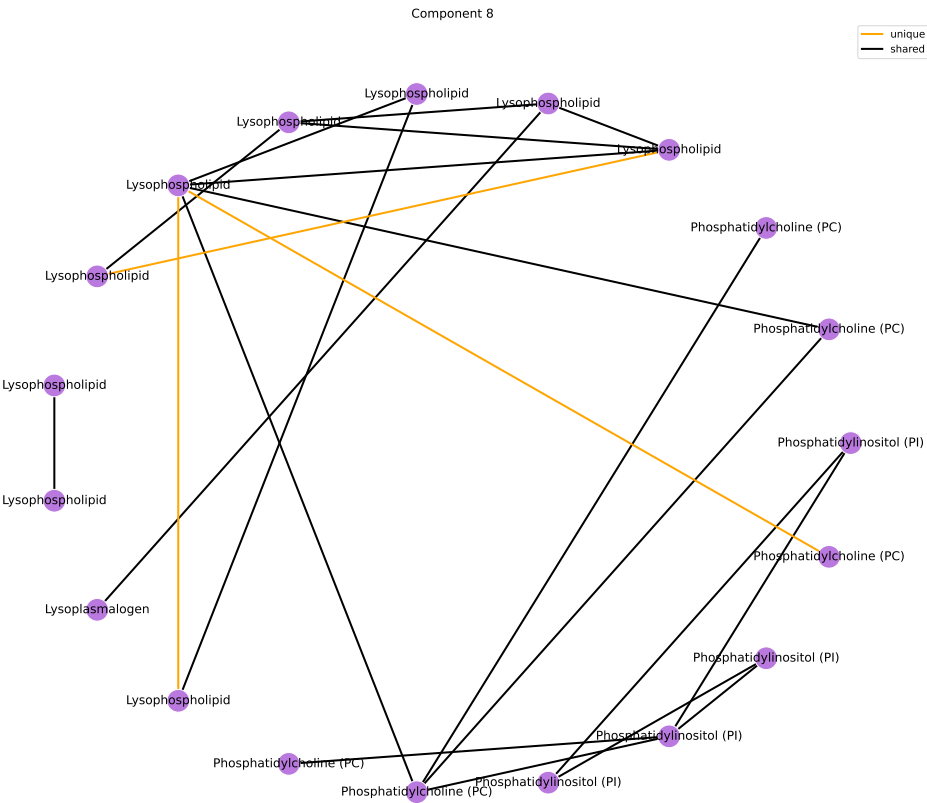
Component 8

Super-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Lipid	363	20	Inf	Inf	3.04e-07	2.74e-06

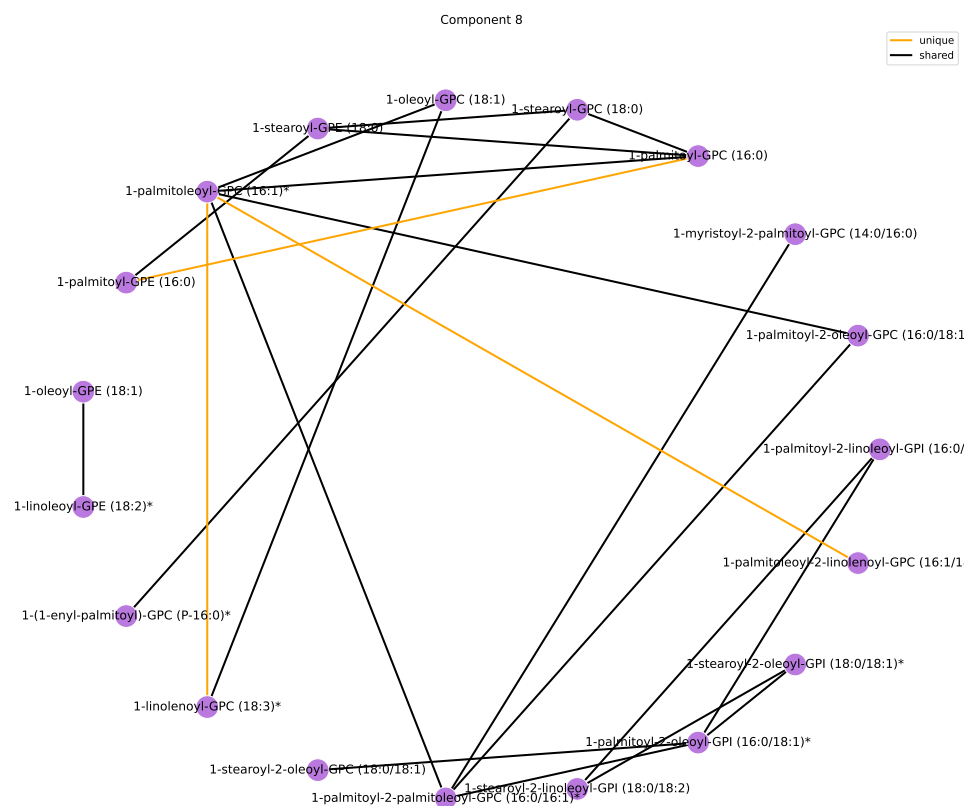
results/006/enrichment/curr-components/8-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Lysophospholipid	24	10	50.402	3.92	1.88e-11	1.96e-09
Phosphatidylinositol (PI)	6	4	88.541	4.483	5.15e-06	0.000268
Phosphatidylcholine (PC)	19	5	17.012	2.834	6.92e-05	0.0024
Lysoplasmalogen	3	1	19.06	2.948	0.0772	1

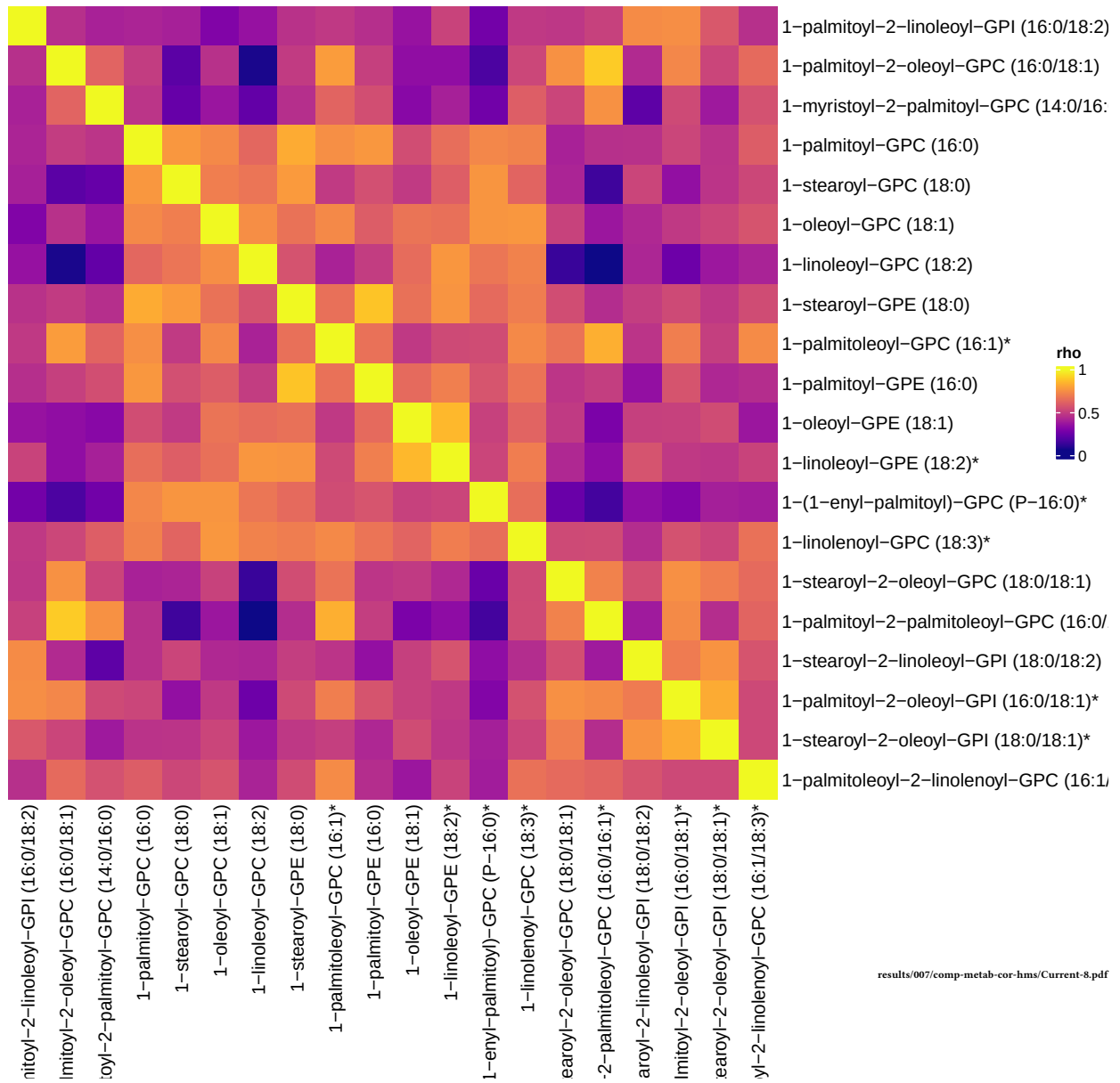
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Current smokers, comp 8

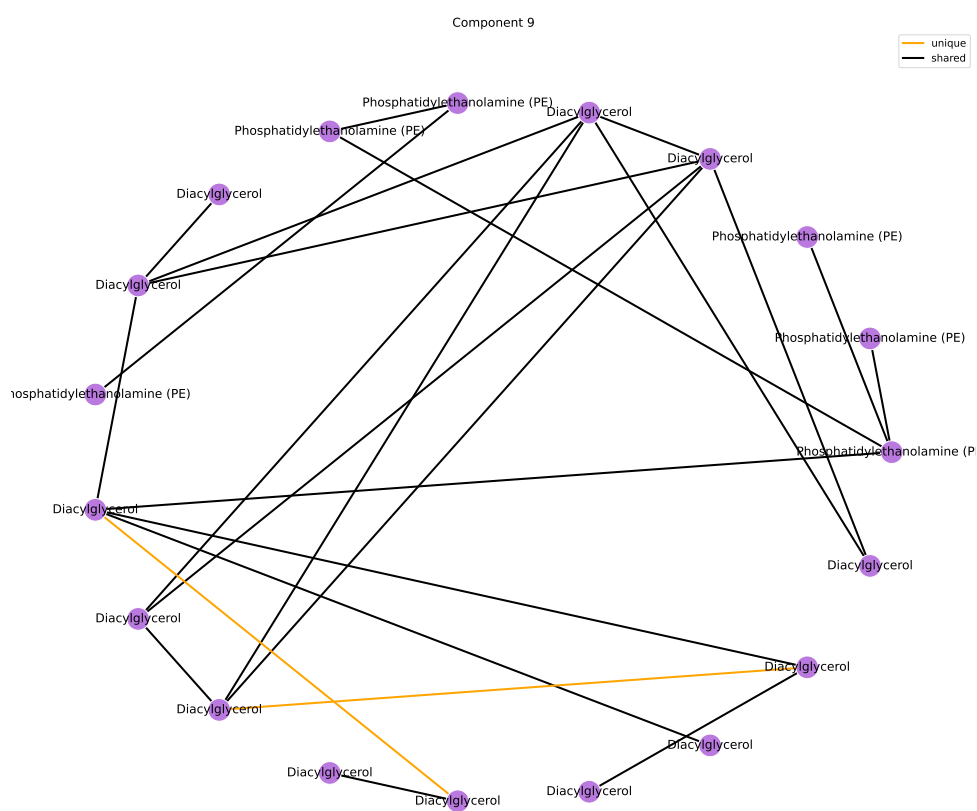


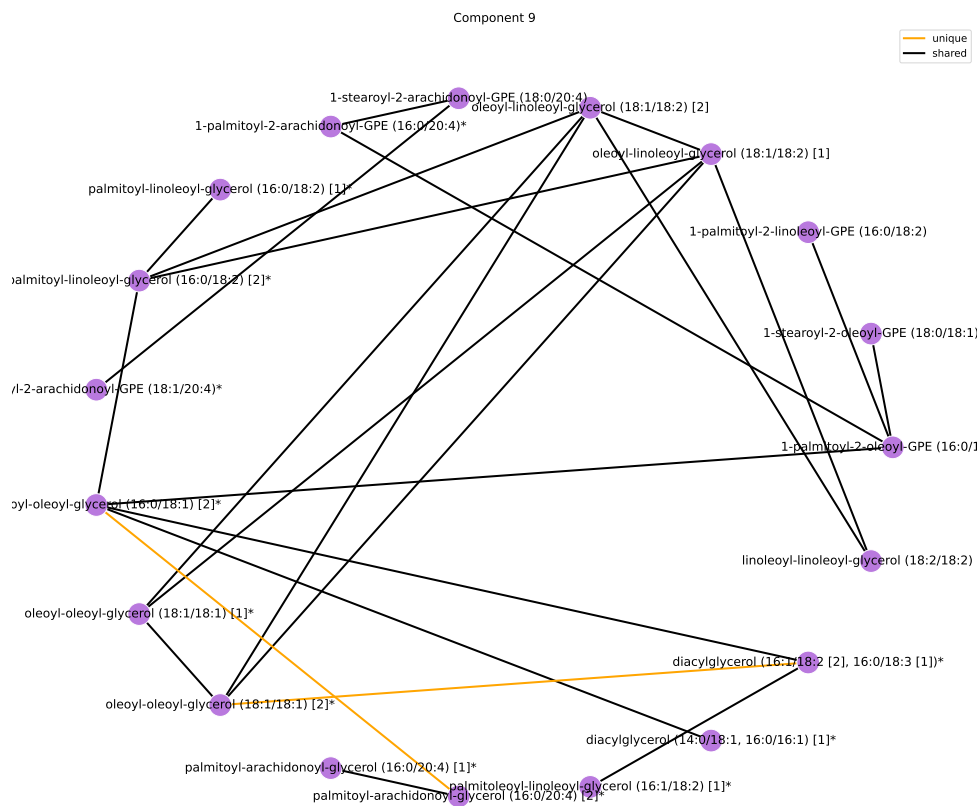
Component 9

Super-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Lipid	363	21	Inf	Inf	1.41e-07	1.27e-06

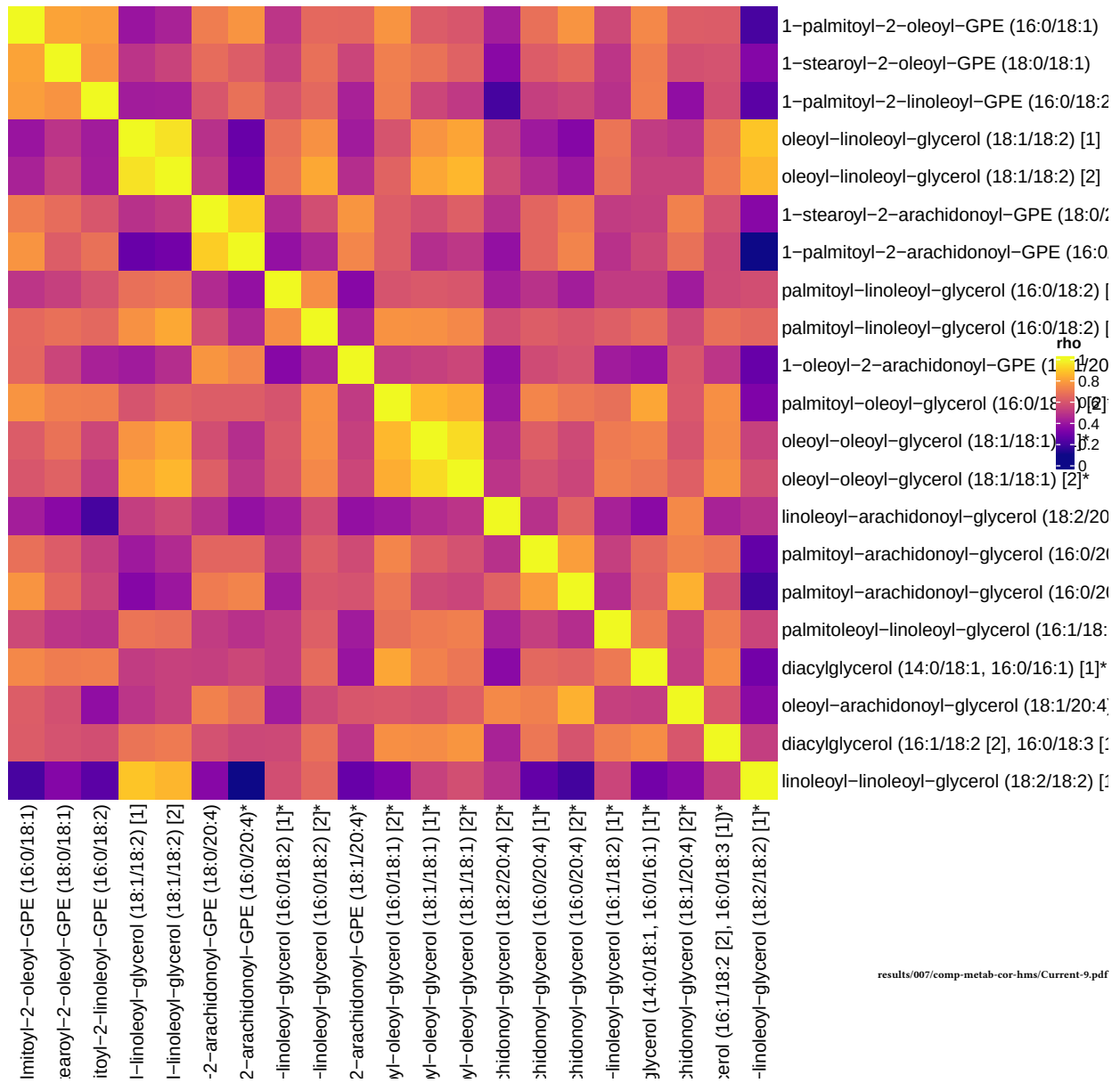
results/006/enrichment/curr-components/9-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	OR	In(OR)	Fisher P	FDR P
Diacylglycerol	19	15	419.751	6.04	1.96e-23	2.04e-21
Phosphatidylethanolamine (PE)	11	6	56.82	4.04	8.9e-08	4.63e-06

[results/006/enrichment/curr-components/9-subpathway.csv](#)[results/005_1/component-graphs/current/sub_pathway-8.pdf](#)



Current smokers, comp 9



Component 10

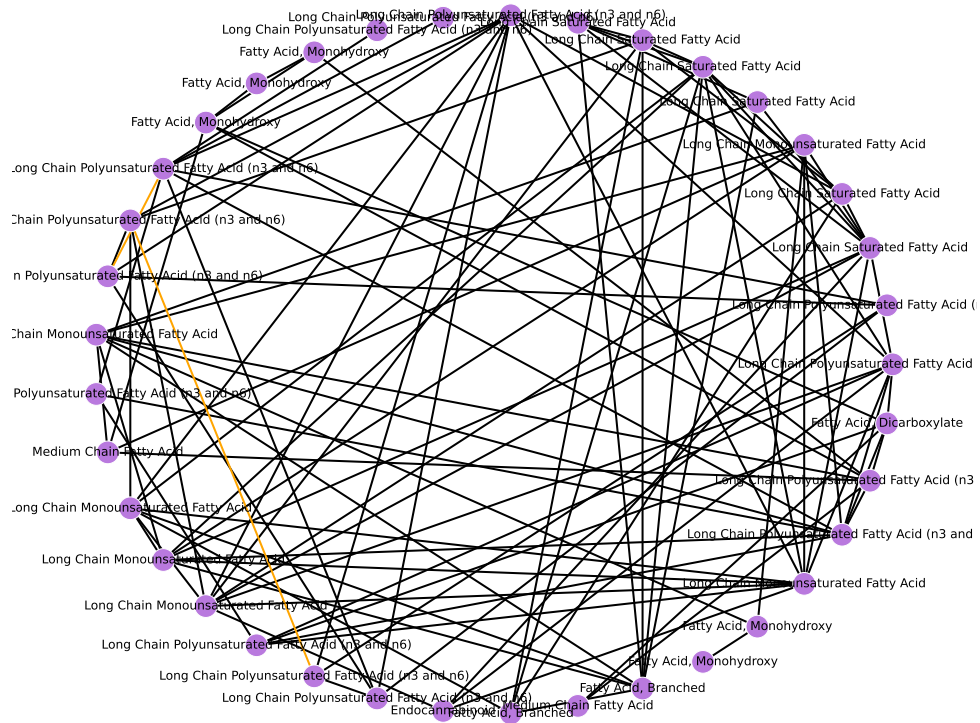
Super-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Lipid	363	40	Inf	Inf	4.96e-14	4.46e-13

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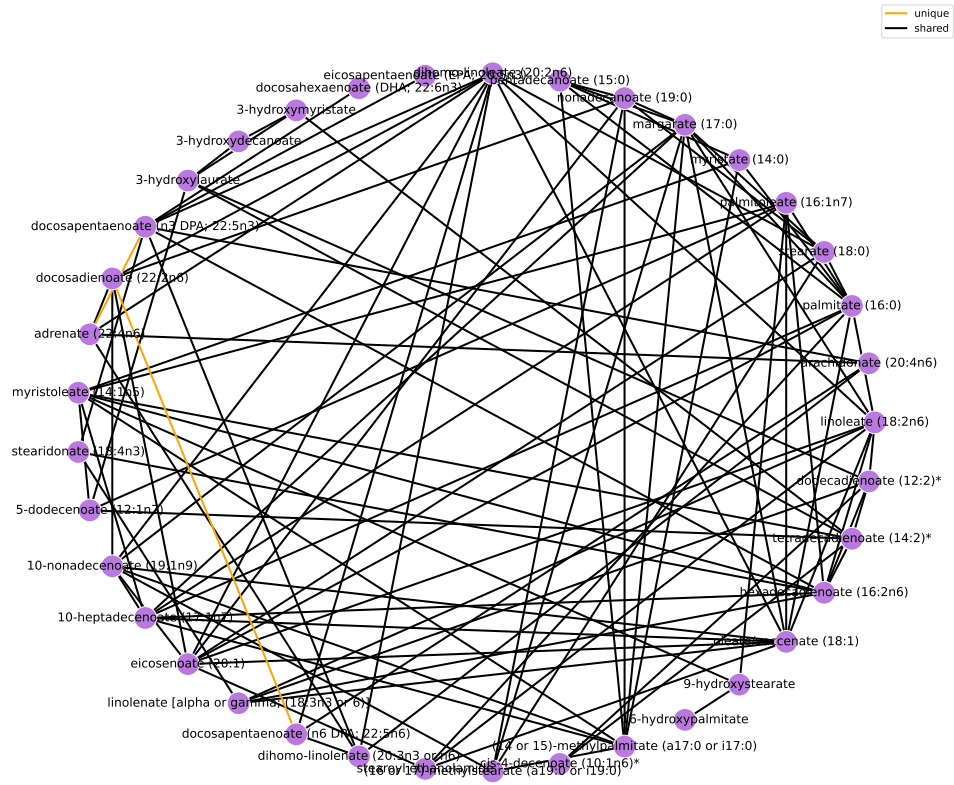
Sub-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Long Chain Polyunsaturated Fatty Acid (n3 and n6)	14	14	Inf	Inf	1.1e-19	1.14e-17
Long Chain Monounsaturated Fatty Acid	7	6	123.581	4.817	1e-07	3.47e-06
Long Chain Saturated Fatty Acid	7	6	123.581	4.817	1e-07	3.47e-06
Fatty Acid, Monohydroxy	20	7	11.405	2.434	3.12e-05	0.000811
Fatty Acid, Branched	2	2	Inf	Inf	0.00272	0.0566
Ketone Bodies	1	1	Inf	Inf	0.0528	0.915
Medium Chain Fatty Acid	8	2	6.212	1.826	0.0622	0.924
Endocannabinoid	5	1	4.559	1.517	0.238	1

results/006/enrichment/curr-components/10-subpathway.csv

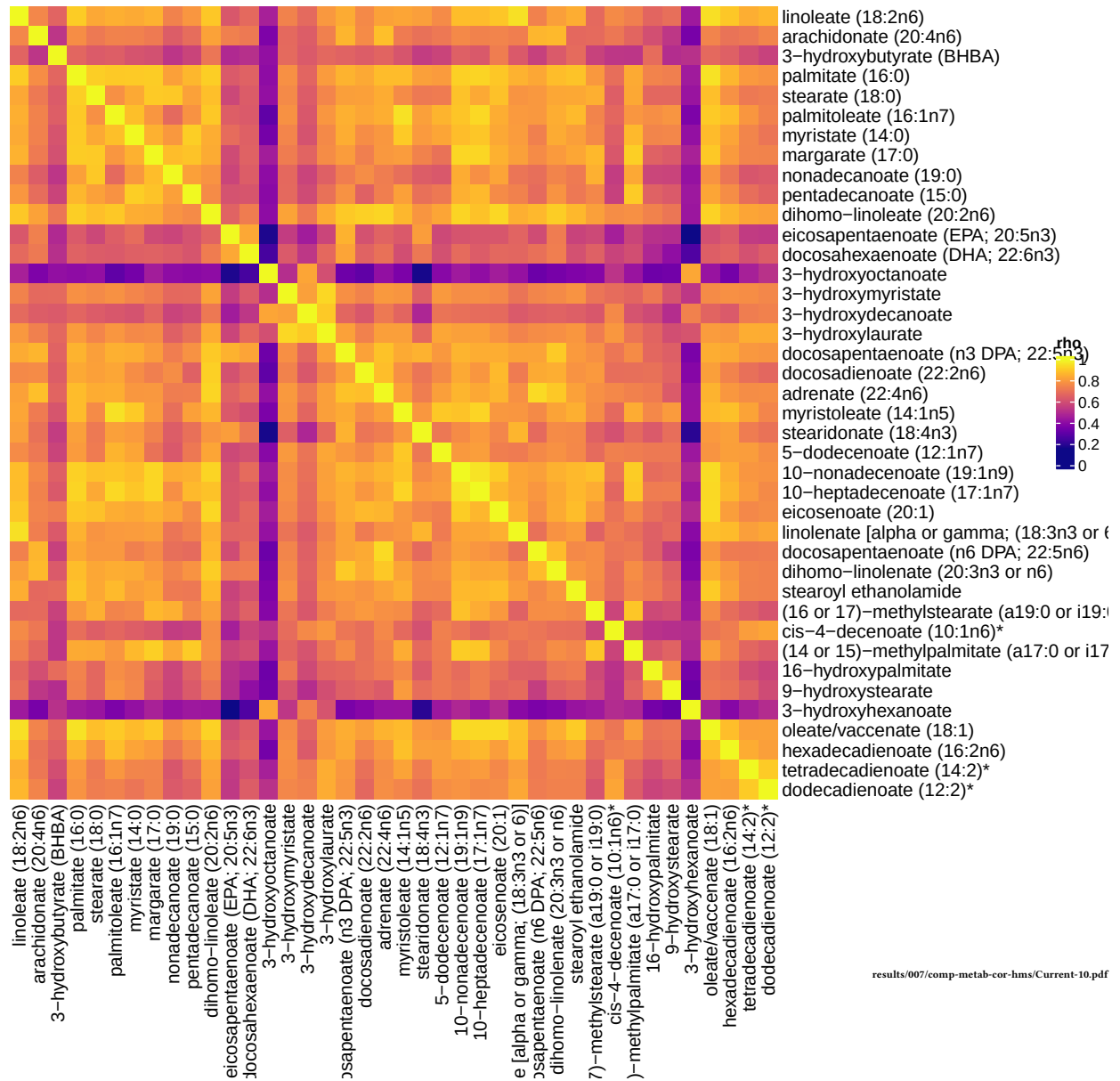
— unique
— shared



Component 10



Current smokers, comp 10



Super-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Lipid	363	2	Inf	Inf	0.229	1

results/006/enrichment/curr-unique-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Ketone Bodies	1	1	Inf	Inf	0.00264	0.275
Sphingomyelins	28	1	26.532	3.278	0.0726	1

results/006/enrichment/curr-unique-subpathway.csv

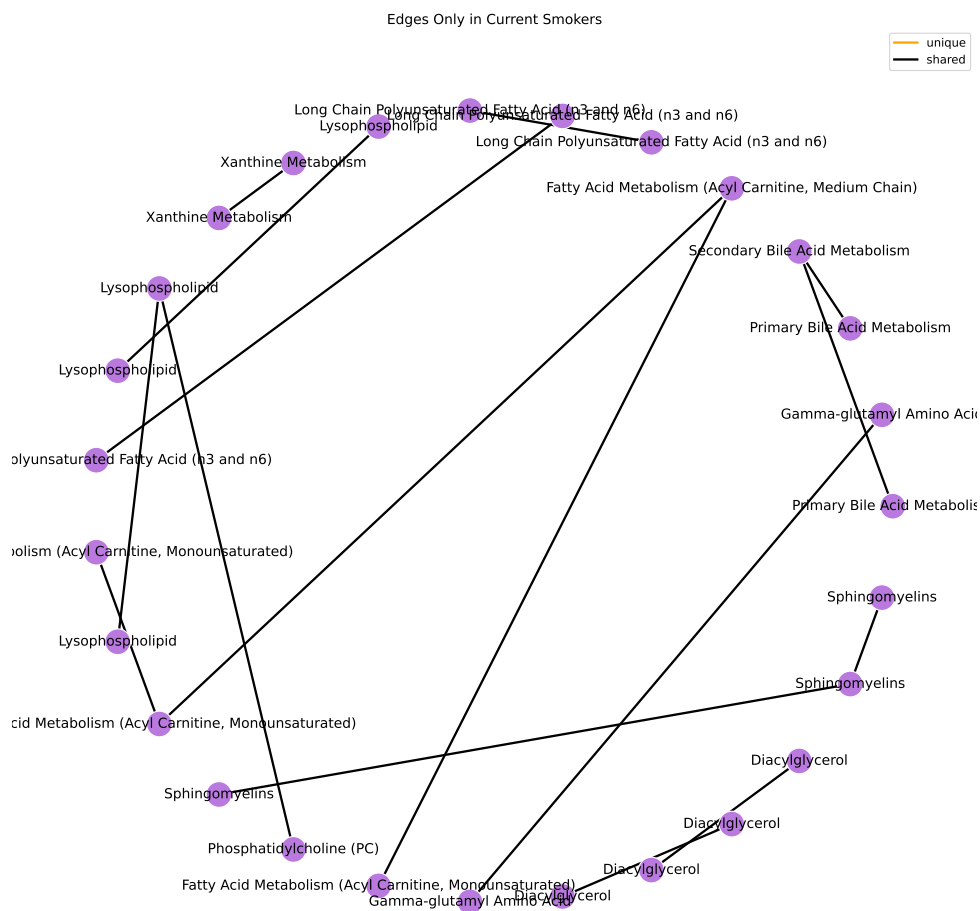
Nodes Only in Current Smokers

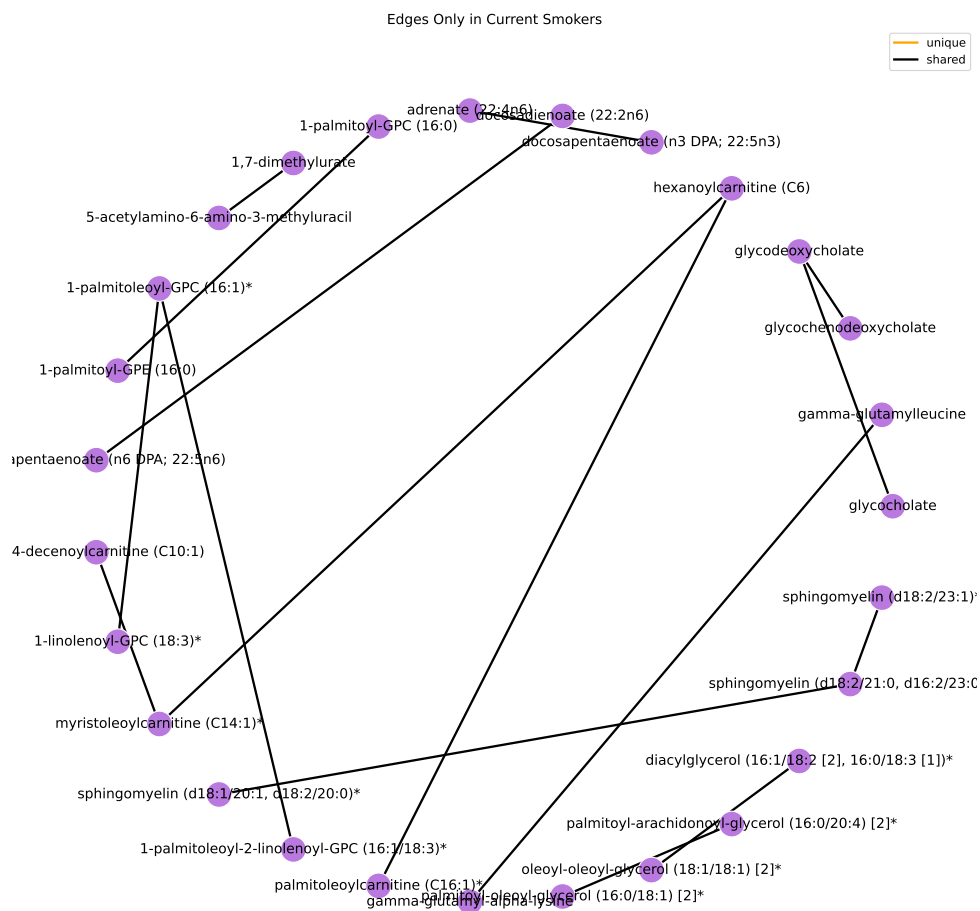


Nodes Only in Current Smokers



results/005_1/chemical_name-node-only-curr-graph.pdf





All Metabolites (not filtering for shared or unique)

Super-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Lipid	363	215	4.577	1.521	2.82e-23	2.54e-22
Peptide	25	11	1.14	0.131	0.45	1

results/006/enrichment/current-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Long Chain Polyunsaturated Fatty Acid (n3 and n6)	14	14	Inf	Inf	3.07e-06	0.000319
Diacylglycerol	19	17	12.903	2.557	1.29e-05	0.000671
Phosphatidylethanolamine (PE)	11	10	14.858	2.699	0.000839	0.0225
Sphingomyelins	28	20	3.786	1.331	0.000866	0.0225
Plasmalogen	11	9	6.654	1.895	0.00671	0.129
Lysophospholipid	24	16	2.989	1.095	0.00868	0.129
Fatty Acid Metabolism (Acyl Carnitine, Monounsaturated)	8	7	10.299	2.332	0.00951	0.129
Fatty Acid Metabolism (Acyl Carnitine, Medium Chain)	5	5	Inf	Inf	0.0112	0.129
Fatty Acid Metabolism (Acyl Carnitine, Polyunsaturated)	5	5	Inf	Inf	0.0112	0.129
Long Chain Monounsaturated Fatty Acid	7	6	8.8	2.175	0.0208	0.197
Long Chain Saturated Fatty Acid	7	6	8.8	2.175	0.0208	0.197
Fatty Acid Metabolism (Acyl Choline)	4	4	Inf	Inf	0.0277	0.222
Pregnenolone Steroids	4	4	Inf	Inf	0.0277	0.222
Fatty Acid Metabolism (Acyl Carnitine, Long Chain Saturated)	8	6	4.393	1.48	0.0546	0.406
Ascorbate and Aldarate Metabolism	3	3	Inf	Inf	0.068	0.442
Lysoplasmalogen	3	3	Inf	Inf	0.068	0.442
Phosphatidylcholine (PC)	19	11	2.022	0.704	0.0995	0.595
Gamma-glutamyl Amino Acid	17	10	2.098	0.741	0.103	0.595
Androgenic Steroids	14	8	1.95	0.668	0.165	0.795
Xanthine Metabolism	14	8	1.95	0.668	0.165	0.795
Fatty Acid, Branched	2	2	Inf	Inf	0.167	0.795
Primary Bile Acid Metabolism	8	5	2.429	0.887	0.187	0.795
Progestin Steroids	4	3	4.36	1.472	0.189	0.795
Phosphatidylinositol (PI)	6	4	2.911	1.068	0.191	0.795
Phospholipid Metabolism	6	4	2.911	1.068	0.191	0.795
Ceramides	11	6	1.747	0.558	0.265	1
Glutathione Metabolism	7	4	1.937	0.661	0.307	1

Phenylalanine Metabolism	7	4	1.937	0.661	0.307	1
Fatty Acid Metabolism (Acyl Carnitine, Hydroxy)	3	2	2.899	1.064	0.365	1
Lactosylceramides (LCER)	3	2	2.899	1.064	0.365	1
Fatty Acid Metabolism (Acyl Carnitine, Short Chain)	1	1	Inf	Inf	0.409	1
Ketone Bodies	1	1	Inf	Inf	0.409	1
Medium Chain Fatty Acid	8	4	1.45	0.372	0.427	1
Fatty Acid, Dicarboxylate	23	10	1.115	0.109	0.479	1

results/006/enrichment/current-subpathway.csv

Former Smokers

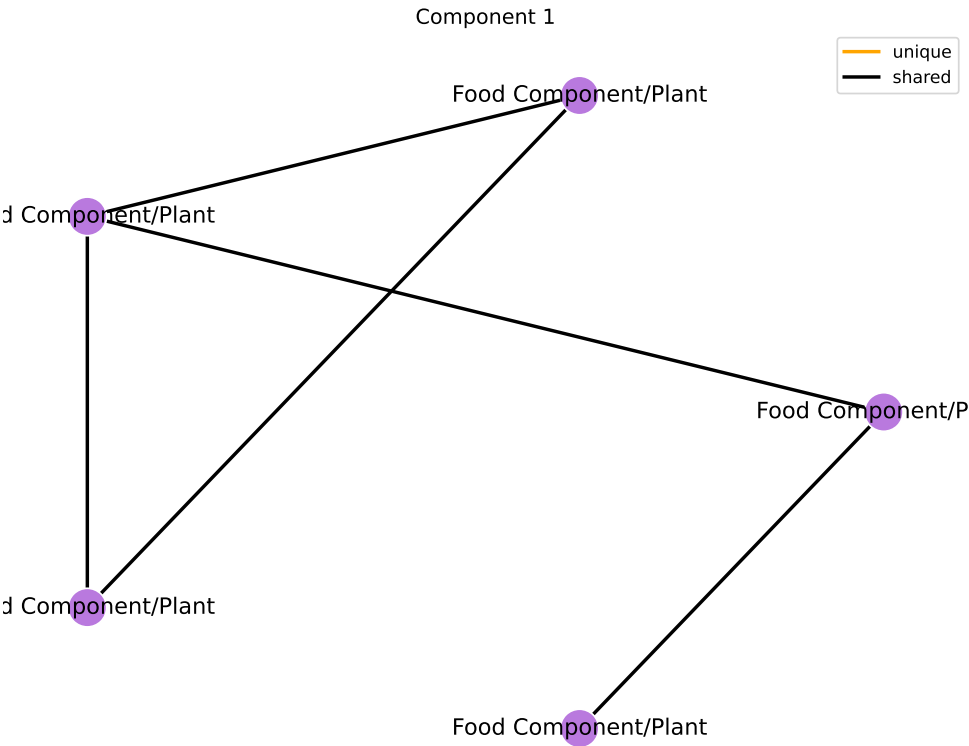
Component 1

Super-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Xenobiotics	96	6	Inf	Inf	3.59e-06	3.23e-05

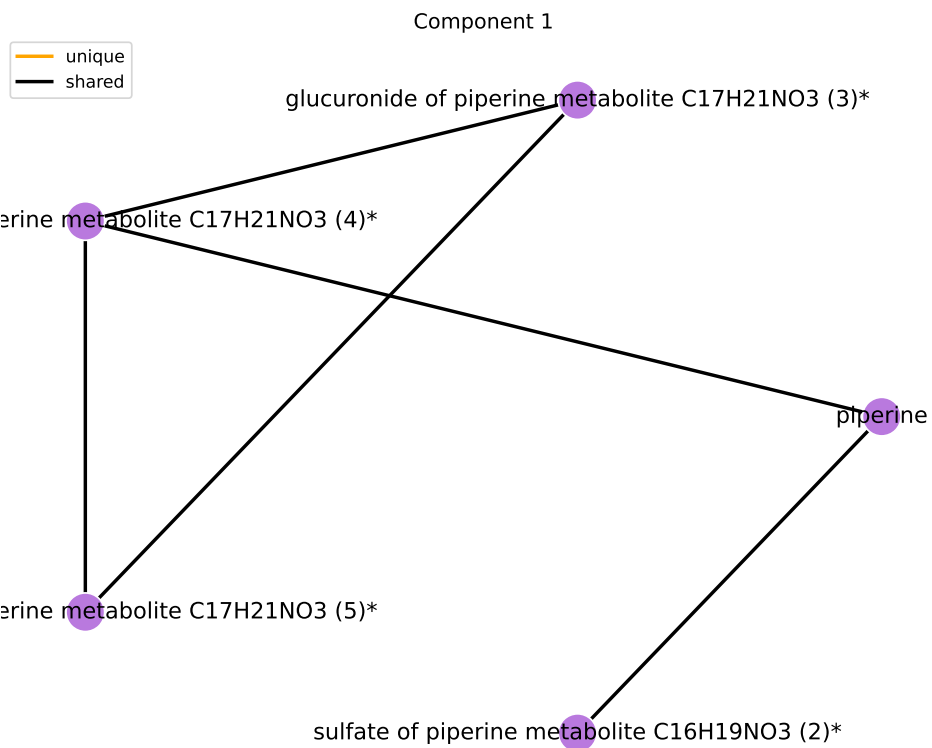
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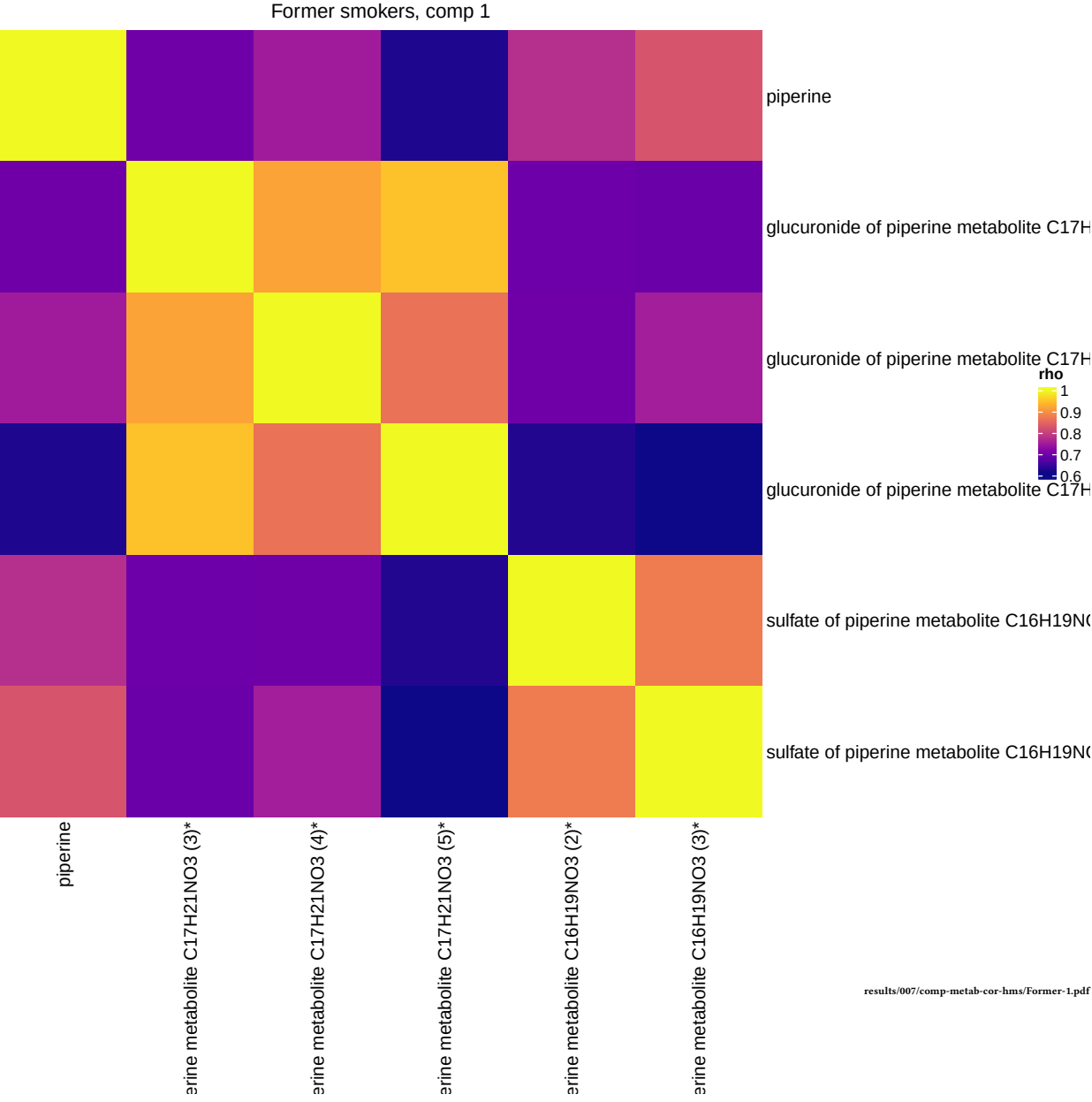
Sub-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Food Component/Plant	38	6	Inf	Inf	1.07e-08	1.11e-06

results/006/enrichment/form-components/1-subpathway.csv



results/005_1/component-graphs/former/sub_pathway-0.pdf





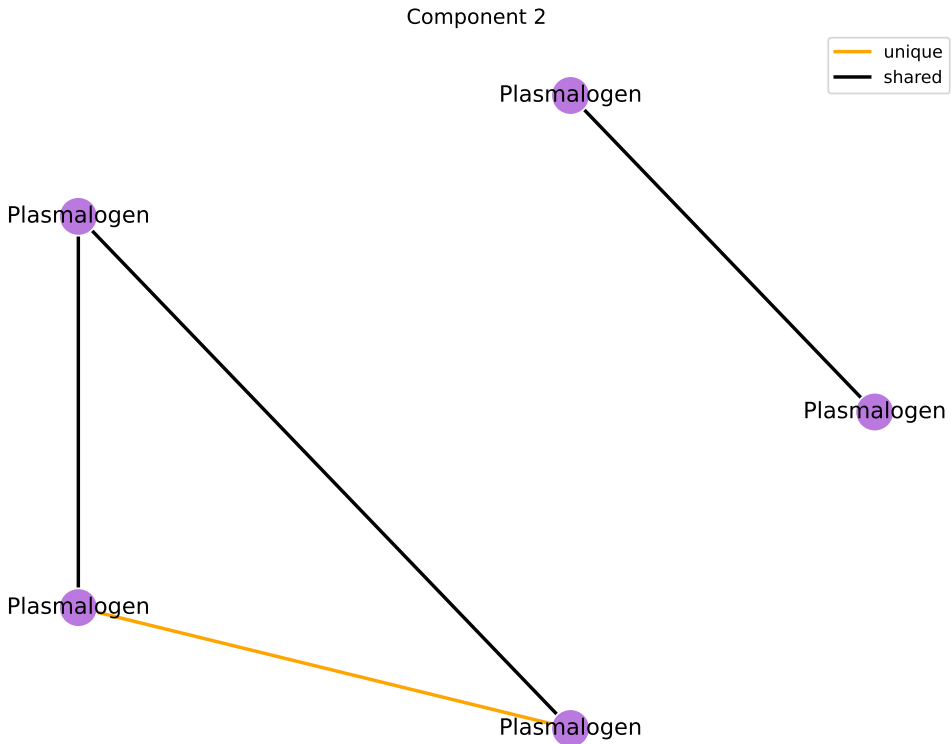
Component 2

Super-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Lipid	363	6	Inf	Inf	0.0118	0.106

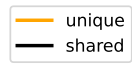
results/006/enrichment/form-components/2-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Plasmalogen	11	6	Inf	Inf	1.79e-12	1.86e-10

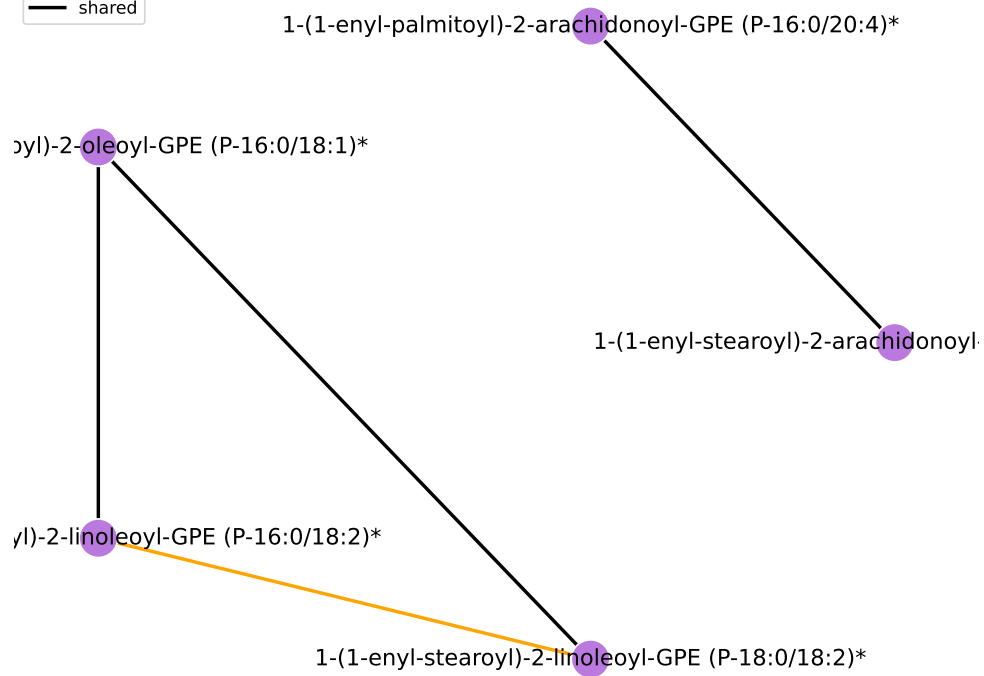
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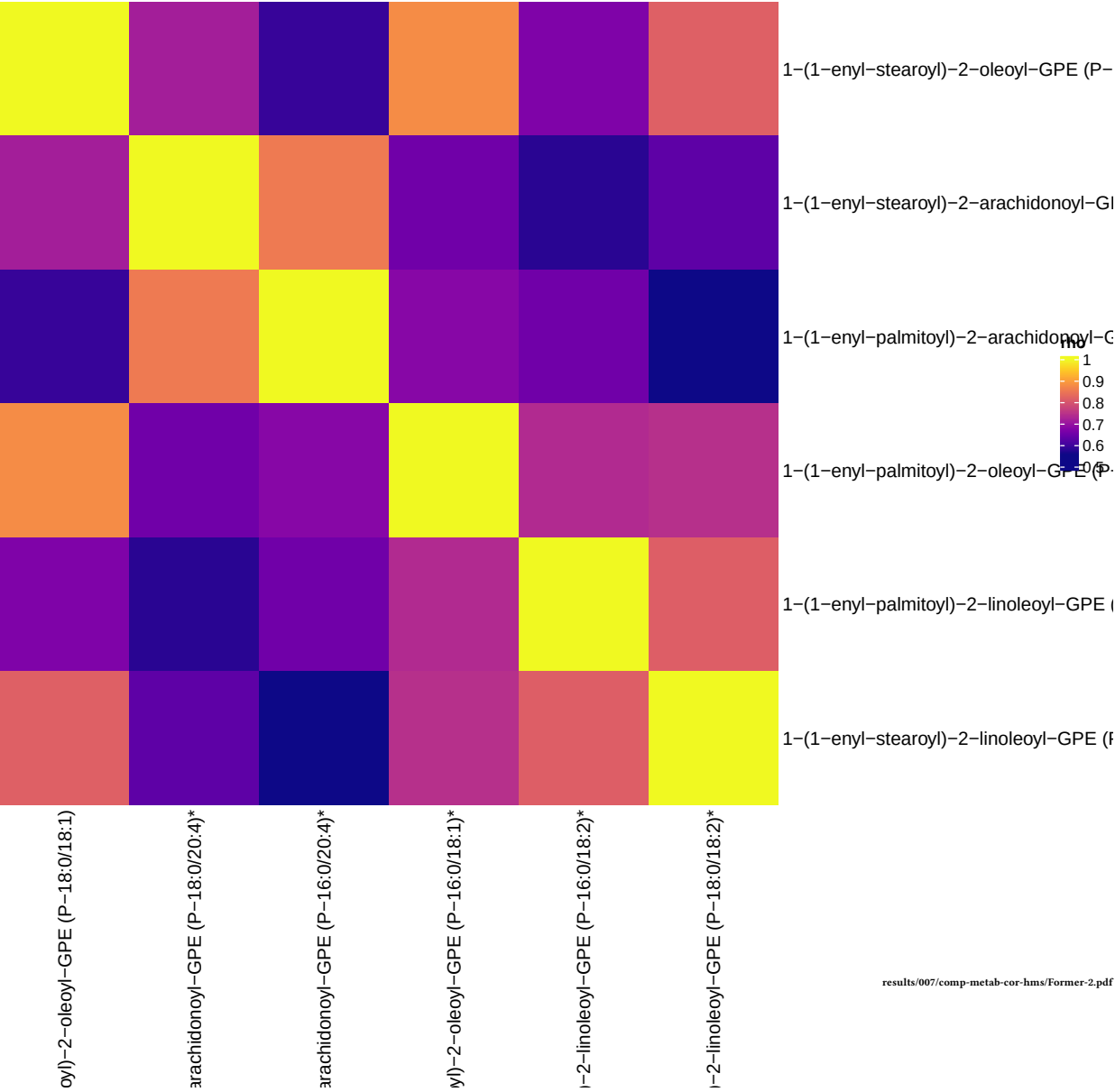
results/005_1/component-graphs/former/sub_pathway-1.pdf



Component 2



Former smokers, comp 2



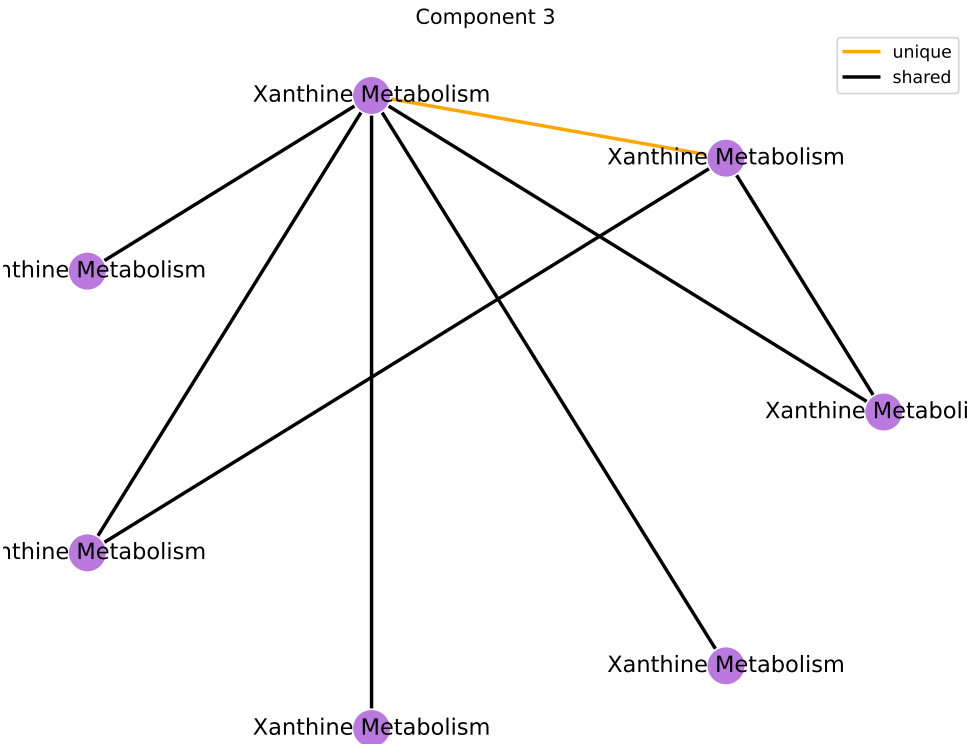
Component 3

Super-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Xenobiotics	96	8	Inf	Inf	5.09e-08	4.58e-07

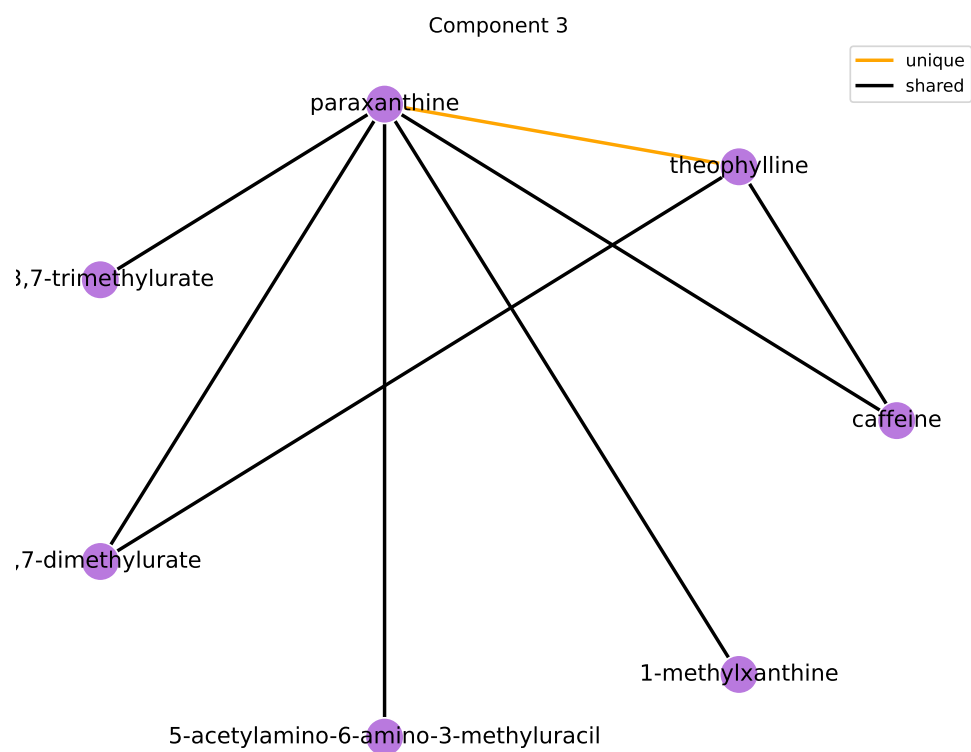
results/006/enrichment/form-components/3-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Xanthine Metabolism	14	8	Inf	Inf	1.15e-15	1.2e-13

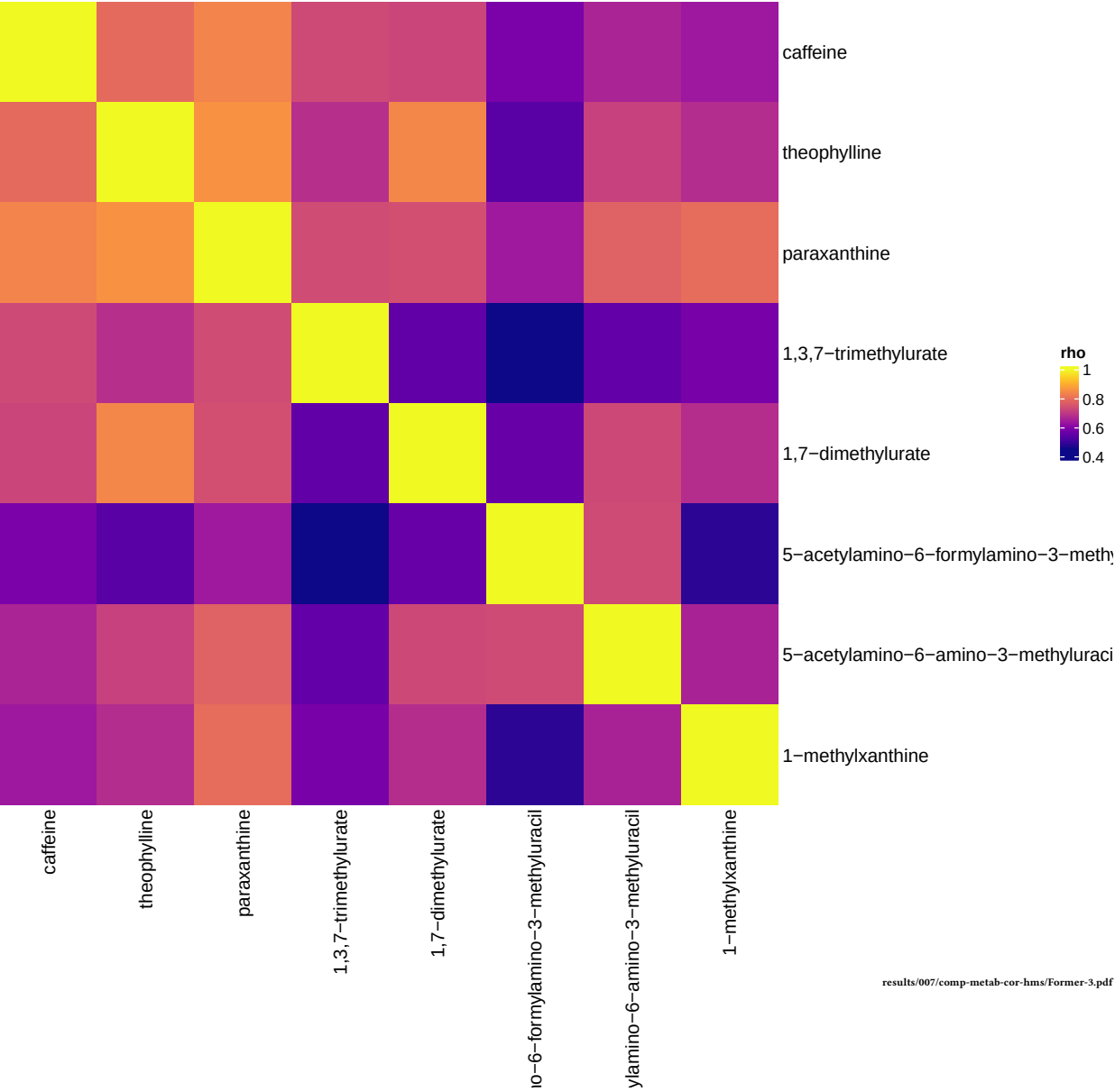
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results/005_1/component-graphs/former/sub_pathway-2.pdf



Former smokers, comp 3



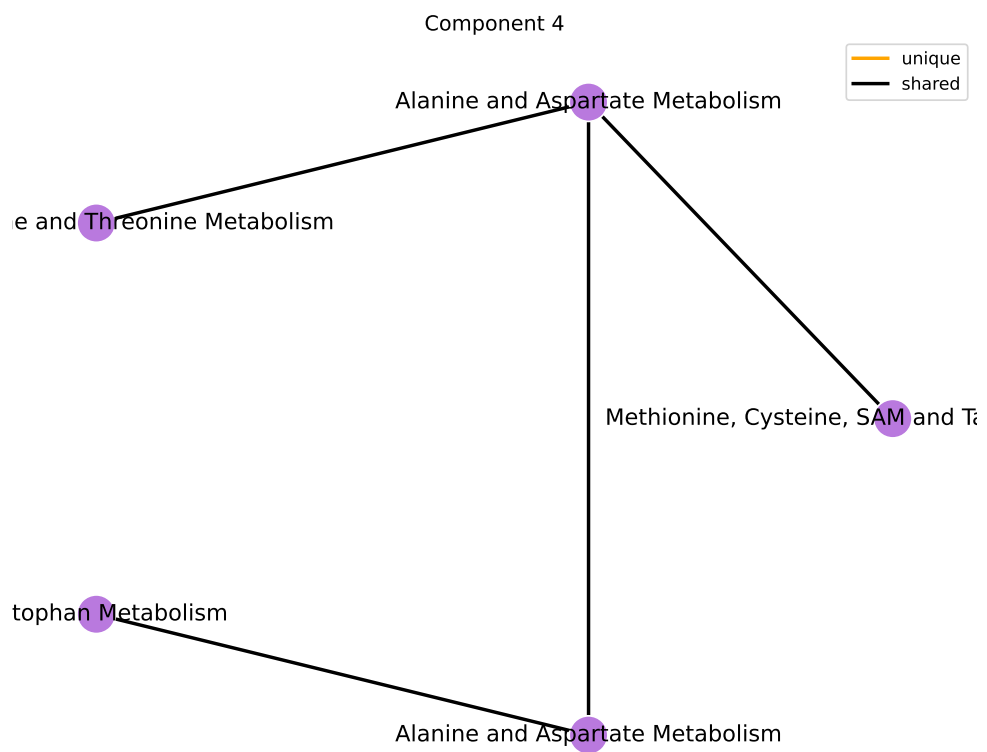
Component 4

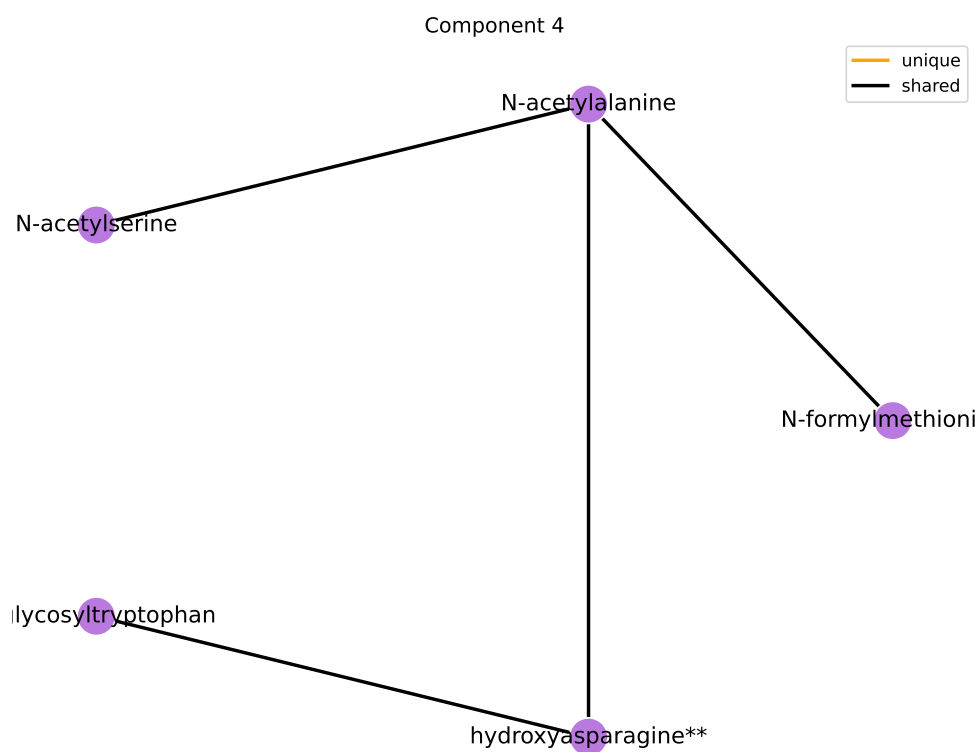
Super-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Nucleotide	32	4	14.512	2.675	0.000704	0.00634
Amino Acid	180	7	5.789	1.756	0.00524	0.0236

results/006/enrichment/form-components/4-superpathway.csv

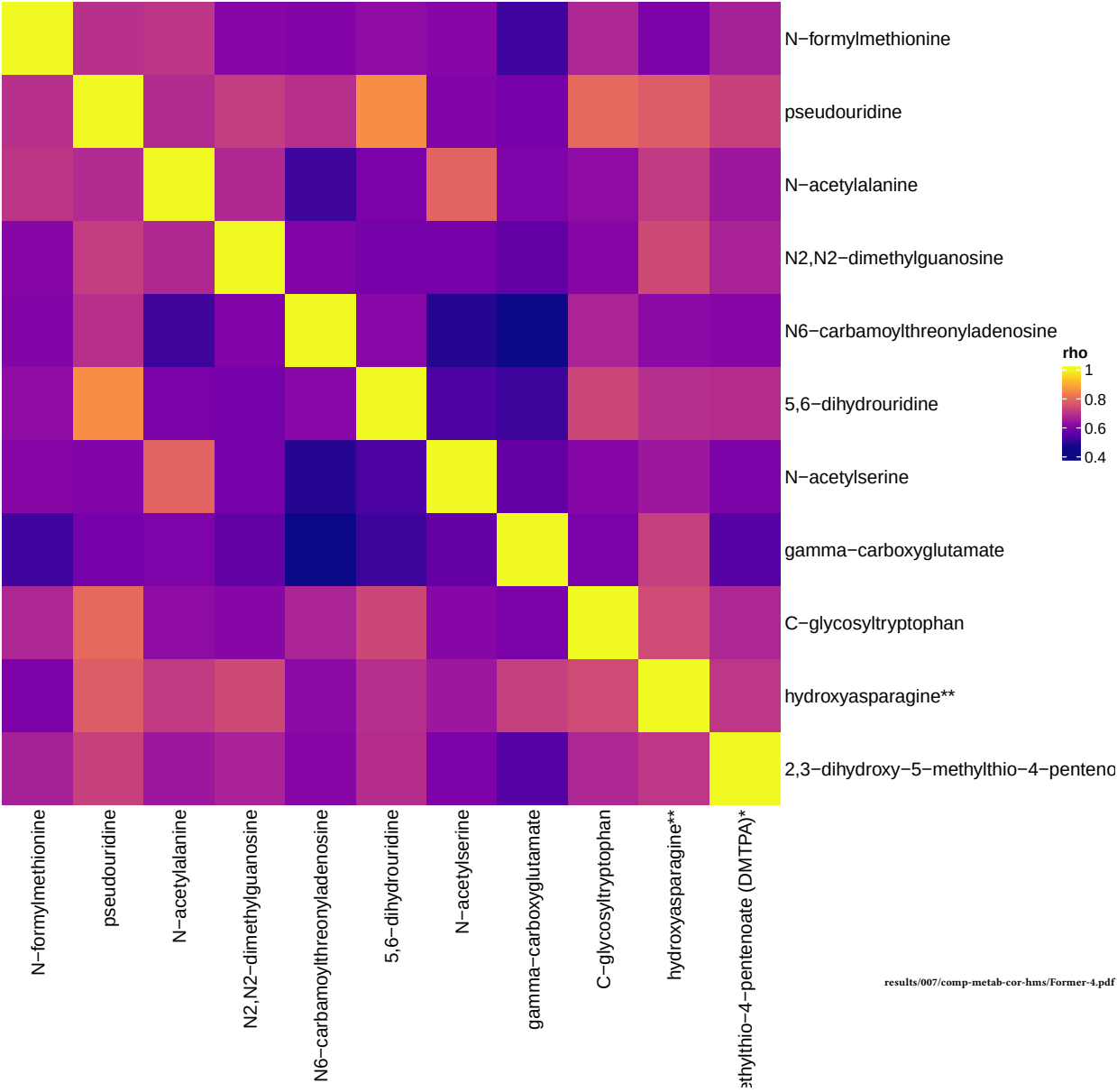
Sub-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Alanine and Aspartate Metabolism	8	2	26.804	3.289	0.00512	0.421
Pyrimidine Metabolism, Uracil containing	10	2	20.157	3.004	0.00809	0.421
Methionine, Cysteine, SAM and Taurine Metabolism	17	2	10.738	2.374	0.0231	0.749
Purine Metabolism, Guanine containing	2	1	71.303	4.267	0.0288	0.749
Purine Metabolism, Adenine containing	4	1	24.264	3.189	0.0569	1
Glycine, Serine and Threonine Metabolism	10	1	8.134	2.096	0.137	1
Glutamate Metabolism	11	1	7.318	1.99	0.149	1
Tryptophan Metabolism	16	1	4.857	1.58	0.21	1

results/006/enrichment/form-components/4-subpathway.csv





Former smokers, comp 4



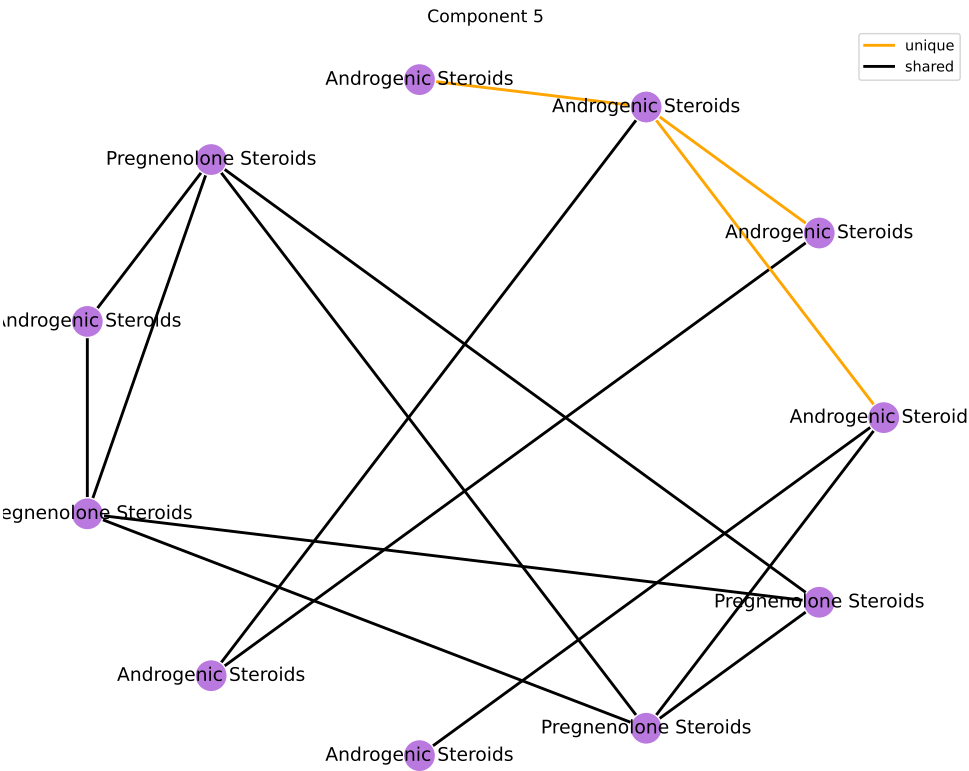
Component 5

Super-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Lipid	363	13	Inf	Inf	6.22e-05	0.00056

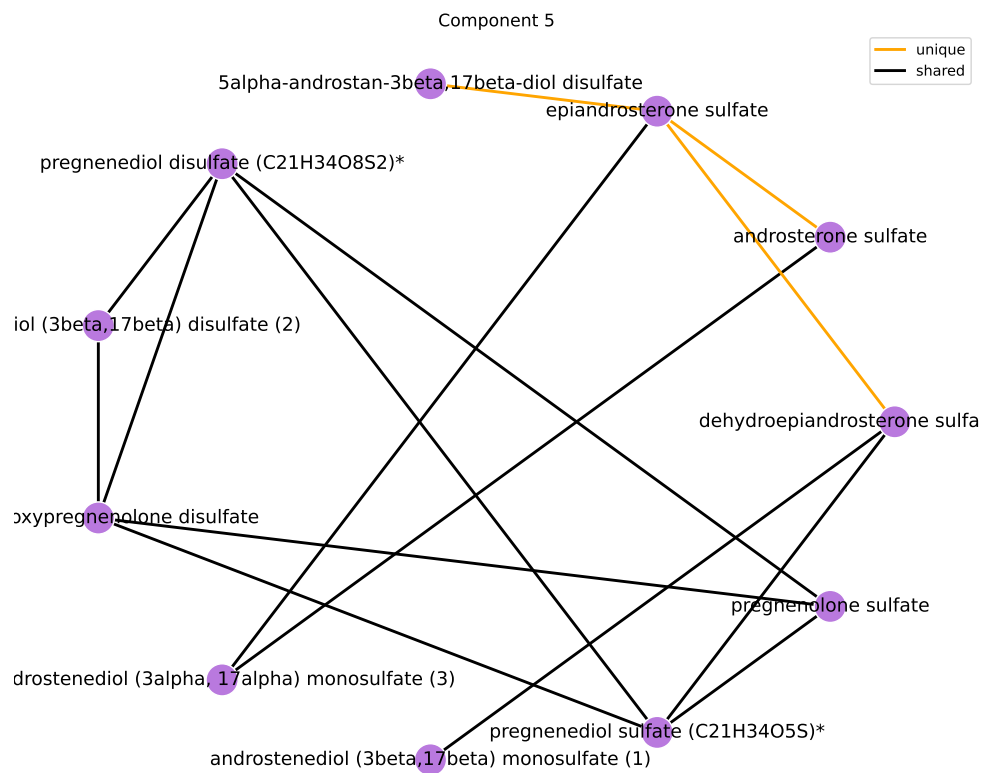
results/006/enrichment/form-components/5-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Androgenic Steroids	14	9	303.398	5.715	6.44e-15	6.7e-13
Pregnenolone Steroids	4	4	Inf	Inf	5.24e-08	2.72e-06

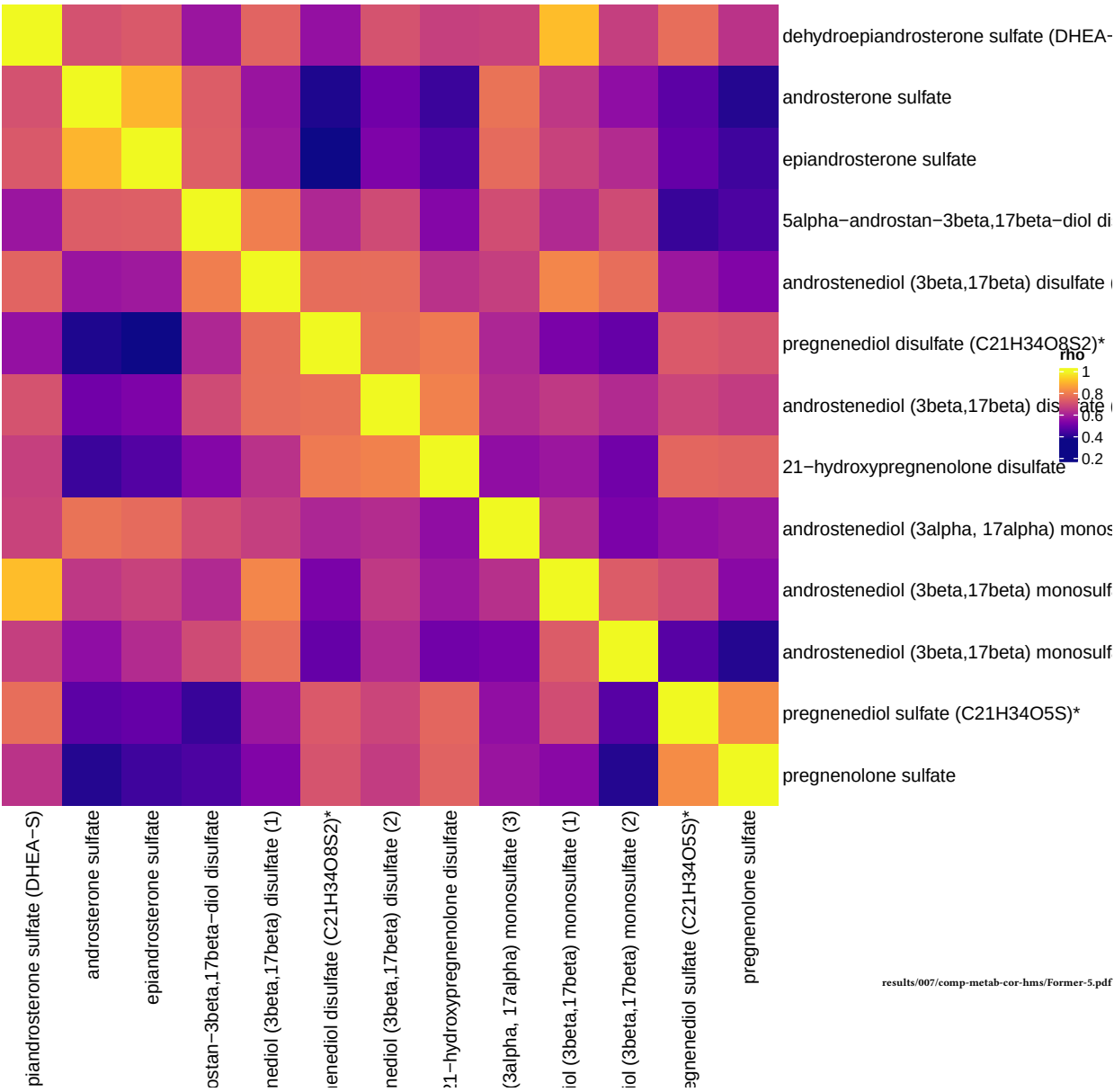
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results/005_1/component-graphs/former/sub_pathway-4.pdf



Former smokers, comp 5



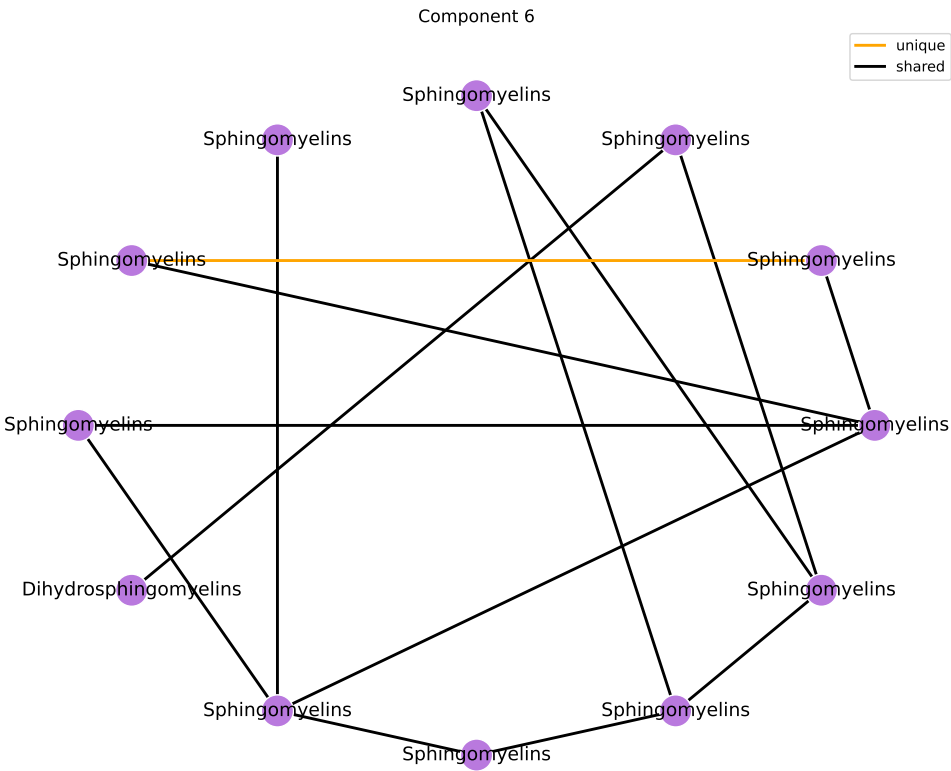
Component 6

Super-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Lipid	363	13	Inf	Inf	6.22e-05	0.00056

results/006/enrichment/form-components/6-superpathway.csv

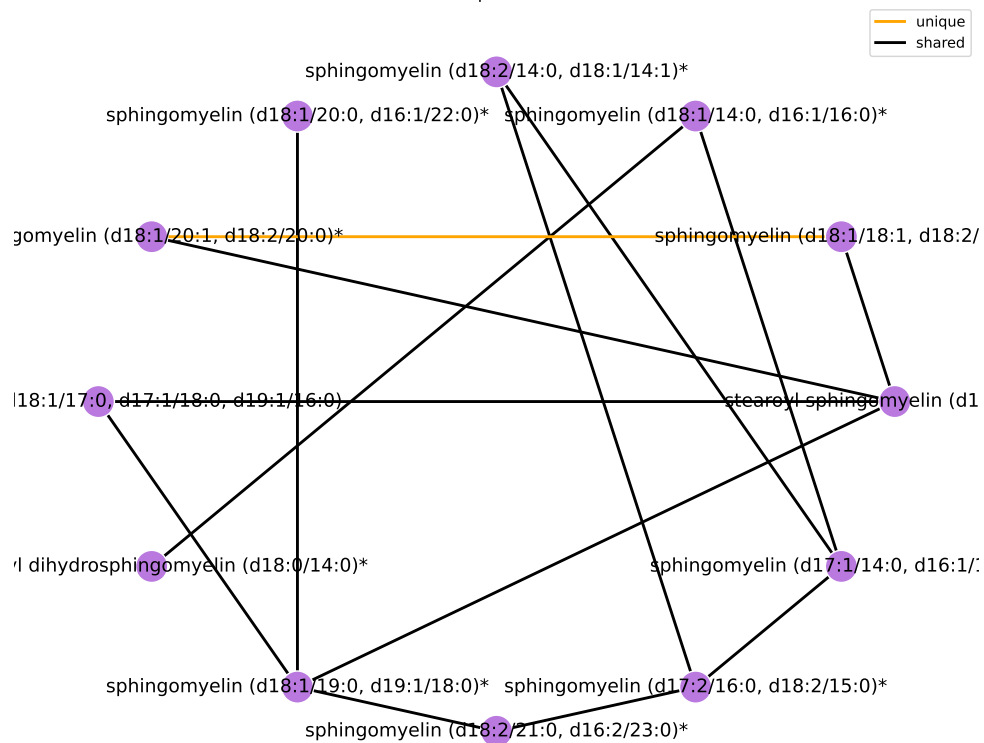
Sub-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Sphingomyelins	28	12	518.406	6.251	5.63e-18	5.86e-16
Dihydrosphingomyelins	5	1	15.216	2.722	0.0831	1

results/006/enrichment/form-components/6-subpathway.csv

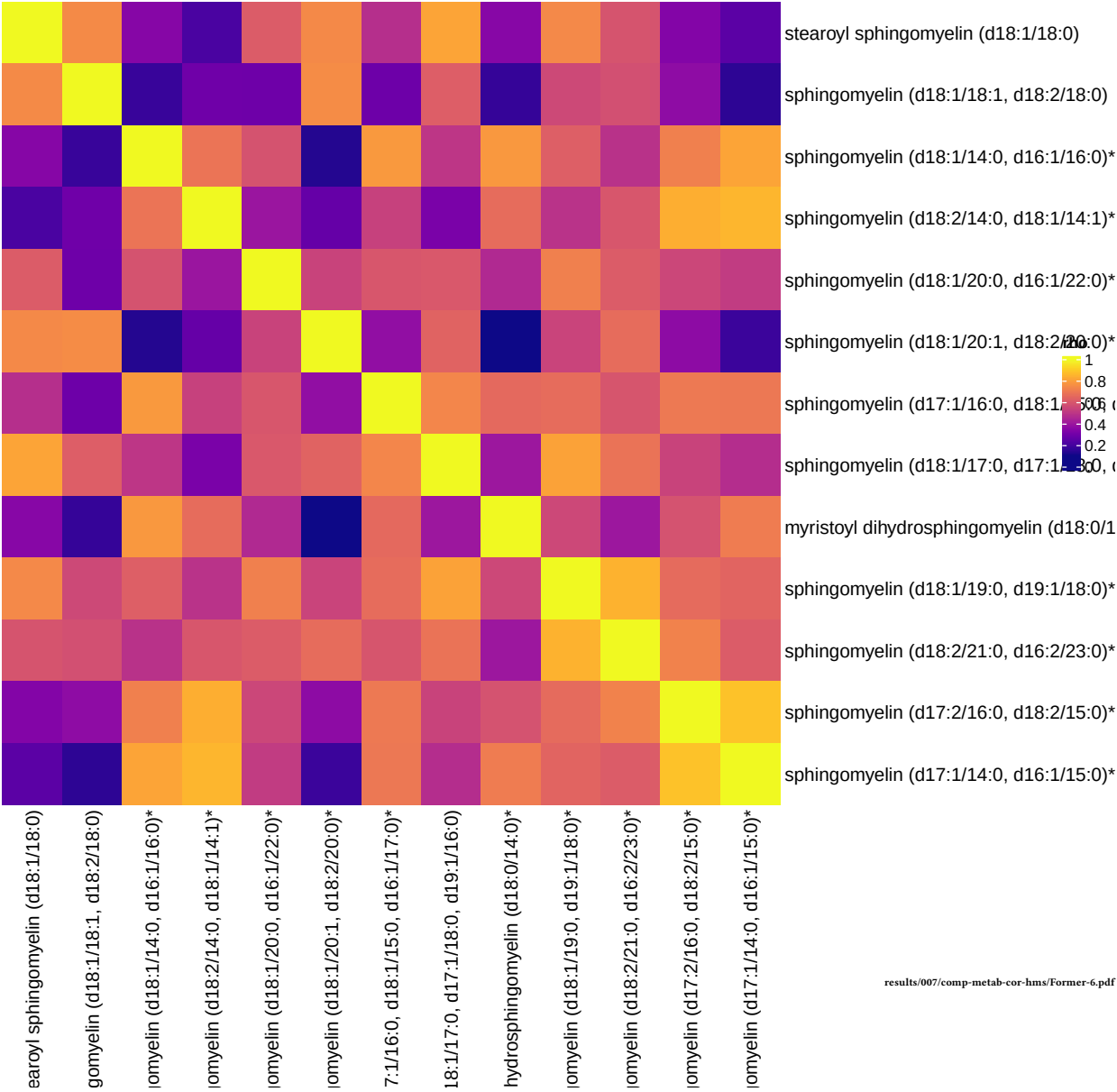


results/005_1/component-graphs/former/sub_pathway-5.pdf

Component 6



Former smokers, comp 6



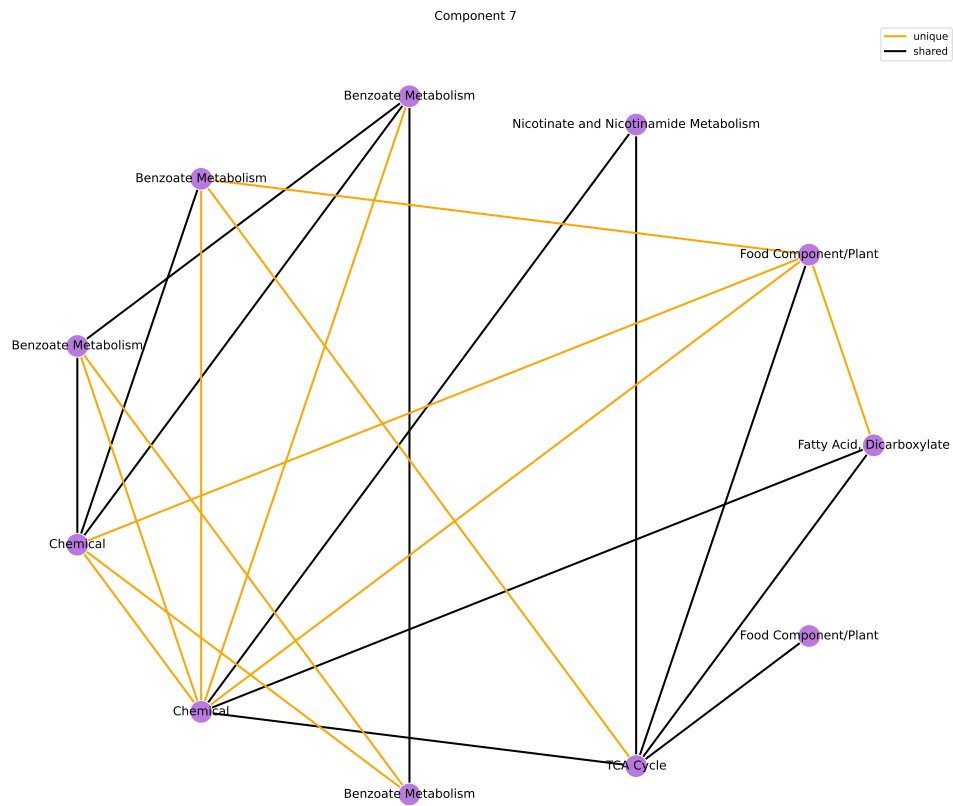
Component 7

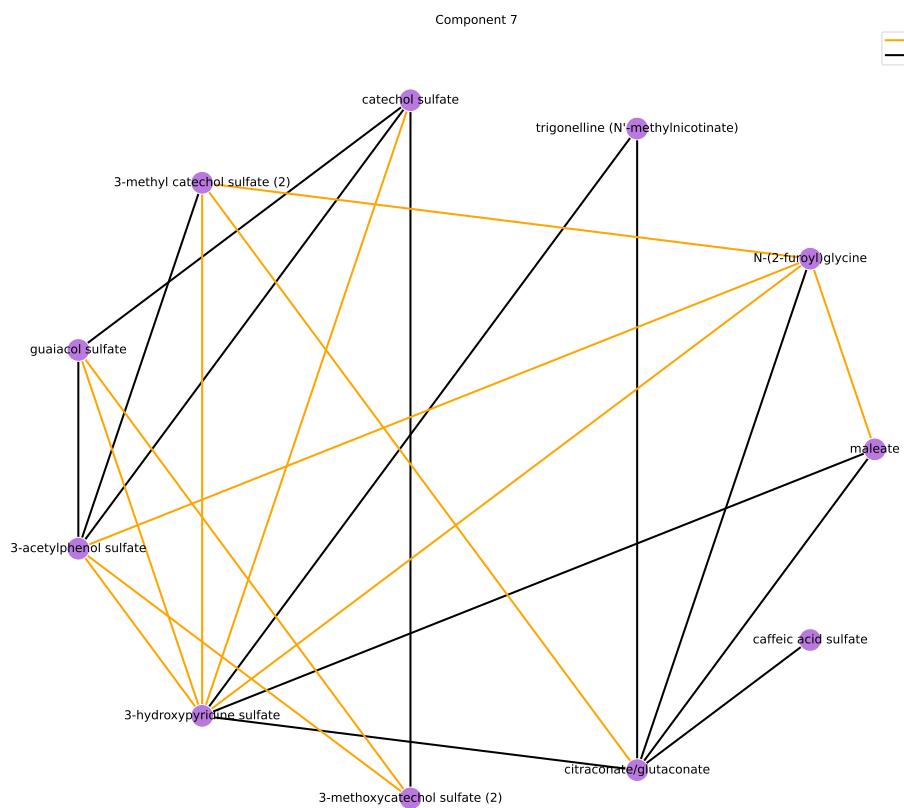
Super-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Xenobiotics	96	12	23.307	3.149	1.06e-08	9.54e-08
Energy	10	1	5.401	1.687	0.193	0.868
Cofactors and Vitamins	25	1	1.992	0.689	0.418	1

results/006/enrichment/form-components/7-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Benzoate Metabolism	21	5	20.312	3.011	3.55e-05	0.00369
Food Component/Plant	38	5	9.691	2.271	0.000708	0.0368
Chemical	18	2	6.444	1.863	0.0525	1
Nicotinate and Nicotinamide Metabolism	5	1	12.157	2.498	0.101	1
TCA Cycle	9	1	6.081	1.805	0.176	1
Tyrosine Metabolism	17	1	3.018	1.105	0.307	1
Fatty Acid, Dicarboxylate	23	1	2.179	0.779	0.392	1

results/006/enrichment/form-components/7-subpathway.csv





Heatmap showing the correlation (rho) between 14 different chemical compounds. The color scale ranges from 0.2 (dark blue) to 1.0 (yellow). The diagonal is all yellow (rho=1). The matrix is symmetric.

Compounds (X-axis labels):

- quinate
- maleate
- N-(2-furoyl)glycine
- trigonelline (N'-methylnicotinate)
- catechol sulfate
- 3-methyl catechol sulfate (2)
- 3-methyl catechol sulfate (1)
- guaiacol sulfate
- 3-acetylphenol sulfate
- 3-hydroxypyridine sulfate
- 4-vinylguaiacol sulfate
- 4-methoxyphenol sulfate
- 3-methoxycatechol sulfate (2)
- citraconate/glutaconate
- caffeic acid sulfate
- 3-ethylcatechol sulfate (1)

Compounds (Y-axis labels):

- quinate
- maleate
- N-(2-furoyl)glycine
- trigonelline (N'-methylnicotinate)
- catechol sulfate
- 3-methyl catechol sulfate (2)
- 3-methyl catechol sulfate (1)
- guaiacol sulfate
- 3-acetylphenol sulfate
- 3-hydroxypyridine sulfate
- 4-vinylguaiacol sulfate
- 4-methoxyphenol sulfate
- 3-methoxycatechol sulfate (2)
- citraconate/glutaconate
- caffeic acid sulfate
- 3-ethylcatechol sulfate (1)

Color scale (rho):

- 1.0 (Yellow)
- 0.8 (Orange)
- 0.6 (Pink)
- 0.4 (Purple)
- 0.2 (Dark Blue)

Source: results/007/comp-metab-cor-hms/Former-7.pdf

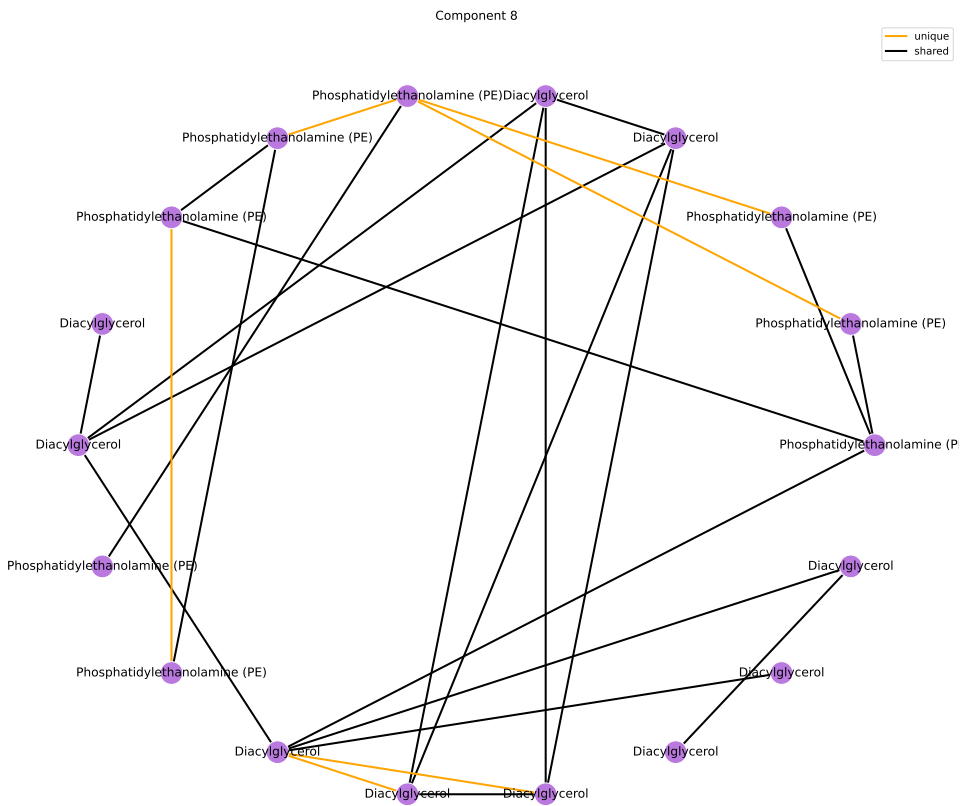
Component 8

Super-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Lipid	363	19	Inf	Inf	6.53e-07	5.88e-06

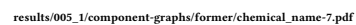
results/006/enrichment/form-components/8-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Diacylglycerol	19	11	119.97	4.787	4.77e-15	4.96e-13
Phosphatidylethanolamine (PE)	11	8	168.335	5.126	4.6e-12	2.39e-10

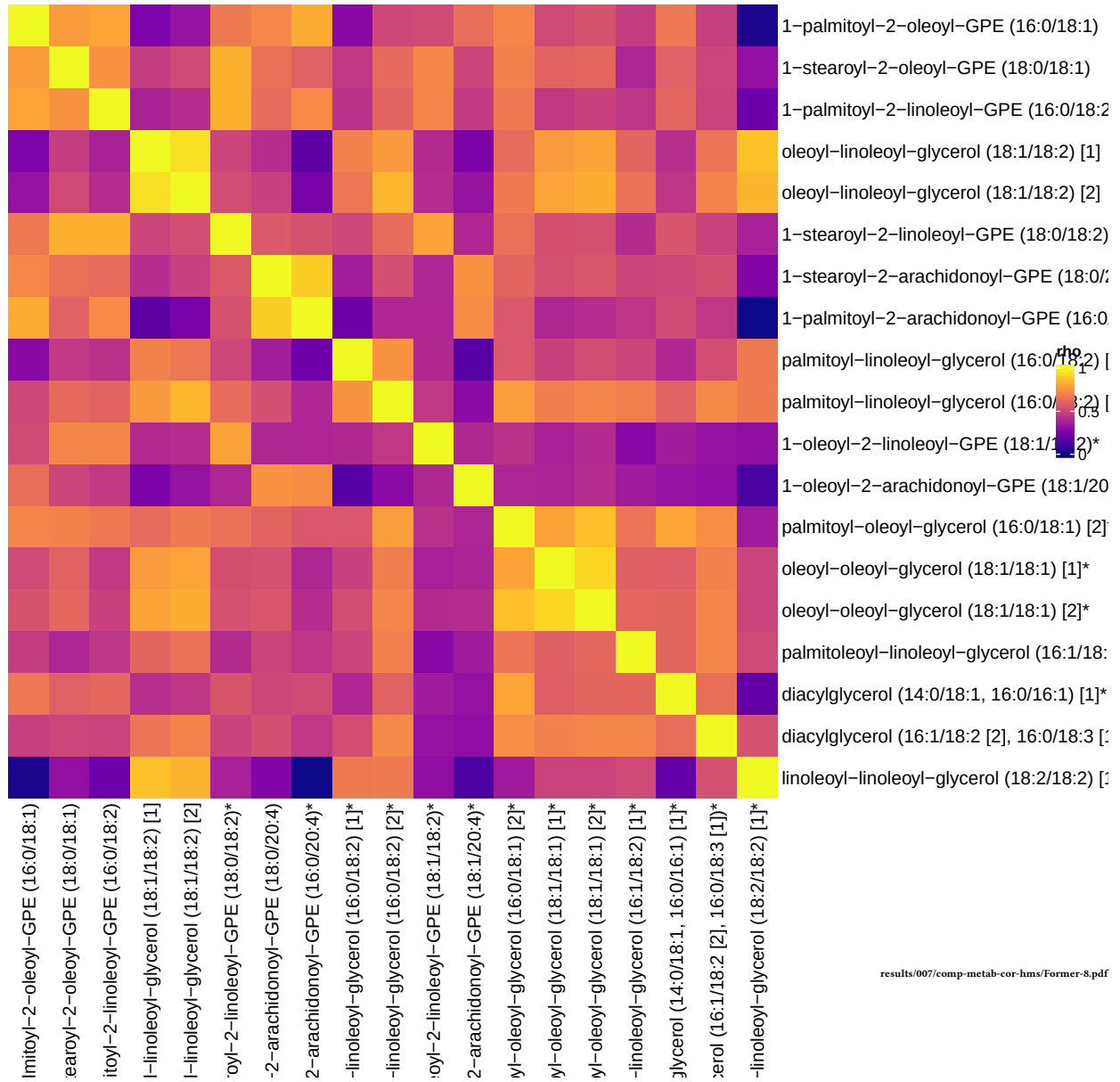
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results/005_1/component-graphs/former/sub_pathway-7.pdf



Former smokers, comp 8



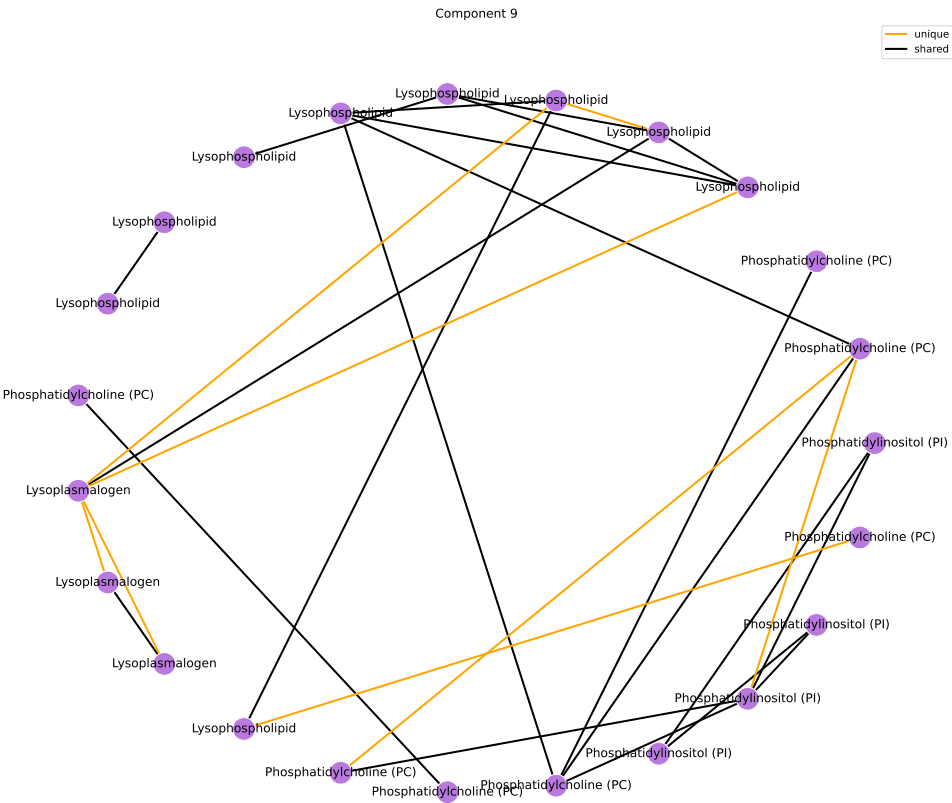
Component 9

Super-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Lipid	363	30	Inf	Inf	1.33e-10	1.2e-09

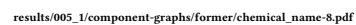
results/006/enrichment/form-components/9-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Lysophospholipid	24	12	38.964	3.663	2.59e-12	2.69e-10
Phosphatidylcholine (PC)	19	10	39.118	3.667	1.37e-10	7.12e-09
Phosphatidylinositol (PI)	6	5	141.28	4.951	4.04e-07	1.4e-05
Lysoplasmalogen	3	3	Inf	Inf	5.62e-05	0.00146

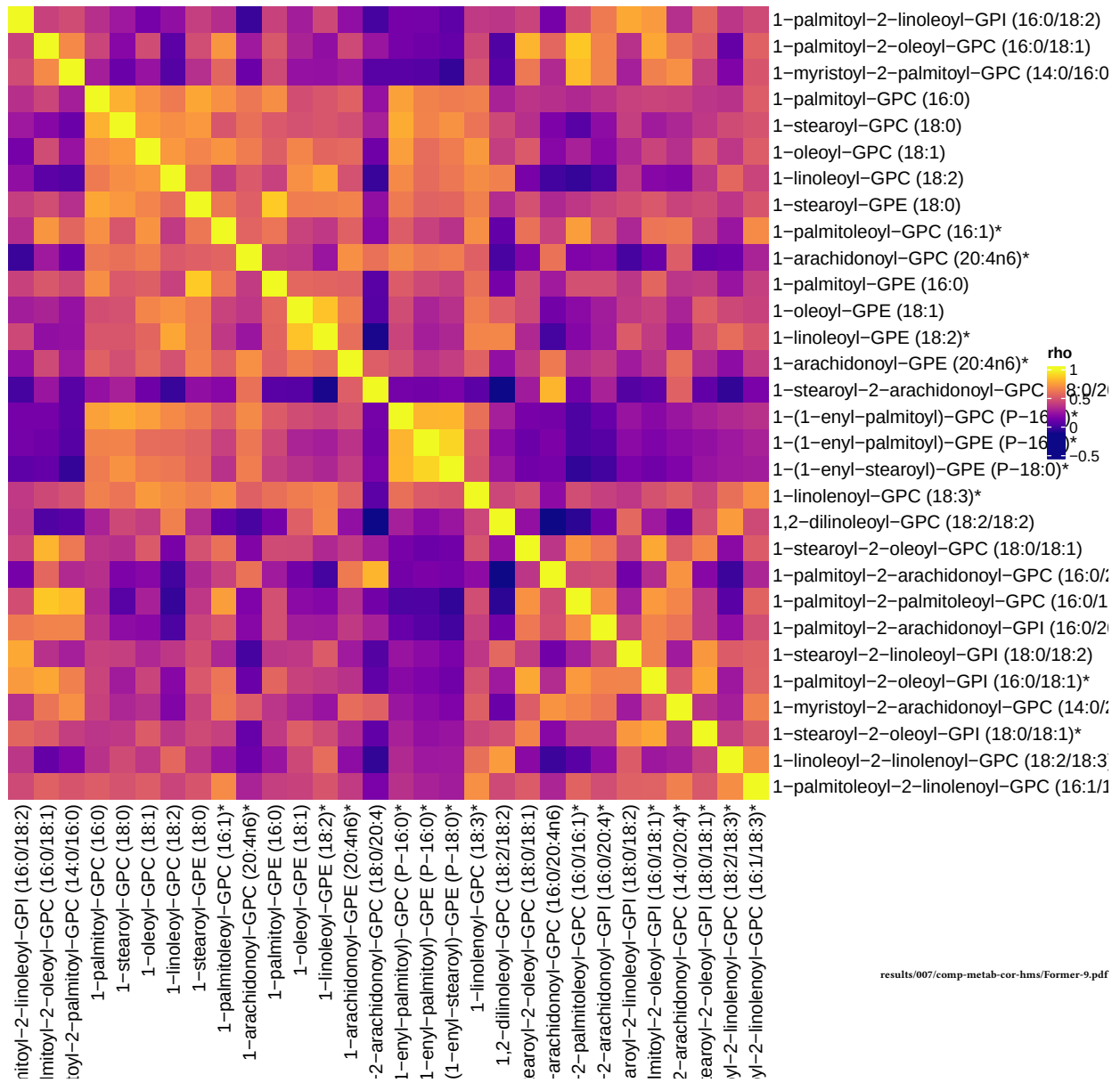
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results/005_1/component-graphs/former/sub_pathway-8.pdf



Former smokers, comp 9



Component 10

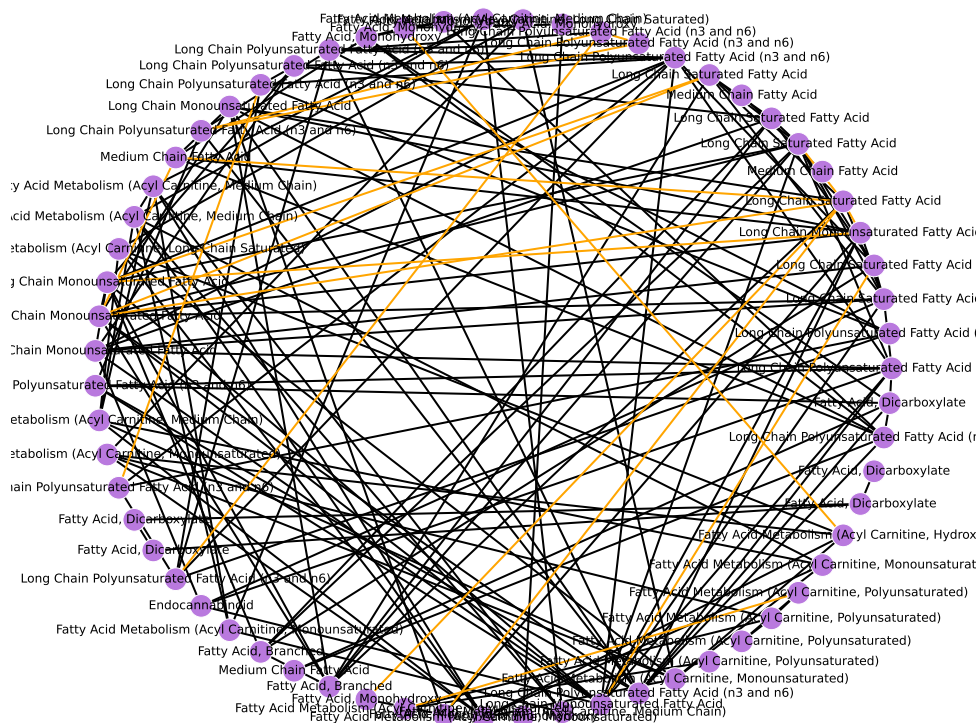
Super-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Lipid	363	69	Inf	Inf	2.09e-24	1.88e-23

results/006/enrichment/form-components/10-superpathway.csv

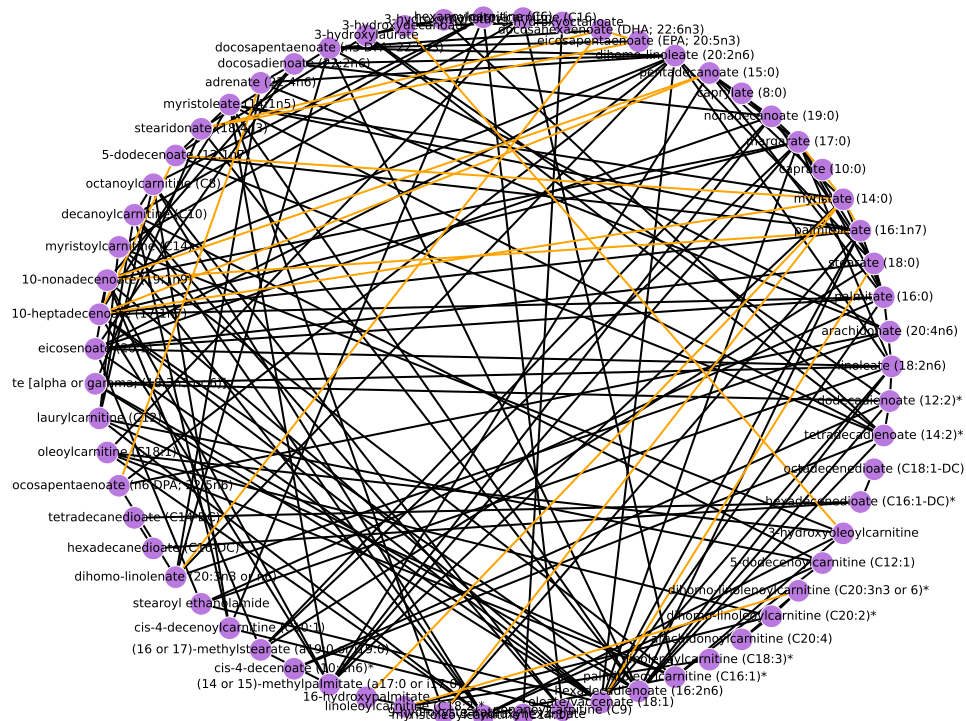
Sub-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Long Chain Polyunsaturated Fatty Acid (n3 and n6)	14	14	Inf	Inf	7.36e-16	7.65e-14
Long Chain Monounsaturated Fatty Acid	7	6	64.656	4.169	3.02e-06	0.000105
Long Chain Saturated Fatty Acid	7	6	64.656	4.169	3.02e-06	0.000105
Fatty Acid Metabolism (Acyl Carnitine, Medium Chain)	5	5	Inf	Inf	5.46e-06	0.000114
Fatty Acid Metabolism (Acyl Carnitine, Polyunsaturated)	5	5	Inf	Inf	5.46e-06	0.000114
Fatty Acid, Monohydroxy	20	8	7.36	1.996	0.000164	0.00284
Fatty Acid Metabolism (Acyl Carnitine, Monounsaturated)	8	5	17.702	2.874	0.000245	0.00318
Medium Chain Fatty Acid	8	5	17.702	2.874	0.000245	0.00318
Fatty Acid, Dicarboxylate	23	7	4.732	1.554	0.00288	0.0333
Fatty Acid, Branched	2	2	Inf	Inf	0.00818	0.0851
Fatty Acid Metabolism (Acyl Carnitine, Long Chain Saturated)	8	3	6.189	1.823	0.029	0.274
Glycerolipid Metabolism	2	1	10.051	2.308	0.174	1
Fatty Acid Metabolism (Acyl Carnitine, Hydroxy)	3	1	5.032	1.616	0.249	1
Endocannabinoid	5	1	2.514	0.922	0.38	1

results/006/enrichment/form-components/10-subpathway.csv

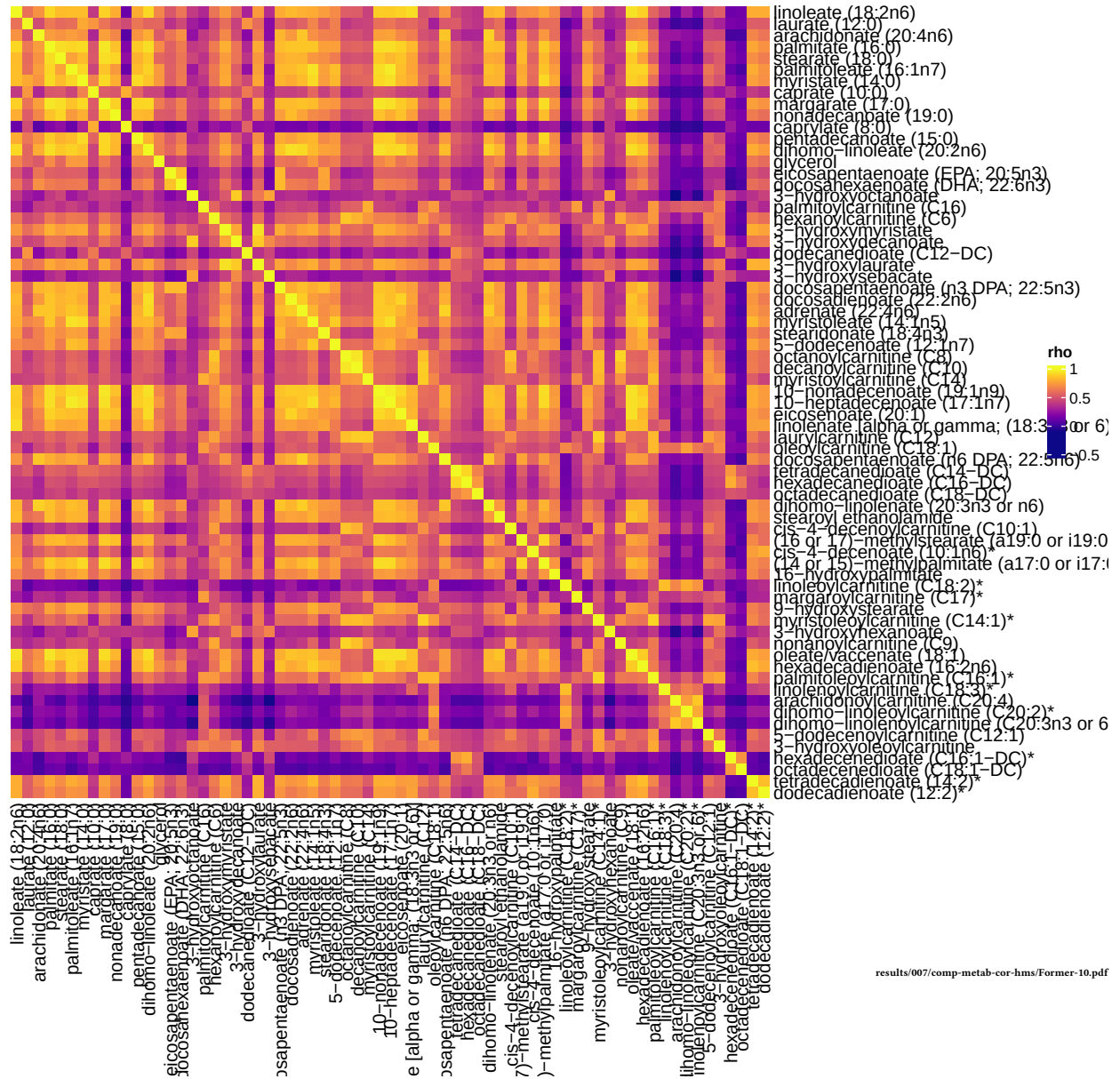
— unique
— shared

[results/005_1/component-graphs/former/sub_pathway-9.pdf](#)

— unique
— shared

[results/005_1/component-graphs/former/chemical_name-9.pdf](#)

Former smokers, comp 10



results/007/comp-metab-cor-hms/Former-10.pdf

Super-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Cofactors and Vitamins	25	10	5.825	1.762	0.00015	0.00135
Partially Characterized Molecules	3	2	16.072	2.777	0.0346	0.156
Xenobiotics	96	13	1.283	0.249	0.267	0.7
Energy	10	2	2.001	0.694	0.311	0.7
Nucleotide	32	4	1.137	0.128	0.492	0.886

results/006/enrichment/form-unique-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Dihydrosphingomyelins	5	3	12.194	2.501	0.0115	0.322

Nicotinate and Nicotinamide Metabolism	5	3	12.194	2.501	0.0115	0.322
Corticosteroids	2	2	Inf	Inf	0.0124	0.322
Vitamin B6 Metabolism	2	2	Inf	Inf	0.0124	0.322
Partially Characterized Molecules	3	2	16.072	2.777	0.0346	0.6
Pyrimidine Metabolism, Orotate containing	3	2	16.072	2.777	0.0346	0.6
Fatty Acid, Monohydroxy	20	5	2.737	1.007	0.0628	0.835
Hemoglobin and Porphyrin Metabolism	4	2	8.042	2.085	0.0642	0.835
Vitamin A Metabolism	5	2	5.361	1.679	0.0993	0.971
Fatty Acid, Dicarboxylate	23	5	2.271	0.82	0.104	0.971
Oxidative Phosphorylation	1	1	Inf	Inf	0.112	0.971
Pantothenate and CoA Metabolism	1	1	Inf	Inf	0.112	0.971
Chemical	18	4	2.321	0.842	0.133	1
Glycolysis, Gluconeogenesis, and Pyruvate Metabolism	6	2	4.019	1.391	0.138	1
Hexosylceramides (HCER)	7	2	3.212	1.167	0.18	1
Benzoate Metabolism	21	4	1.904	0.644	0.201	1
Drug - Topical Agents	2	1	7.958	2.074	0.212	1
Glycerolipid Metabolism	2	1	7.958	2.074	0.212	1
Purine Metabolism, Guanine containing	2	1	7.958	2.074	0.212	1
Alanine and Aspartate Metabolism	8	2	2.674	0.984	0.223	1
Tyrosine Metabolism	17	3	1.721	0.543	0.295	1
Glutamate Metabolism	11	2	1.776	0.574	0.355	1
Phosphatidylcholine (PC)	19	3	1.501	0.406	0.361	1
Progestin Steroids	4	1	2.654	0.976	0.379	1
Purine Metabolism, Adenine containing	4	1	2.654	0.976	0.379	1
Fatty Acid Metabolism (also BCAA Metabolism)	5	1	1.989	0.688	0.449	1
Androgenic Steroids	14	2	1.327	0.283	0.478	1

results/006/enrichment/form-unique-subpathway.csv

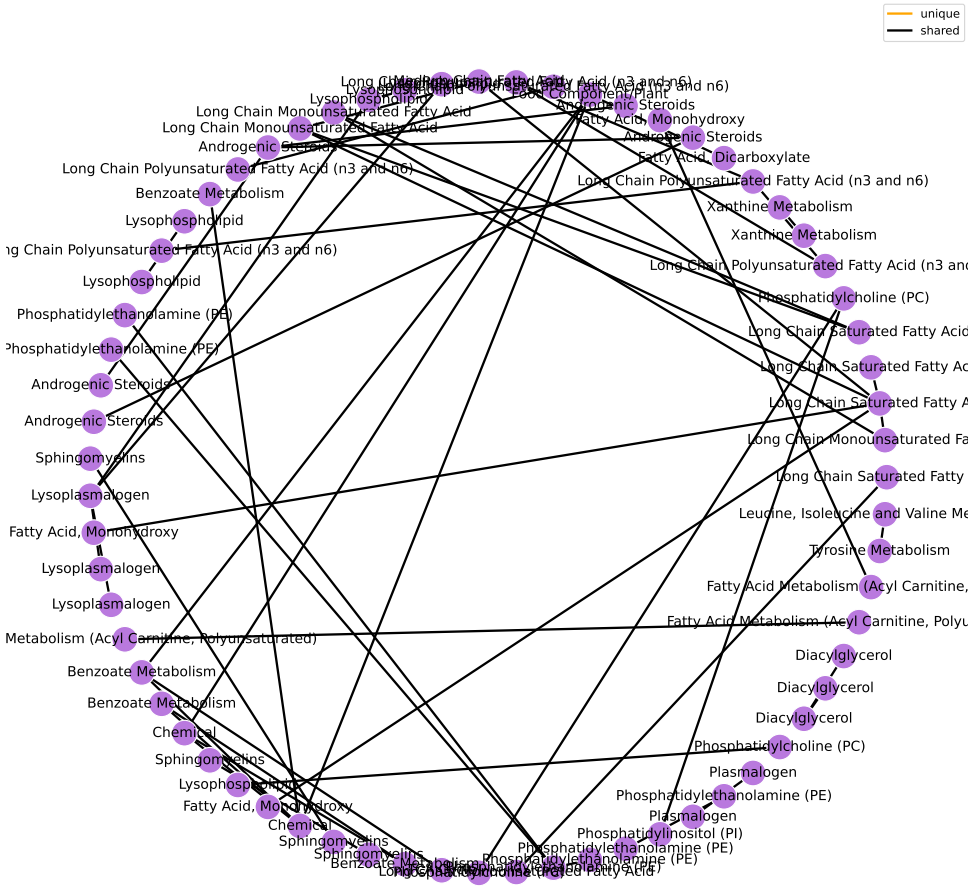
Nodes Only in Former Smokers



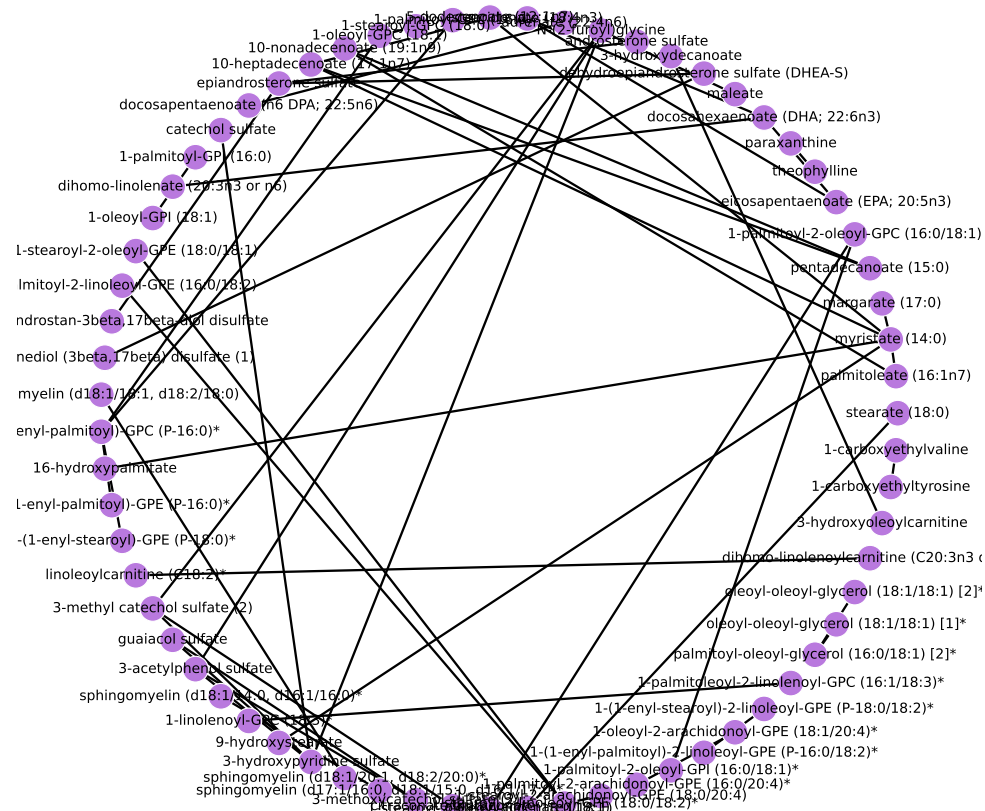
Nodes Only in Former Smokers



Edges Only in Former Smokers

[results/005_1/sub_pathway-edge-only-form-graph.pdf](#)

— unique
— shared



All Metabolites (not filtering for shared or unique)

Super-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Lipid	363	251	3.985	1.383	2.75e-20	2.47e-19
Cofactors and Vitamins	25	17	2.016	0.701	0.0739	0.333

results/006/enrichment/former-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Long Chain Polyunsaturated Fatty Acid (n3 and n6)	14	14	Inf	Inf	9.05e-05	0.00941
Diacylglycerol	19	17	8.189	2.103	0.000554	0.0236
Phosphatidylethanolamine (PE)	11	11	Inf	Inf	0.00068	0.0236
Lysophospholipid	24	19	3.652	1.295	0.00498	0.129
Plasmalogen	11	10	9.483	2.25	0.0078	0.146
Sphingomyelins	28	21	2.884	1.059	0.00949	0.146
Long Chain Saturated Fatty Acid	7	7	Inf	Inf	0.00981	0.146
Fatty Acid Metabolism (Acyl Carnitine, Medium Chain)	5	5	Inf	Inf	0.037	0.374
Fatty Acid Metabolism (Acyl Carnitine, Polyunsaturated)	5	5	Inf	Inf	0.037	0.374
Phosphatidylcholine (PC)	19	14	2.657	0.977	0.0431	0.374
Fatty Acid Metabolism (Acyl Carnitine, Long Chain Saturated)	8	7	6.588	1.885	0.0432	0.374
Fatty Acid Metabolism (Acyl Carnitine, Monounsaturated)	8	7	6.588	1.885	0.0432	0.374
Fatty Acid Metabolism (Acyl Choline)	4	4	Inf	Inf	0.0717	0.485
Pregnenolone Steroids	4	4	Inf	Inf	0.0717	0.485
Progestin Steroids	4	4	Inf	Inf	0.0717	0.485
Long Chain Monounsaturated Fatty Acid	7	6	5.633	1.729	0.0746	0.485
Androgenic Steroids	14	10	2.354	0.856	0.112	0.685
Phosphatidylinositol (PI)	6	5	4.683	1.544	0.127	0.688
Fatty Acid, Dicarboxylate	23	15	1.77	0.571	0.137	0.688
Ascorbate and Aldarate Metabolism	3	3	Inf	Inf	0.139	0.688
Lysoplasmalogen	3	3	Inf	Inf	0.139	0.688
Dihydrosphingomyelins	5	4	3.737	1.318	0.211	0.954
Nicotinate and Nicotinamide Metabolism	5	4	3.737	1.318	0.211	0.954
Glutathione Metabolism	7	5	2.336	0.848	0.257	1
Corticosteroids	2	2	Inf	Inf	0.268	1
Fatty Acid, Branched	2	2	Inf	Inf	0.268	1
Vitamin B6 Metabolism	2	2	Inf	Inf	0.268	1
Fatty Acid, Monohydroxy	20	12	1.405	0.34	0.305	1

Hemoglobin and Porphyrin Metabolism	4	3	2.796	1.028	0.34	1
Gamma-glutamyl Amino Acid	17	10	1.335	0.289	0.37	1
Phospholipid Metabolism	6	4	1.865	0.623	0.378	1
Benzoate Metabolism	21	12	1.245	0.219	0.394	1
Medium Chain Fatty Acid	8	5	1.554	0.441	0.404	1
Primary Bile Acid Metabolism	8	5	1.554	0.441	0.404	1
Xanthine Metabolism	14	8	1.243	0.218	0.45	1

results/006/enrichment/former-subpathway.csv

Intersection of Both

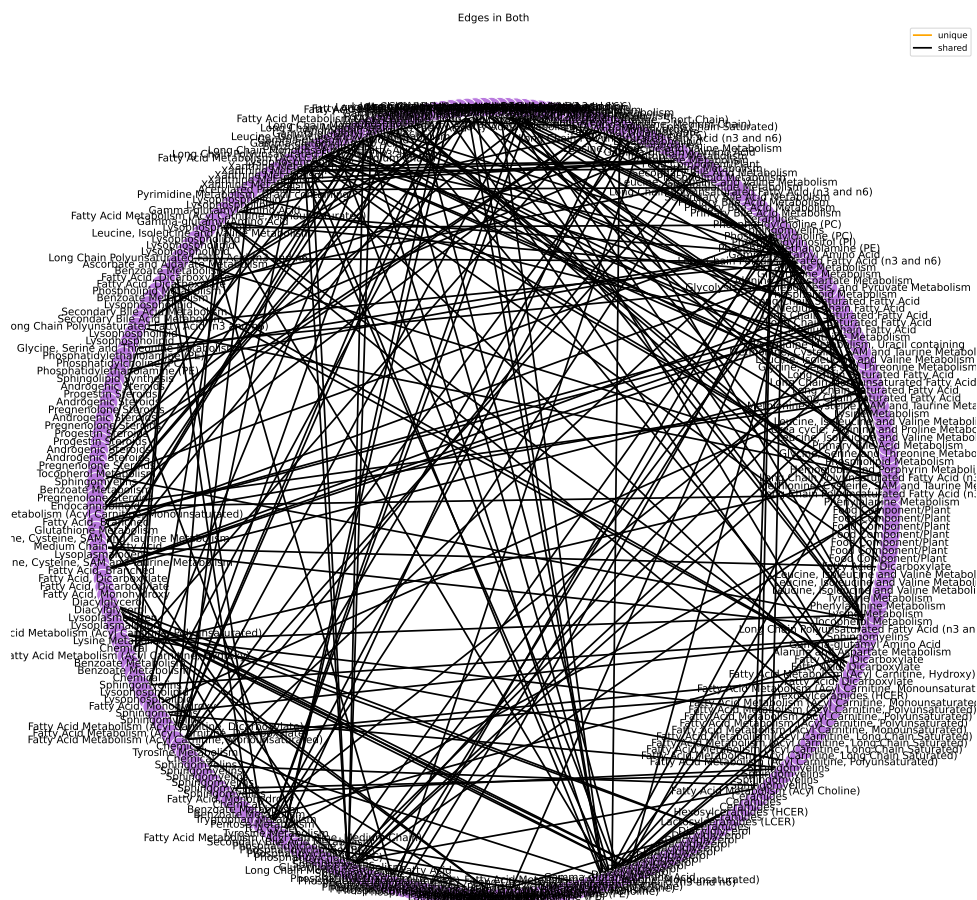
Super-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Lipid	363	215	4.577	1.521	2.82e-23	2.54e-22
Peptide	25	11	1.14	0.131	0.45	1

results/006/enrichment/intersection-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Long Chain Polyunsaturated Fatty Acid (n3 and n6)	14	14	Inf	Inf	3.07e-06	0.000319
Diacylglycerol	19	17	12.903	2.557	1.29e-05	0.000671
Phosphatidylethanolamine (PE)	11	10	14.858	2.699	0.000839	0.0225
Sphingomyelins	28	20	3.786	1.331	0.000866	0.0225
Plasmalogen	11	9	6.654	1.895	0.00671	0.129
Lysophospholipid	24	16	2.989	1.095	0.00868	0.129
Fatty Acid Metabolism (Acyl Carnitine, Monounsaturated)	8	7	10.299	2.332	0.00951	0.129
Fatty Acid Metabolism (Acyl Carnitine, Medium Chain)	5	5	Inf	Inf	0.0112	0.129
Fatty Acid Metabolism (Acyl Carnitine, Polyunsaturated)	5	5	Inf	Inf	0.0112	0.129
Long Chain Monounsaturated Fatty Acid	7	6	8.8	2.175	0.0208	0.197
Long Chain Saturated Fatty Acid	7	6	8.8	2.175	0.0208	0.197
Fatty Acid Metabolism (Acyl Choline)	4	4	Inf	Inf	0.0277	0.222
Pregnenolone Steroids	4	4	Inf	Inf	0.0277	0.222
Fatty Acid Metabolism (Acyl Carnitine, Long Chain Saturated)	8	6	4.393	1.48	0.0546	0.406
Ascorbate and Aldarate Metabolism	3	3	Inf	Inf	0.068	0.442
Lysoplasmalogen	3	3	Inf	Inf	0.068	0.442
Phosphatidylcholine (PC)	19	11	2.022	0.704	0.0995	0.595
Gamma-glutamyl Amino Acid	17	10	2.098	0.741	0.103	0.595
Androgenic Steroids	14	8	1.95	0.668	0.165	0.795
Xanthine Metabolism	14	8	1.95	0.668	0.165	0.795
Fatty Acid, Branched	2	2	Inf	Inf	0.167	0.795
Primary Bile Acid Metabolism	8	5	2.429	0.887	0.187	0.795
Progestin Steroids	4	3	4.36	1.472	0.189	0.795
Phosphatidylinositol (PI)	6	4	2.911	1.068	0.191	0.795
Phospholipid Metabolism	6	4	2.911	1.068	0.191	0.795
Ceramides	11	6	1.747	0.558	0.265	1
Glutathione Metabolism	7	4	1.937	0.661	0.307	1

Phenylalanine Metabolism	7	4	1.937	0.661	0.307	1
Fatty Acid Metabolism (Acyl Carnitine, Hydroxy)	3	2	2.899	1.064	0.365	1
Lactosylceramides (LCER)	3	2	2.899	1.064	0.365	1
Fatty Acid Metabolism (Acyl Carnitine, Short Chain)	1	1	Inf	Inf	0.409	1
Ketone Bodies	1	1	Inf	Inf	0.409	1
Medium Chain Fatty Acid	8	4	1.45	0.372	0.427	1
Fatty Acid, Dicarboxylate	23	10	1.115	0.109	0.479	1

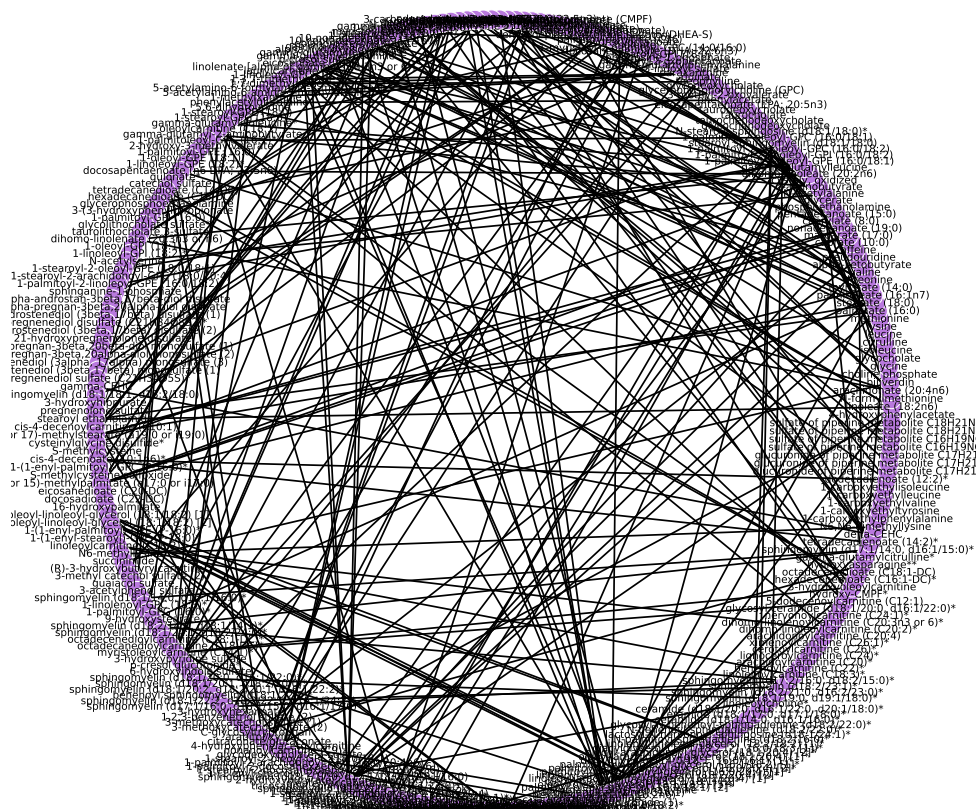
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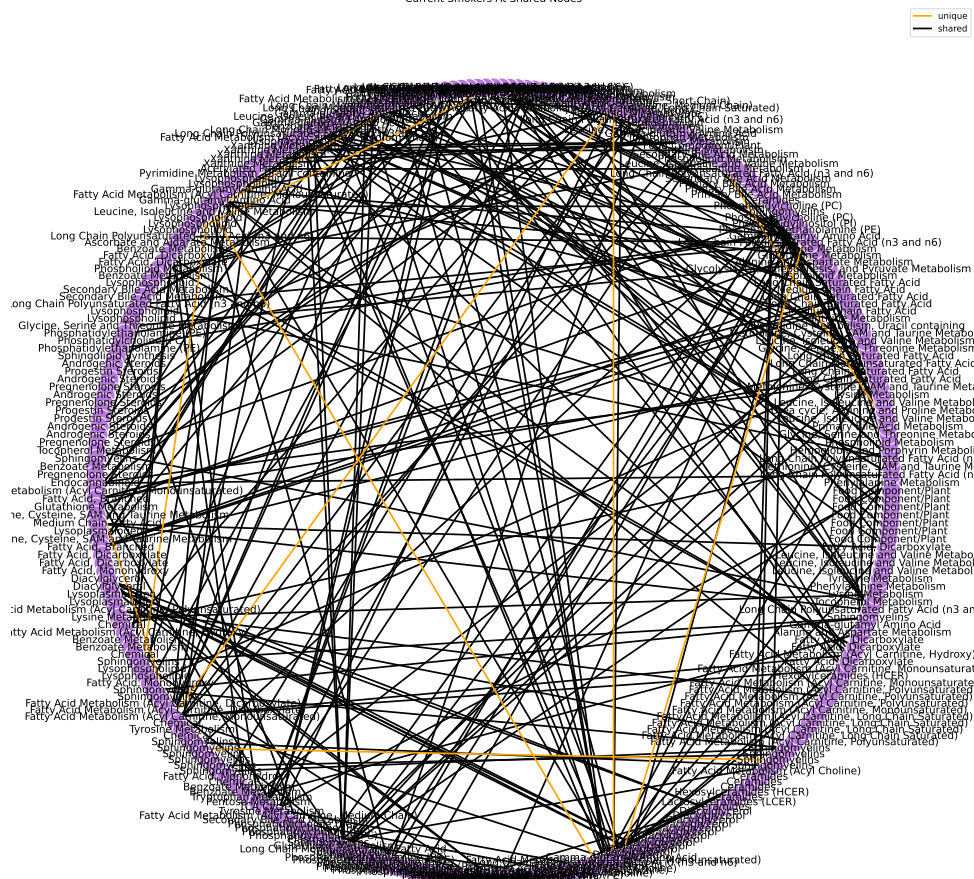
— unique
— shared

[results/005_1/sub_pathway-edge-shared-graph.pdf](#)

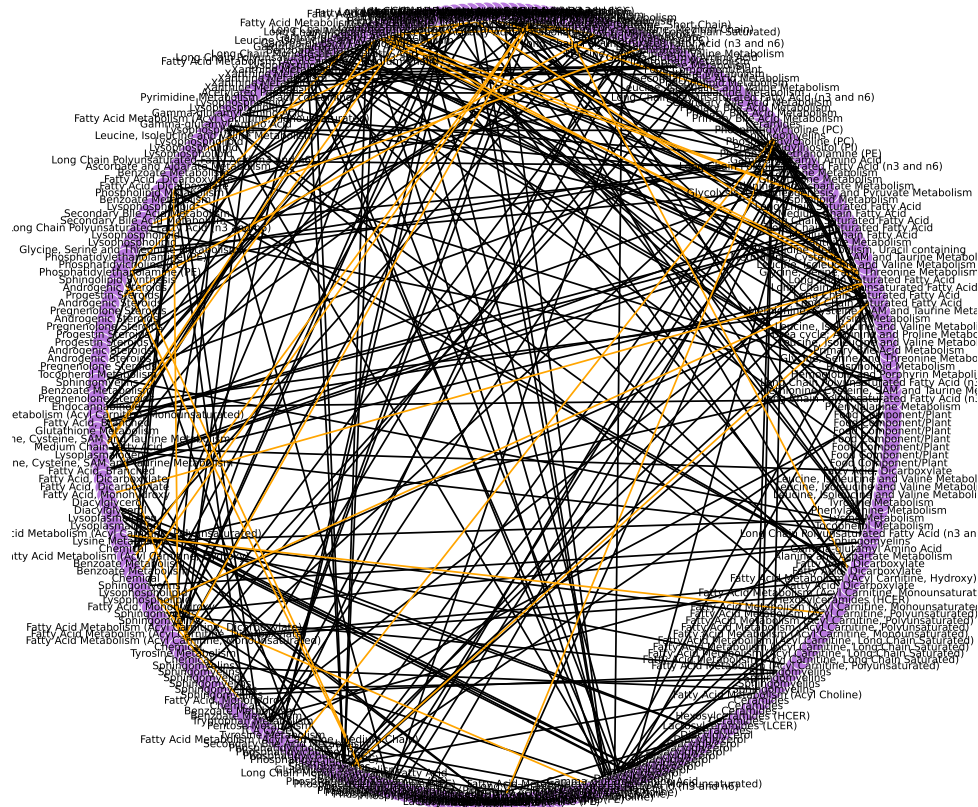
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Current Smokers At Shared Nodes

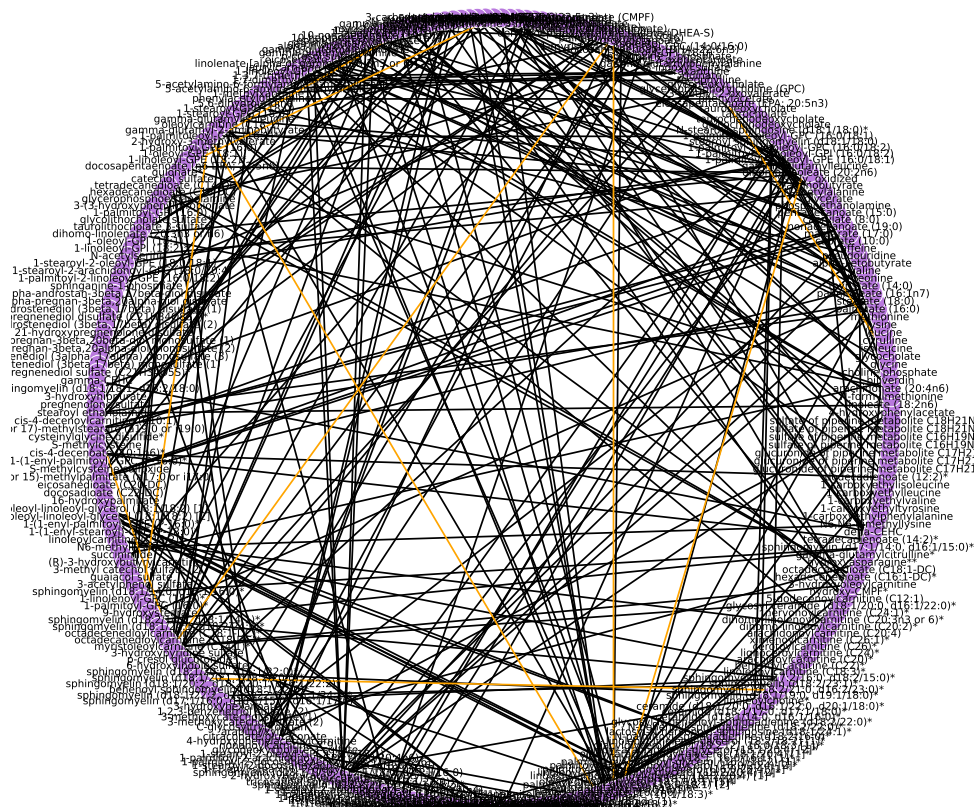


— unique
— shared

[results/005_1/sub_pathway-node-shared-form-graph.pdf](#)

Current Smokers At Shared Nodes

— unique
— shared



results/005_1/chemical_name-node-shared-curr-graph.pdf

Unique Edges (in orange)

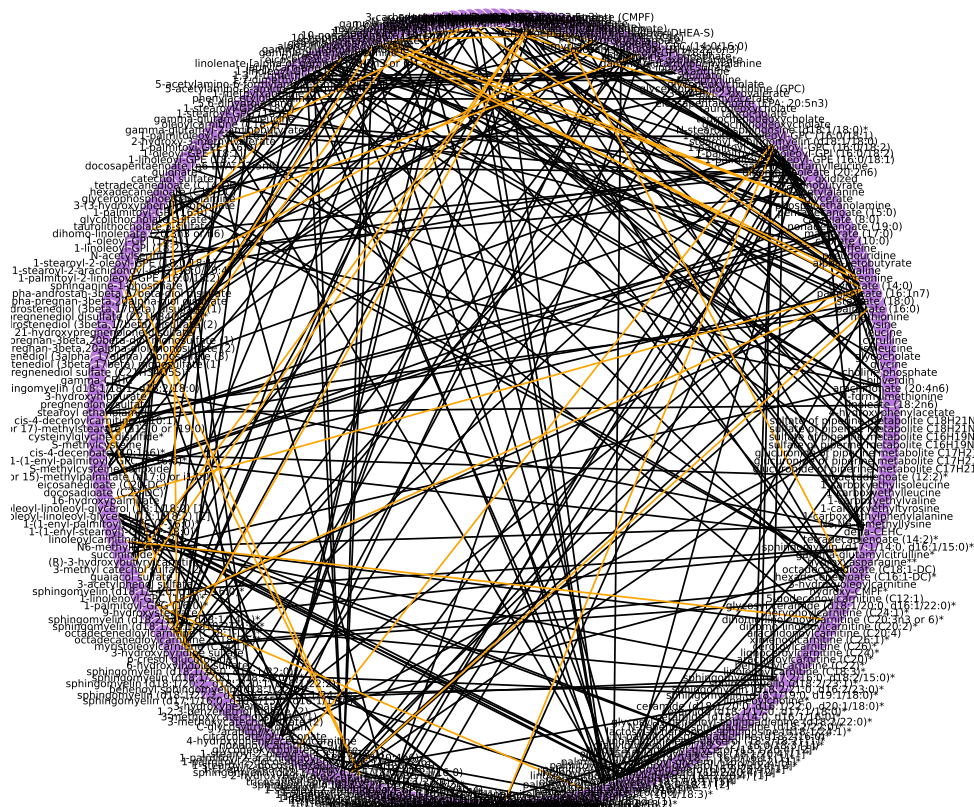
Primary Bile Acid Metabolism glycocholate	Secondary Bile Acid Metabolism glycodeoxycholate
Gamma-glutamyl Amino Acid gamma-glutamylleucine	Gamma-glutamyl Amino Acid gamma-glutamyl-alpha-lysine
Primary Bile Acid Metabolism glycochenodeoxycholate	Secondary Bile Acid Metabolism glycodeoxycholate
Fatty Acid Metabolism (Acyl Carnitine, Medium Chain) hexanoylcarnitine (C6)	Fatty Acid Metabolism (Acyl Carnitine, Monounsaturated) myristoleoylcarnitine (C14:1)*
Fatty Acid Metabolism (Acyl Carnitine, Medium Chain) hexanoylcarnitine (C6)	Fatty Acid Metabolism (Acyl Carnitine, Monounsaturated) palmitoleoylcarnitine (C16:1)*

Long Chain Polyunsaturated Fatty Acid (n3 and n6) docosapentaenoate (n3 DPA; 22:5n3)	Long Chain Polyunsaturated Fatty Acid (n3 and n6) adrenate (22:4n6)
Long Chain Polyunsaturated Fatty Acid (n3 and n6) docosadienoate (22:2n6)	Long Chain Polyunsaturated Fatty Acid (n3 and n6) docosapentaenoate (n6 DPA; 22:5n6)
Lysophospholipid 1-palmitoyl-GPC (16:0)	Lysophospholipid 1-palmitoyl-GPE (16:0)
Xanthine Metabolism 1,7-dimethylurate	Xanthine Metabolism 5-acetylamino-6-amino-3-methyluracil
Lysophospholipid 1-palmitoleoyl-GPC (16:1)*	Lysophospholipid 1-linolenoyl-GPC (18:3)*
Lysophospholipid 1-palmitoleoyl-GPC (16:1)*	Phosphatidylcholine (PC) 1-palmitoleoyl-2-linolenoyl-GPC (16:1/18:3)*
Fatty Acid Metabolism (Acyl Carnitine, Monounsaturated) cis-4-decenoylcarnitine (C10:1)	Fatty Acid Metabolism (Acyl Carnitine, Monounsaturated) myristoleoylcarnitine (C14:1)*
Sphingomyelins sphingomyelin (d18:1/20:1, d18:2/20:0)*	Sphingomyelins sphingomyelin (d18:2/21:0, d16:2/23:0)*
Diacylglycerol palmitoyl-oleoyl-glycerol (16:0/18:1) [2]*	Diacylglycerol palmitoyl-arachidonoyl-glycerol (16:0/20:4) [2]*
Diacylglycerol oleoyl-oleoyl-glycerol (18:1/18:1) [2]*	Diacylglycerol diacylglycerol (16:1/18:2 [2], 16:0/18:3 [1])*
Sphingomyelins sphingomyelin (d18:2/21:0, d16:2/23:0)*	Sphingomyelins sphingomyelin (d18:2/23:1)*

results/005/curr-metabs/unique-edges-at-shared-nodes.txt

Former Smokers At Shared Nodes

— unique
— shared



results/005_1/chemical_name-node-shared-form-graph.pdf

Unique Edges (in orange)

Long Chain Saturated Fatty Acid stearate (18:0)	Long Chain Monounsaturated Fatty Acid oleate/vaccenate (18:1)
Long Chain Monounsaturated Fatty Acid palmitoleate (16:1n7)	Long Chain Saturated Fatty Acid myristate (14:0)
Long Chain Monounsaturated Fatty Acid palmitoleate (16:1n7)	Long Chain Monounsaturated Fatty Acid 10-nonadecenoate (19:1n9)
Long Chain Saturated Fatty Acid myristate (14:0)	Long Chain Saturated Fatty Acid margarate (17:0)
Long Chain Saturated Fatty Acid myristate (14:0)	Medium Chain Fatty Acid 5-dodecenoate (12:1n7)

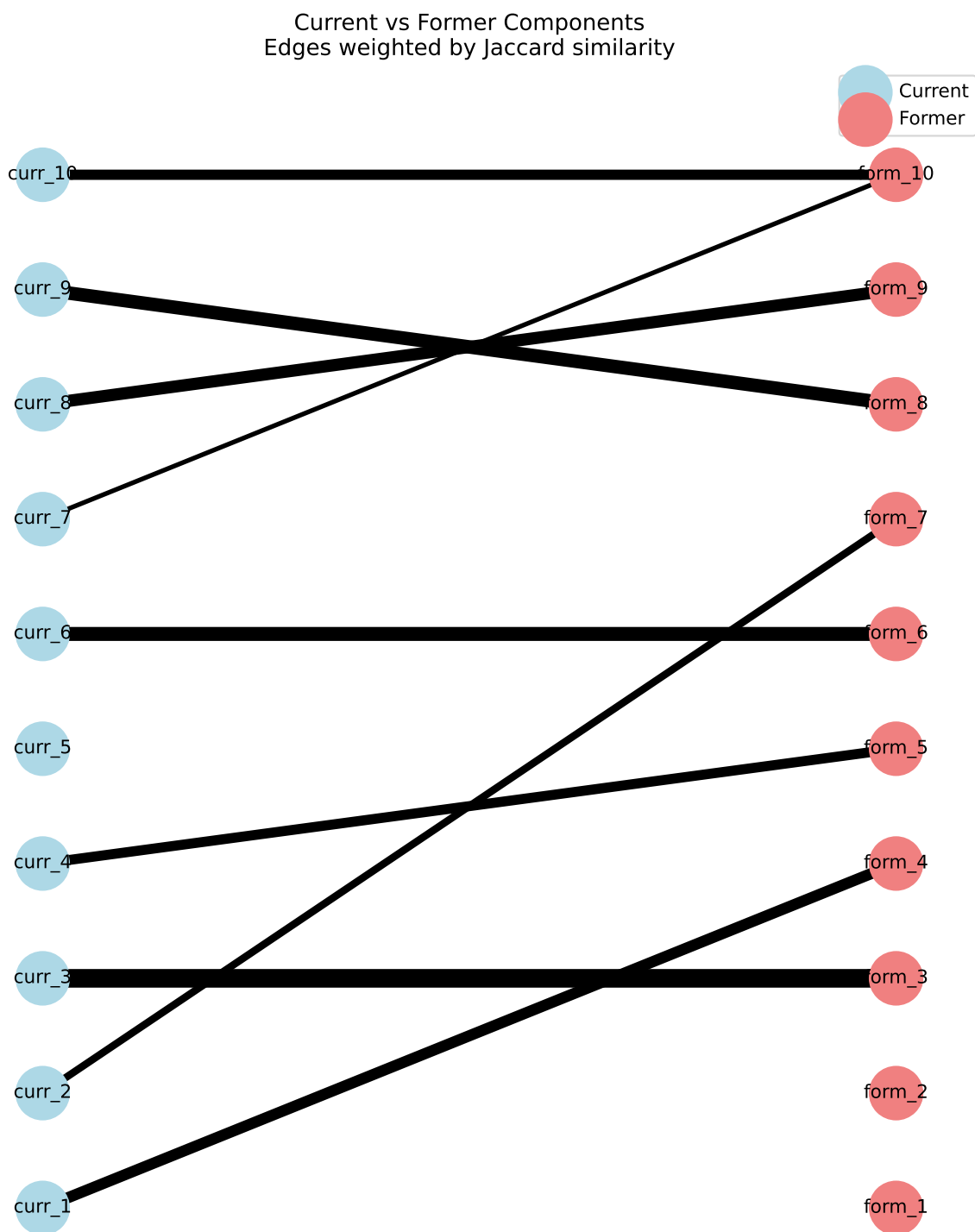
Long Chain Saturated Fatty Acid myristate (14:0)	Long Chain Monounsaturated Fatty Acid 10-heptadecenoate (17:1n7)
Long Chain Saturated Fatty Acid myristate (14:0)	Fatty Acid, Monohydroxy 16-hydroxypalmitate
Long Chain Saturated Fatty Acid myristate (14:0)	Fatty Acid, Monohydroxy 9-hydroxystearate
Long Chain Saturated Fatty Acid pentadecanoate (15:0)	Long Chain Monounsaturated Fatty Acid 10-nonadecenoate (19:1n9)
Long Chain Saturated Fatty Acid pentadecanoate (15:0)	Long Chain Monounsaturated Fatty Acid 10-heptadecenoate (17:1n7)
Phosphatidylcholine (PC) 1-palmitoyl-2-oleoyl-GPC (16:0/18:1)	Phosphatidylcholine (PC) 1-stearoyl-2-oleoyl-GPC (18:0/18:1)
Phosphatidylcholine (PC) 1-palmitoyl-2-oleoyl-GPC (16:0/18:1)	Phosphatidylinositol (PI) 1-palmitoyl-2-oleoyl-GPI (16:0/18:1)*
Long Chain Polyunsaturated Fatty Acid (n3 and n6) eicosapentaenoate (EPA; 20:5n3)	Long Chain Polyunsaturated Fatty Acid (n3 and n6) docosahexaenoate (DHA; 22:6n3)
Long Chain Polyunsaturated Fatty Acid (n3 and n6) eicosapentaenoate (EPA; 20:5n3)	Long Chain Polyunsaturated Fatty Acid (n3 and n6) stearidonate (18:4n3)
Xanthine Metabolism theophylline	Xanthine Metabolism paraxanthine
Long Chain Polyunsaturated Fatty Acid (n3 and n6) docosahexaenoate (DHA; 22:6n3)	Long Chain Polyunsaturated Fatty Acid (n3 and n6) stearidonate (18:4n3)
Long Chain Polyunsaturated Fatty Acid (n3 and n6) docosahexaenoate (DHA; 22:6n3)	Long Chain Polyunsaturated Fatty Acid (n3 and n6) dihomo-linolenate (20:3n3 or n6)
Fatty Acid, Dicarboxylate maleate	Food Component/Plant N-(2-furoyl)glycine
Androgenic Steroids dehydroepiandrosterone sulfate (DHEA-S)	Androgenic Steroids epiandrosterone sulfate
Androgenic Steroids dehydroepiandrosterone sulfate (DHEA-S)	Androgenic Steroids androstenediol (3beta,17beta) disulfate (1)
Fatty Acid, Monohydroxy 3-hydroxydecanoate	Fatty Acid Metabolism (Acyl Carnitine, Hydroxy) 3-hydroxyoleoylcarnitine
Androgenic Steroids androsterone sulfate	Androgenic Steroids epiandrosterone sulfate
Food Component/Plant N-(2-furoyl)glycine	Benzoate Metabolism 3-methyl catechol sulfate (2)
Food Component/Plant N-(2-furoyl)glycine	Chemical 3-acetylphenol sulfate
Food Component/Plant N-(2-furoyl)glycine	Chemical 3-hydroxypyridine sulfate
Long Chain Polyunsaturated Fatty Acid (n3 and n6) adrenate (22:4n6)	Long Chain Polyunsaturated Fatty Acid (n3 and n6) docosapentaenoate (n6 DPA; 22:5n6)
Medium Chain Fatty Acid 5-dodecenoate (12:1n7)	Long Chain Monounsaturated Fatty Acid 10-heptadecenoate (17:1n7)
Lysophospholipid 1-palmitoyl-GPC (16:0)	Lysoplasmalogen 1-(1-enyl-palmitoyl)-GPC (P-16:0)*

Lysophospholipid 1-stearoyl-GPC (18:0)	Lysophospholipid 1-oleoyl-GPC (18:1)
Lysophospholipid 1-oleoyl-GPC (18:1)	Lysoplasmalogen 1-(1-enyl-palmitoyl)-GPC (P-16:0)*
Androgenic Steroids epiandrosterone sulfate	Androgenic Steroids 5alpha-androstan-3beta,17beta-diol disulfate
Benzoate Metabolism catechol sulfate	Chemical 3-hydroxypyridine sulfate
Lysophospholipid 1-palmitoyl-GPI (16:0)	Lysophospholipid 1-oleoyl-GPI (18:1)
Phosphatidylethanolamine (PE) 1-stearoyl-2-oleoyl-GPE (18:0/18:1)	Phosphatidylethanolamine (PE) 1-stearoyl-2-linoleoyl-GPE (18:0/18:2)*
Phosphatidylethanolamine (PE) 1-palmitoyl-2-linoleoyl-GPE (16:0/18:2)	Phosphatidylethanolamine (PE) 1-stearoyl-2-linoleoyl-GPE (18:0/18:2)*
Sphingomyelins sphingomyelin (d18:1/18:1, d18:2/18:0)	Sphingomyelins sphingomyelin (d18:1/20:1, d18:2/20:0)*
Lysoplasmalogen 1-(1-enyl-palmitoyl)-GPC (P-16:0)*	Lysoplasmalogen 1-(1-enyl-palmitoyl)-GPE (P-16:0)*
Lysoplasmalogen 1-(1-enyl-palmitoyl)-GPC (P-16:0)*	Lysoplasmalogen 1-(1-enyl-stearoyl)-GPE (P-18:0)*
Fatty Acid Metabolism (Acyl Carnitine, Polyunsaturated) linoleoylcarnitine (C18:2)*	Fatty Acid Metabolism (Acyl Carnitine, Polyunsaturated) dihomo-linolenoylcarnitine (C20:3n3 or 6)*
Benzoate Metabolism 3-methyl catechol sulfate (2)	Chemical 3-hydroxypyridine sulfate
Benzoate Metabolism 3-methyl catechol sulfate (2)	TCA Cycle citrate/alpha-ketoglutarate
Benzoate Metabolism guaiacol sulfate	Chemical 3-hydroxypyridine sulfate
Benzoate Metabolism guaiacol sulfate	Benzoate Metabolism 3-methoxycatechol sulfate (2)
Chemical 3-acetylphenol sulfate	Chemical 3-hydroxypyridine sulfate
Chemical 3-acetylphenol sulfate	Benzoate Metabolism 3-methoxycatechol sulfate (2)
Sphingomyelins sphingomyelin (d18:1/14:0, d16:1/16:0)*	Sphingomyelins sphingomyelin (d17:1/16:0, d18:1/15:0, d16:1/17:0)*
Lysophospholipid 1-linolenoyl-GPC (18:3)*	Phosphatidylcholine (PC) 1-palmitoleoyl-2-linolenoyl-GPC (16:1/18:3)*
Phosphatidylethanolamine (PE) 1-stearoyl-2-linoleoyl-GPE (18:0/18:2)*	Phosphatidylethanolamine (PE) 1-stearoyl-2-arachidonoyl-GPE (18:0/20:4)
Phosphatidylethanolamine (PE) 1-palmitoyl-2-arachidonoyl-GPE (16:0/20:4)*	Phosphatidylethanolamine (PE) 1-oleoyl-2-arachidonoyl-GPE (18:1/20:4)*
Plasmalogen 1-(1-enyl-palmitoyl)-2-linoleoyl-GPE (P-16:0/18:2)*	Plasmalogen 1-(1-enyl-stearoyl)-2-linoleoyl-GPE (P-18:0/18:2)*
Diacylglycerol palmitoyl-oleoyl-glycerol (16:0/18:1) [2]*	Diacylglycerol oleoyl-oleoyl-glycerol (18:1/18:1) [1]*

Diacylglycerol palmitoyl-oleoyl-glycerol (16:0/18:1) [2]*	Diacylglycerol oleoyl-oleoyl-glycerol (18:1/18:1) [2]*
Tyrosine Metabolism 1-carboxyethyltyrosine	Leucine, Isoleucine and Valine Metabolism 1-carboxyethylvaline

results/005/form-metabs/unique-edges-at-shared-nodes.txt

Smoker Component Pairs



Current Smoker Component Metabolites

Component 1: N-acetylserine, N-acetylalanine, N-formylmethionine, 5,6-dihydrouridine, hydroxyasparagine**, C-glycosyltryptophan, pseudouridine,

Component 2: 3-hydroxypyridine sulfate, trigonelline (N'-methylnicotinate), citraconate/glutaconate, maleate, N-(2-furoyl)glycine, caffeic acid sulfate, quinate,

Component 3: 5-acetylamino-6-amino-3-methyluracil, 1-methylxanthine, theophylline, 1,7-dimethylurate, paraxanthine, caffeine, 1,3,7-trimethylurate, 5-acetylamino-6-formylamino-3-methyluracil,

Component 4: pregnenolone sulfate, pregnenediol disulfate (C₂₁H₃₄O₈S₂)*, 5alpha-androstan-3beta,17beta-diol disulfate, 21-hydroxypregnenolone disulfate, pregnenediol sulfate (C₂₁H₃₄O₅S)*, androstenediol (3beta,17beta) disulfate (2), dehydroepiandrosterone sulfate (DHEA-S), androstenediol (3beta,17beta) monosulfate (1), androstenediol (3beta,17beta) disulfate (1),

Component 5: valine, gamma-glutamylleucine, gamma-glutamyl-epsilon-lysine, isoleucine, gamma-glutamyl-alpha-lysine, leucine, gamma-glutamylphenylalanine, gamma-glutamylmethionine, methionine, lysine, gamma-glutamylisoleucine*,

Component 6: sphingomyelin (d18:1/20:0, d16:1/22:0)*, stearoyl sphingomyelin (d18:1/18:0), sphingomyelin (d18:1/20:1, d18:2/20:0)*, sphingomyelin (d18:1/18:1, d18:2/18:0), sphingomyelin (d18:2/23:1)*, sphingomyelin (d18:1/22:2, d18:2/22:1, d16:1/24:2)*, sphingomyelin (d18:2/14:0, d18:1/14:1)*, sphingomyelin (d18:2/21:0, d16:2/23:0)*, sphingomyelin (d17:1/14:0, d16:1/15:0)*, sphingomyelin (d18:1/14:0, d16:1/16:0)*, sphingomyelin (d18:1/20:2, d18:2/20:1, d16:1/22:2)*, sphingomyelin (d18:2/18:1)*, sphingomyelin (d18:1/19:0, d19:1/18:0)*, myristoyl dihydrosphingomyelin (d18:0/14:0)*, sphingomyelin (d18:1/17:0, d17:1/18:0, d19:1/16:0), sphingomyelin (d17:2/16:0, d18:2/15:0)*, sphingomyelin (d17:1/16:0, d18:1/15:0, d16:1/17:0)*,

Component 7: nonanoylcarnitine (C₉), arachidonoylcarnitine (C_{20:4}), myristoylcarnitine (C₁₄), oleoylcarnitine (C_{18:1}), dihomo-linolenoylcarnitine (C_{20:3n3} or 6)*, cis-4-decenoylcarnitine (C_{10:1}), 3-hydroxyoleoylcarnitine, palmitoylcarnitine (C₁₆), myristoleoylcarnitine (C_{14:1})*, hexanoylcarnitine (C₆), dihomo-linoleoylcarnitine (C_{20:2})*, 5-dodecenoylcarnitine (C_{12:1}), octanoylcarnitine (C₈), linoleoylcarnitine (C_{18:2})*, linolenoylcarnitine (C_{18:3})*, palmitoleoylcarnitine (C_{16:1})*, laurylcarnitine (C₁₂), decanoylcarnitine (C₁₀),

Component 8: 1-(1-enyl-palmitoyl)-GPC (P-16:0)*, 1-palmitoyl-2-linoleoyl-GPI (16:0/18:2), 1-palmitoyl-2-oleoyl-GPI (16:0/18:1)*, 1-stearoyl-GPC (18:0), 1-stearoyl-GPE (18:0), 1-linoleoyl-GPE (18:2)*, 1-palmitoyl-GPE (16:0), 1-palmitoleoyl-2-linolenoyl-GPC (16:1/18:3)*, 1-palmitoyl-GPC (16:0), 1-stearoyl-2-oleoyl-GPC (18:0/18:1), 1-oleoyl-GPC (18:1), 1-palmitoyl-2-oleoyl-GPC (16:0/18:1), 1-stearoyl-2-linoleoyl-GPI (18:0/18:2), 1-myristoyl-2-palmitoyl-GPC (14:0/16:0), 1-linolenoyl-GPC (18:3)*, 1-stearoyl-2-oleoyl-GPI (18:0/18:1)*, 1-palmitoyl-2-palmitoleoyl-GPC (16:0/16:1)*, 1-oleoyl-GPE (18:1), 1-palmitoleoyl-GPC (16:1)*, 1-linoleoyl-GPC (18:2),

Component 9: oleoyl-linoleoyl-glycerol (18:1/18:2) [2], 1-oleoyl-2-arachidonoyl-GPE (18:1/20:4)*, palmitoyl-linoleoyl-glycerol (16:0/18:2) [1]*, 1-stearoyl-2-arachidonoyl-GPE (18:0/20:4), palmitoyl-linoleoyl-glycerol (16:0/18:2) [2]*, 1-stearoyl-2-oleoyl-GPE (18:0/18:1), oleoyl-oleoyl-glycerol (18:1/18:1) [1]*, 1-palmitoyl-2-oleoyl-GPE (16:0/18:1), palmitoyl-arachidonoyl-glycerol (16:0/20:4) [2]*, diacylglycerol (16:1/18:2 [2], 16:0/18:3 [1])* , 1-palmitoyl-2-linoleoyl-GPE (16:0/18:2), diacylglycerol (14:0/18:1, 16:0/16:1) [1]*, 1-palmitoyl-2-arachidonoyl-GPE (16:0/20:4)*, oleoyl-oleoyl-glycerol (18:1/18:1) [2]*, linoleoyl-arachidonoyl-glycerol (18:2/20:4) [2]*, palmitoyl-arachidonoyl-glycerol (16:0/20:4) [1]*, palmitoleoyl-linoleoyl-glycerol (16:1/18:2) [1]*, oleoyl-linoleoyl-glycerol (18:1/18:2) [1], palmitoyl-oleoyl-glycerol (16:0/18:1) [2]*, linoleoyl-linoleoyl-glycerol (18:2/18:2) [1]*, oleoyl-arachidonoyl-glycerol (18:1/20:4) [2]*,

Component 10: dihomo-linolenate (20:3n3 or n6), dodecadienoate (12:2)*, docosapentaenoate (n6 DPA; 22:5n6), 3-hydroxymyristate, palmitoleate (16:1n7), dihomo-linoleate (20:2n6), oleate/vaccenate (18:1), docosadienoate (22:2n6), arachidonate (20:4n6), 10-nonadecenoate (19:1n9), docosahexaenoate (DHA; 22:6n3), linolenate [alpha or gamma; (18:3n3 or 6)], docosapentaenoate (n3 DPA; 22:5n3), 3-hydroxybutyrate (BHBA), (14 or 15)-methylpalmitate (a17:0 or i17:0), myristoleate (14:1n5), nonadecanoate (19:0), stearate (18:0), cis-4-decenoate (10:1n6)*, eicosapentaenoate (EPA; 20:5n3), hexadecadienoate (16:2n6), palmitate (16:0), 3-hydroxylaurate, stearoyl ethanolamide, (16 or 17)-methylstearate (a19:0 or i19:0), 9-hydroxystearate, 10-heptadecenoate (17:1n7), myristate (14:0), 3-hydroxyhexanoate, 16-hydroxypalmitate, linoleate (18:2n6), stearidonate (18:4n3), pentadecanoate (15:0), 3-hydroxydecanoate, eicosenoate (20:1), margarate (17:0), 5-dodecenoate (12:1n7), tetradecadienoate (14:2)*, adrenate (22:4n6), 3-hydroxyoctanoate,

Former Smoker Component Metabolites

Component 1: sulfate of piperine metabolite C₁₆H₁₉NO₃ (2)*, glucuronide of piperine metabolite C₁₇H₂₁NO₃ (4)*, glucuronide of piperine metabolite C₁₇H₂₁NO₃ (3)*, glucuronide of piperine metabolite C₁₇H₂₁NO₃ (5)*, piperine, sulfate of piperine metabolite C₁₆H₁₉NO₃ (3)*,

Component 2: 1-(1-enyl-palmitoyl)-2-oleoyl-GPE (P-16:0/18:1)*, 1-(1-enyl-palmitoyl)-2-arachidonoyl-GPE (P-16:0/20:4)*, 1-(1-enyl-palmitoyl)-2-linoleoyl-GPE (P-16:0/18:2)*, 1-(1-enyl-stearoyl)-2-arachidonoyl-GPE (P-18:0/20:4)*, 1-(1-enyl-stearoyl)-2-linoleoyl-GPE (P-18:0/18:2)*, 1-(1-enyl-stearoyl)-2-oleoyl-GPE (P-18:0/18:1),

Component 3: 5-acetylamino-6-amino-3-methyluracil, 1-methylxanthine, theophylline, 1,7-dimethylurate, paraxanthine, caffeine, 1,3,7-trimethylurate, 5-acetylamino-6-formylamino-3-methyluracil,

Component 4: N-acetylserine, N-acetylalanine, N-formylmethionine, gamma-carboxyglutamate, 5,6-dihydrouridine, hydroxyasparagine**, C-glycosyltryptophan, N6-carbamoylthreonyladenine, 2,3-dihydroxy-5-methylthio-4-pentenoate (DMTPA)*, N2,N2-dimethylguanosine, pseudouridine,

Component 5: pregnenolone sulfate, 5alpha-androstan-3beta,17beta-diol disulfate, pregnenediol disulfate (C21H34O8S2)*, androsterone sulfate, epiandrosterone sulfate, 21-hydroxypregnenolone disulfate, androstenediol (3alpha, 17alpha) monosulfate (3), pregnenediol sulfate (C21H34O5S)*, androstenediol (3beta,17beta) disulfate (2), dehydroepiandrosterone sulfate (DHEA-S), androstenediol (3beta,17beta) monosulfate (1), androstenediol (3beta,17beta) monosulfate (2), androstenediol (3beta,17beta) disulfate (1),

Component 6: sphingomyelin (d18:1/20:0, d16:1/22:0)*, stearoyl sphingomyelin (d18:1/18:0), sphingomyelin (d18:1/20:1, d18:2/20:0)*, sphingomyelin (d18:1/18:1, d18:2/18:0), sphingomyelin (d18:2/14:0, d18:1/14:1)*, sphingomyelin (d18:2/21:0, d16:2/23:0)*, sphingomyelin (d17:1/14:0, d16:1/15:0)*, sphingomyelin (d18:1/14:0, d16:1/16:0)*, myristoyl dihydrosphingomyelin (d18:0/14:0)*, sphingomyelin (d18:1/19:0, d19:1/18:0)*, sphingomyelin (d18:1/17:0, d17:1/18:0, d19:1/16:0), sphingomyelin (d17:2/16:0, d18:2/15:0)*, sphingomyelin (d17:1/16:0, d18:1/15:0, d16:1/17:0)*,

Component 7: 3-methyl catechol sulfate (1), 3-acetylphenol sulfate, guaiacol sulfate, catechol sulfate, 3-methoxycatechol sulfate (2), 3-hydroxypyridine sulfate, 3-ethylcatechol sulfate (1), 4-vinylguaiacol sulfate, trigonelline (N'-methylnicotinate), 4-methoxyphenol sulfate, citraconate/glutaconate, maleate, N-(2-furoyl)glycine, caffeic acid sulfate, 3-methyl catechol sulfate (2), quinate,

Component 8: oleoyl-linoleoyl-glycerol (18:1/18:2) [2], 1-oleoyl-2-arachidonoyl-GPE (18:1/20:4)*, palmitoyl-linoleoyl-glycerol (16:0/18:2) [1]*, 1-stearoyl-2-arachidonoyl-GPE (18:0/20:4), palmitoyl-linoleoyl-glycerol (16:0/18:2) [2]*, 1-stearoyl-2-oleoyl-GPE (18:0/18:1), 1-stearoyl-2-linoleoyl-GPE (18:0/18:2)*, oleoyl-oleoyl-glycerol (18:1/18:1) [1]*, 1-palmitoyl-2-oleoyl-GPE (16:0/18:1), diacylglycerol (16:1/18:2 [2], 16:0/18:3 [1])* , 1-palmitoyl-2-linoleoyl-GPE (16:0/18:2), diacylglycerol (14:0/18:1, 16:0/16:1) [1]*, 1-palmitoyl-2-arachidonoyl-GPE (16:0/20:4)*, oleoyl-oleoyl-glycerol (18:1/18:1) [2]*, 1-oleoyl-2-linoleoyl-GPE (18:1/18:2)*, palmitoleoyl-linoleoyl-glycerol (16:1/18:2) [1]*, oleoyl-linoleoyl-glycerol (18:1/18:2) [1], palmitoyl-oleoyl-glycerol (16:0/18:1) [2]*, linoleoyl-linoleoyl-glycerol (18:2/18:2) [1]*,

Component 9: 1-(1-enyl-palmitoyl)-GPC (P-16:0)*, 1-palmitoyl-2-linoleoyl-GPI (16:0/18:2), 1-palmitoyl-2-oleoyl-GPI (16:0/18:1)*, 1-stearoyl-GPC (18:0), 1-stearoyl-GPE (18:0), 1-(1-enyl-stearoyl)-GPE (P-18:0)*, 1-linoleoyl-GPE (18:2)*, 1-palmitoyl-GPE (16:0), 1-palmitoleoyl-2-linolenoyl-GPC (16:1/18:3)*, 1-(1-enyl-palmitoyl)-GPE (P-16:0)*, 1-linoleoyl-2-linolenoyl-GPC (18:2/18:3)*, 1-palmitoyl-GPC (16:0), 1-stearoyl-2-arachidonoyl-GPC (18:0/20:4), 1-arachidonoyl-GPE (20:4n6)*, 1-arachidonoyl-GPC (20:4n6)*, 1-stearoyl-2-oleoyl-GPC (18:0/18:1), 1,2-dilinoeloyl-GPC (18:2/18:2), 1-palmitoyl-2-arachidonoyl-GPI (16:0/20:4)*, 1-oleoyl-GPC (18:1), 1-palmitoyl-2-oleoyl-GPC (16:0/18:1), 1-stearoyl-2-linoleoyl-GPI (18:0/18:2), 1-myristoyl-2-palmitoyl-GPC (14:0/16:0), 1-linolenoyl-GPC (18:3)*, 1-stearoyl-2-oleoyl-GPI (18:0/18:1)*, 1-palmitoyl-2-palmitoleoyl-GPC (16:0/16:1)*, 1-oleoyl-GPE (18:1), 1-myristoyl-2-arachidonoyl-GPC (14:0/20:4)*, 1-palmitoyl-2-arachidonoyl-GPC (16:0/20:4n6), 1-palmitoleoyl-GPC (16:1)*, 1-linoleoyl-GPC (18:2),

Component 10: dihomolinenate (20:3n3 or n6), hexadecanedioate (C16-DC), dodecadienoate (12:2)*, docosapentaenoate (n6 DPA; 22:5n6), 3-hydroxymyristate, palmitoleate (16:1n7), margaroylcarnitine (C17)*, dihomolinoate (20:2n6), oleate/vaccenate (18:1), docosadienoate (22:2n6), arachidonate (20:4n6), hexadecenedioate (C16:1-DC)*, 10-nonadecenoate (19:1n9), docosahexaenoate (DHA; 22:6n3), linolenate [alpha or gamma; (18:3n3 or 6)], docosapentaenoate (n3 DPA; 22:5n3), myristoleoylcarnitine (C14:1)*, palmitoylcarnitine (C16), dihomolinoeloylcarnitine (C20:2)*, glycerol, palmitoleoylcarnitine (C16:1)*, (14 or 15)-methylpalmitate (a17:0 or i17:0), myristoleate (14:1n5), arachidonoylcarnitine (C20:4), nonadecanoate (19:0), stearate (18:0), cis-4-decenoate (10:1n6)*, eicosapentaenoate (EPA; 20:5n3), caprate (10:0), dihomolinoeloylcarnitine (C20:3n3 or 6)*, 3-hydroxyoleoylcarnitine, hexanoylcarnitine (C6), hexadecadienoate (16:2n6), palmitate (16:0), decanoylcarnitine (C10), 3-hydroxylaurate, tetradecanedioate (C14-DC), octadecenedioate (C18:1-DC), stearoyl ethanolamide, cis-4-decenoylcarnitine (C10:1), (16 or 17)-methylstearate (a19:0 or i19:0), 9-hydroxystearate, 10-heptadecenoate (17:1n7), myristate (14:0), 3-hydroxyhexanoate, octanoylcarnitine (C8), 5-dodecenoylcarnitine (C12:1), linoleoylcarnitine (C18:2)*, linolenoylcarnitine (C18:3)*, 16-hydroxypalmitate, linoleate (18:2n6), dodecanedioate (C12-DC), stearidonate (18:4n3), laurylcarnitine (C12),

pentadecanoate (15:0), nonanoylcarnitine (C9), 3-hydroxydecanoate, eicosenoate (20:1), margarate (17:0), myristoylcarnitine (C14), oleoylcarnitine (C18:1), 3-hydroxysebacate, 5-dodecenoate (12:1n7), caprylate (8:0), tetradecadienoate (14:2)*, adrenate (22:4n6), octadecanedioate (C18-DC), 3-hydroxyoctanoate, laurate (12:0),